Uppsala Studies in Human-Computer Interaction

8

Design for Societal Values

Co-Designing Cultural Interventions with Societal Value Stakeholders

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UPPSALA
UNIVERSITET

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Abstract

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Co-design endeavors with representatives from the public sector, NGOs, and other societal value stakeholders are common in social innovation projects. Value considerations are crucial for ensuring that such projects succeed and avoid breakdowns. Yet there is limited HCI research on how societal values are mediated in these collaborations. This thesis investigates this within the domain of designing cultural interventions in public settings.

The thesis adopts a Research through Design (RtD) approach, with three main design projects as empirical cases. The projects were followed over several years, from initiation through to implementation, and are analyzed through the lens of how societal values are mediated in design processes. The analysis is structured around three sensitizing constructs: Value Elicitation, Value Translation, and Value Manifestation.

The findings show that societal value stakeholders possess high value competence and a strong aptitude for engaging in value conversations, but were unfamiliar with interaction design practices. The study demonstrates that a relational approach, characterized by sustained informal conversations and creative, playful, and reflective methods, lowered prestige, built trust, and enabled complex value discussions. As values became manifested in sketches and prototypes, the value conversations gained increasing granularity. New societal value considerations also emerged throughout the design process and beyond implementation. Designing for societal value mediation was found to require continuous deliberation and interpretative work, rather than relying on fixed design requirements.

The thesis contributes insights into co-design with societal value stakeholders in social innovation projects. It positions designers as societal value mediators, both by conceptualizing this role theoretically and by proposing concrete methods for eliciting, translating, and manifesting societal values in the design of cultural interventions in public settings.

Keywords: Research through Design, Value Sensitive Design, Societal Values, Design Processes

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To Sigrid, whom I value most of all

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List of Papers

This thesis is based on the following papers, which are referred to in the text by their Roman numerals.

I. Johansson, K., & Back J. (2024). Daddy, You Can Be the Fox. Designing Hybrid Playgrounds for Nature Integrated Storytelling. In *Proceedings of the 13th Nordic Conference on Human-Computer Interaction*. (Article 27). Association for Computing Machinery. https://doi.org/10.1145/3679318.3685360

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IV. Johansson, K., Bandelin, J., & Back, J. (2025). The Quest for Learning. Pedagogical and Interaction Design Perspectives on Quest Rooms. In *Proceedings of the 19th European Conference on Games Based Learning*. (1125-1133). https://doi.org/10.34190/ecgbl.19.2.4035

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Johansson, K., & Morganti, M. (2026). Larp with Erasmus: A Quantitative Study of Live Action Roleplay Projects Funded by EUs Erasmus+ Program Between 2014-2022. *International Journal of Role-Playing*, 19, 103–121. https://doi.org/10.33063/ijrp.vi19.1254

Márquez Segura, E., Spiel, K., Johansson, K., Back, J., Toups, P., Hammer, J., Waern, A., Tanenbaum, T., & Isbister, K. (2019). Larping (Live Action Role Playing) as an Embodied Design Research Method. In *Companion Publication of the 2019 on Designing Interactive Systems Conference* (DIS '19). (389–392). Association for Computing Machinery. https://doi.org/10.1145/3301019.3320002

Paananen, S., Johansson, K., Bongo, L., Sayffaerth, C., Papachristos, E., Suoheimo, M., & Häkkilaä, J. Interactive Storytelling for Communities, Heritage and Public Engagement (2024). In *Adjunct Proceedings of the 2024 Nordic Conference on Human-Computer Interaction* (NordiCHI'24). (Article 36). Association for Computing Machinery. https://doi.org/10.1145/3677045.3685451

Waern, A., Rajkowska, P., Johansson, K., Back, J., Spence, J., & Sundnes Løvlie, A. (2020). Sensitizing Scenarios: Sensitizing Designer Teams to Theory. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (CHI '20). Association for Computing Machinery. https://doi.org/10.1145/3313831.3376620

Contributions to Articles

Article I. I contributed by writing most of the text, conducting the analysis, developing the topic for the article, and carrying out the annotations. Jon Back and I participated in the full design project and the data gathering, and Jon provided feedback on the text.

Article II. All the three authors participated in the design process and in the redesign events after the inauguration. Jon Back and I co-wrote the article. Jonas Hangvar provided input to the text and contributed particularly with technological insights. Jonas was also responsible for the coder summer camp and the related data collection.

Article III. I wrote the text, made the analysis, and facilitated the workshops. Anna Lindström contributed by conducting the value interviews, designing the probes and making the annotated illustrations.

Article IV. I led the writing of the article, with support from the other authors. Jon Back was the main RtD researcher for the first and second iterations of the design. I was a co-designer of the third iteration. I participated in user studies and interviews across all three iterations. Jacob Bandelin and I did the theorization, with support from Jon.

Article V. I led the writing of the article, conducted the literature review, and contributed one of the design exemplars. Most co-writing was done with Raquel Robinson. The author team together planned the article structure, conducted the analysis, contributed design exemplars, and gave feedback. Katherine Isbister took the lead in providing feedback during the final stages.

Article VI. Alexandros Rouchitsas and Didem Gürdür Broo planned and facilitated the workshop. I participated in the workshop and contributed insights from it. Alexandros and Didem took the lead on the writing, I contributed with the writing on design thinking, futuring, sustainability, and value perspectives. We all engaged in the analysis of the workshop.

Images and Credits

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Abbreviations

RtD	Research through Design
HCI	Human-Computer Interaction
SVS	Societal Value Stakeholders
VSD	Value Sensitive Design
DfV	Design for Values
PD	Participatory Design
NGO	Non-Governmental Organization

Chapter 1. Introduction

As has long been argued in design-related research, moral values can be actively manifested in and mediated through design (Latour, 1994; Flanagan et al. 2008; Papanek, 1985; Verbeek, 2011; Winner, 1980). Finding methods to incorporate such values into design processes has engaged HCI scholars for decades (Ehn et al., 1987; Friedman, 1996; Sengers et al., 2005; Stone, 2021). However, it has been argued that the research seldom distinguishes between diverse types of values, and that neither values nor stakeholders are clearly conceptualized (Manders-Huits 2011; Stone 2021). There is also a lack of studies that engage in real-world design work and investigate the relationship between value-oriented design approaches and actual design decisions (Friedman and Hendry, 2019).

This thesis conceptualizes a particular range of values – societal values – such as increasing a sense of community or promoting cultural heritage awareness. The thesis places societal values within the context of HCI and design, and investigates how conversations about societal values can be carried out in design projects and manifested in design solutions. It does so by following several long-term real-world design projects using a Research through Design (RtD) approach. In particular, the thesis examines co-design processes involving representatives of organizations that pursue societal values, such as stakeholders from the public sector, social enterprises, and non-governmental organizations (NGOs). These organizations are referred to in the thesis as societal value stakeholders (SVS).

A specific type of design endeavor that engages with societal values is the design of cultural interventions, such as museum installations, cultural events, and public artworks. These types of interventions are gaining increasing attention as policy tools for related to the values pursued by public sector organizations and NGOs (Yúdice, 2003). According to Durrer et al. (2003) local cultural place-making can also be seen as a form of policymaking, involving multiple stakeholders, and such cultural policymaking is rapidly changing due to increasing digitalization. As with other forms of digitalization, this means public sector organizations and NGOs that develop such interventions are increasingly engaging with design methods and collaborating with designers and technology developers. This requires design processes to be adapted so

that the values pursued by the organizations are meaningfully embedded within the resulting designs (Brown and Wyatt, 2010; Komatsu Cipriani et al., 2020; Poleac, 2023).

Design as a profession grew out of the design of marketable products (Margolin, 1995), but some design projects, such as those examined in this thesis, do not have a commercial or consumer focus. Instead, they seek to pursue moral values (Armstrong et al., 2014). Public sector organizations and NGOs engage in design projects based on the values their organizations seek to promote, thus operating according to a different logic from profit-driven stakeholders (Komatsu Cipriani et al., 2020). It is becoming increasingly common for these types of organizations to collaborate with designers and use design methods such as design thinking. However, since these methods have been developed for use in commercial contexts (Starostka et al., 2022; Junginger, 2017; Lewis et al., 2020), they might not be fully suitable and may require adaption in order to foster non-commercial values (Flaig et al., 2025; Komatsu et al., 2016). Research suggests that design collaborations with stakeholders whose intentions and priorities differ from those of commercial actors require adaptations to the design process to accommodate these differences (Flaig et al., 2025). Multisectoral collaborations, such as between designers and SVS, can contribute to positive societal transformation if processes are adapted to account for different stakeholder perspectives (Kuruvilla et al., 2018).

Furthermore, the value priorities of SVS may differ from those of users (Chen et al., 2022), and because SVS representatives often lack designerly training, they can struggle to articulate their concerns within design processes (Sanders and Strappers, 2008; Starostka et al., 2022). Applying methods that rely on clearly defined requirements and measurability risks undermining the soft values that are central to such organizations (Wæraas, 2018). At the same time, designers, typically lack profound understanding of contextual values when designing with and for public stakeholders, which can result in unsuccessful designs (Melles et al., 2011). Studies suggest that co-design with public stakeholders needs to focus on public policies and broader societal needs in ways that current design practices are not well adapted to (Poleac, 2023). A study of museum innovation projects demonstrated that differences in value priorities between SVS, in this case museum staff, and designers can lead to project breakdowns (Rajkowska, 2022).

Designing with SVS organizations constitutes a substantial domain of practice. The public sector directly employs one-tenth of the world's workforce and engages many more through public funding (International Labour Organization, 2024). Across OECD countries, government expenditure amounts, on average, to 46.3% of GDP (Organization for Economic Co-operation and

Development, n.d.). A report from the British Design Council states that 88% of public sector organizations recognize design as playing an important role in their practice, while also identifying a lack of design expertise within their organizations (Design Council, 2021). Beyond governmental organizations, there are an estimated 10 million NPOs and NGOs globally (Jesus Rios et al., 2023), and a study among European NGOs showed that 87.9% had participated in service design initiatives (Gesierich et al., 2024). A third type of mission-driven organization is the social enterprise – an organization that engages in economic activity but uses its profits primarily to generate social impact (Organization for Economic Co-operation and Development, 1999). According to the World Economic Forum (Bonnici, 2024), there are more than 10 million social enterprises globally, collectively employing over 200 million people.

Design for Social Innovation is a large and growing sector that often engages with design (Manzini, 2015). It can be defined as non-commercial collaborative projects aimed at developing new products, services, and models that improve society and enhance its capacity to act sustainably (Murray et al., 2010). The EU is a major investor in social innovation projects, and the EU Social Innovation Fund has a budget of 197 million euros (European Commission, 2025).

The scale of the domain is in itself a motivation for researching how to design with SVS, particularly given its connection to the allocation of public resources. There are also deeper moral implications. Studying design practices within such collaborations can be improved, contributing to positive societal change.

As SVS increasingly engage in design collaborations, there is a need to consider how their value aims can be realized within such projects. Research suggests that, in order to successfully pursue desired values in social innovation design projects, designers need to align with project values (Brown and Wyatt, 2010), bridge different stakeholder perspectives (Cibin et al., 2020), facilitate conversations about values and design (Manzini, 2015), and re-evaluate which design competencies are needed within such projects (Komatsu et al., 2016). However, there is limited knowledge of how to achieve this in practice, and the development of such specific value-related competencies is sufficiently emphasized in design education and practice (Manzini, 2015).

While there is extensive research on values and design, current value-oriented design research within HCI lacks a clear conceptualization of different types of values (Manders-Huits, 2011; Stone, 2021). One distinction that has gained attention outside HCI is that between human values and societal values. OECD (2021) defines societal values as shared principles that shape

social order and are incorporated in institutional structures, policy documents, and democratic practices. In this thesis, societal values are conceptualized in relation to design and HCI based on a collection of related concepts, such as collective values (Schwartz, 1992), public values (Moore, 2021), and the Swedish concept of värdegrund (Brytting, 2010).

Societal values can align with, but may also transgress or conflict with, the needs and values of individuals, and this has implications for value-oriented design. Users and individual values are a common focus within current value-oriented design research, such as Human-Centered Design (Norman, 2005) and Participatory Design (Bødker, 2022). Within Value Sensitive Design, values are defined in terms of what people need, often understood through their roles as users (Friedman and Hendry, 2019). It has been argued that user perspectives are promoted in design endeavors at the expense of other perspectives, such as societal perspectives (Norman, 2005; Kensing and Blomberg, 1998), and that an overly narrow focus on users can become problematic, as design methods need to be adapted when addressing complex societal issues (van der Bijl-Brouwer, 2016).

Design projects can become arenas for both the formulation and implementation of public values (Junginger, 2017), making designers participants in policymaking processes. Co-designing with SVS entails particular considerations, including the possibility that their value priorities might not align with those of end users (Chen et al., 2022), the need to address complex societal values (Wæraas, 2018), and the fact that representatives of such organizations typically have a high degree of moral value competence (Kavathatzopoulos, 2009). All of these factors have implications for co-design processes involving SVS.

Despite the impact of SVS, their increasing engagement with design, and their distinctive value orientations, the question of how to carry out value-oriented design collaborations with such stakeholders has received limited attention within HCI, particularly in relation to the design of cultural interventions. This thesis addresses that gap through a Research through Design approach and is guided by the research question stated below.

How can societal values be mediated in co-design projects with societal value stakeholders, particularly in the design of cultural interventions?

This is an important area of study because a lack of understanding of these perspectives can result in less successful designs, value considerations being lost during design processes, and even project breakdowns.

As articulated in the research question, the focus of this thesis is on societal value stakeholders rather than end users. This does not imply that an end-user focus is unimportant in such processes, or that end users were not consulted in the design projects included in the thesis. However, end-user involvement and analysis of how the resulting design solutions are received by the public are considered outside the scope of the thesis. An important limitation of this work is therefore its focus on the perspectives of SVS.

The term "co-design project" in this thesis refers to projects in which designers collaborate with non-designers as members of a shared project team. A designer is defined, for the purposes of this thesis, as a person who has received formal training in design and who applies design methods in a professional capacity. This can include designers working in design or tech companies, as well as Research through Design (RtD) researchers working in academia. This definition builds on the difference between professional and diffuse designers, as described by Manzini (2015).

The thesis is a compilation thesis based on a collection of RtD projects conducted within social innovation collaborations involving museums, municipalities, and non-governmental organizations. The main case studies are situated within a Swedish context.

Based on insights from several real-world co-design projects, the thesis presents a process-oriented framework for societal value mediation. These contributions can support future research on value-oriented design, while also providing practical insights on approaches and skills for designers working in collaboration with societal value stakeholders.

Design Project Overview

This section provides a brief overview of the design projects. In all projects, I have been actively involved as both a designer and researcher throughout their entire duration, from the formulation of the project applications to implementation. My contributions have included attending weekly meetings, organizing design workshops, developing prototypes, and a range of other activities. Each project lasted around three to four years and was conducted within a Swedish setting.

Playful Internet of Things: PlayIT, 2018–2021

PlayIT was an innovation project focused on the Playful Internet of Things (Playful IoT). The project explored the concept of Playful IoT through the design and development of a biodiversity-themed playground. The

playground was inaugurated in October 2020 and has been open to the public ever since. The project was funded by the Swedish National Innovation Agency, the Swedish Energy Agency, and the Swedish Research Council for Sustainable Development through a funding call focused on IoT for societal benefit.

Values in focus: Playful IoT, malleable play, education for sustainable development, and biodiversity awareness.

Core project team:
A municipality, a social enterprise focused on playful education for sustainable development, a design and technology company, and HCI researchers from two universities. Representatives:

1. Municipality – City Gardener
2. Municipality – Playground Development Project Lead
3. Initial design and tech company – Marketing Manager
4. Design and tech company – Developer
5. Design and tech company – CEO
6. University 1 – Lead RtD Researcher
7. Social Enterprise – Educator/Designer (myself)
8. Social Enterprise – Lead, CEO, Senior Sustainability Consultant
9. University 2 – HCI Researcher

Social innovation aims, according to the project proposal:

> Within the project, we explore play and game design in IoT-enhanced play equipment, with a focus on outdoor play and technology-enhanced playground equipment in public environments. We are building communication not only for IoT functionality, but also for open-ended and malleable play values. [...] focus is on the power of the digital as a changeable material to create environments that are adaptable, changeable, and updatable for landowners with changing needs, but also adaptable at the individual level to create temporary ownership of the physical place. By exploring these parts of the playful Internet of Things, we believe we can create play that stimulates children's creativity and imagination while teaching participants to interact, collaborate and negotiate, giving power back to individuals to influence their immediate environment. These are all important qualities to learn in a democratic society. [...] The digitally enhanced play is a powerful pedagogical tool and the experiences created in the project will be enhanced by also implementing ESD - Education for Sustainable Development, and experiential learning. (PlayIT project group, 2018) (author's translation)

Design outcomes: An IoT playground containing two interactive installations: the Magical Eggs and the Story Hut (Figure 1). The installations provide

sound and light feedback, with main interactions taking place through scripted scenarios that are activated during play. The Magical Eggs include scenarios in which they function as music instruments or invite children to chase light patterns across the eggs. In the Story Hut, children can listen to daily changing stories about local wildlife. Both installations are connected to a digital platform that allows existing scenarios to be reprogrammed and new ones added over time.

Publications connected to the project:
Article I – Daddy, You Can Be the Fox. Designing Hybrid Playgrounds for Nature Integrated Storytelling (Johansson and Back, 2024)
Article II – Let's Play Something New! Designing for Digital Malleability in Outdoor Playgrounds (Back, Johansson, Hangvar, 2023)
Supporting publication – Value Driven Design for Playful Technology Enhanced Installations in Public Settings (Back, Johansson, Wireband, 2021)



Figure 1: The IoT playground developed in the PlayIT project. Left: View of the forest glade containing the playground installations. Middle: One of the magical eggs. Right: The Story Hut.

ABC for Meaningful Meetings, 2021–2025

ABC for Meaningful Meetings (ABC) was an innovation project focused on social sustainability in the built environment and on designing for meaningful meetings and a stronger sense of community in urban areas. The project probed a range of design solutions and included several interventions, all situated within a newly developed district of a Swedish city. The project was funded by the Swedish National Innovation Agency through a funding call focused on Social Sustainability in the Built Environment.

Values in focus: Community, social sustainability.

Core project team:
A public real estate company, a sustainability NGO, and researchers in psychology and HCI. Representatives:

1. Public Real Estate Company – Community Manager
2. Public Real Estate Company – Sustainability Manager
3. Public Real Estate Company – Communications Officer
4. University 1 – HCI Researcher
5. University 1 – HCI PhD Candidate (me)
6. University 2 – Researcher in psychology
7. Sustainability NGO – Project Lead
8. Sustainability NGO – Representative (myself, for the first half year, then the role was filled by others)

Social innovation aims, according to the project proposal:

> The project addresses the societal challenge of involuntary loneliness, and more specifically, the barrier to well-being that involuntary loneliness poses. The project aims to create knowledge about how physical meeting places can be better designed and used to increase social interaction and strengthen the experience of community and belonging, factors that lead to increased well-being. Using both research and proven experience, the actual design of the physical space in people's immediate environments as well as activities and working methods, is investigated, and how these in combination with each other, can create the right conditions for meaningful meetings and community feeling. (ABC project group, 2021, 2) (author's translation)

Design outcomes:
The project included several design strands. Of particular relevance to this thesis is the design of an interactive outdoor art installation for the co-creation of soundscapes (Figure 2a). Other strands included a sustainability festival, the development of an evidence-based observation protocol for measuring engagement during social gatherings (Figure 2b), a platform for community dialogue, and the development of a handbook for facilitating meaningful meetings.

Publications connected to the project:
Article III – Valuable Design Conversations. Value Perspectives on Co-Designing a Public Interactive Artwork with Mission-Driven Stakeholders (Johansson and Lindström, manuscript)



Figure 2. The ABC project. Left: The interactive art installation. Right: Testing the observation protocol during the sustainability festival organized as part of the project.

Castle Museum Project, 2023–2025

The Castle Museum Project was a collaborative project between HCI researchers and a design team from the municipality to develop a new castle museum. The collaboration was funded by the university's collaboration unit. It began during the initial conceptualization of the museum, when the venue consisted only of empty rooms. Two design concepts were developed and then implemented as permanent installations within the museum exhibition. As researchers, we also participated in design discussions regarding the museum's overall interaction design and its interactive installations.

Values in focus: Cultural heritage awareness, promoting local history, innovative approaches to cultural heritage mediation.

Core project team:
The Castle Museum – a municipality-run museum – and HCI researchers. Representatives:

1. University 1 – HCI researcher (me)
2. Municipality – Digital Producer for the Museum
3. Municipality – Project Manager for the Museum
4. Municipality – Museum Educator

Social innovation aims, according to the project proposal:

> Explore the possibilities of co-creating an innovation lab for cultural heritage mediation in the castle premises, where prototypes can be co-created, tested, evaluated, and developed in collaboration between cultural heritage actors, researchers, and visitors. (Slottslabbet project group, 2022) (author's translation)

Design outcomes:
The Archive Quest – An escape room-inspired installation in which visitors uncover historical secrets and understand while learning how knowledge about the past is constructed (Figure 3, left and middle).
The Silver Throne – A technology-augmented throne that plays authentic quotations from the 17th-century Swedish monarch Queen Kristina when visitors sit on it (Figure 3, right).

Publications connected to the project:
Article IV – The Quest for Learning. Pedagogical and Interaction Design Perspectives on Quest Rooms (Johansson, Bandelin, Back, 2025)
Supporting article: Interactive Storytelling for Communities, Heritage, and Public Engagement (Paananen et al., 2024)



Figure 3: Castle Museum project design outcomes. Left and middle: The Archive Quest. Right: The Silver Throne.

Supporting Design Project: GIFT, 2017–2019

In addition to the main projects listed above, the research presented in this thesis is also connected to an earlier EU Horizon project called GIFT. The project explored hybrid museum experiences that combine physical and digital elements to create personal encounters with cultural heritage. The project brought together museums, researchers, an artist collective, and a design company. The overarching GIFT project contained many different sub-projects and design strands. My role in this project was more limited, and I contributed as a designer and research assistant in one of the sub-projects. I was involved in one of the design processes, designing and facilitating playful sensitizing scenarios that were used in innovation processes with museum staff and designer teams to foster shared understanding and explore values. My work was

specifically focused on a museum installation in Serbia, providing insights into the thesis from a design context outside Sweden.

Publications connected to the project:
Article V – Why Larp? A Synthesis Paper on Live Action Roleplay in Relation to HCI Research and Practice (Johansson et al., 2024)
Supporting publication: Sensitizing Scenarios: Sensitizing Designer Teams to Theory (Waern et al., 2020)

Thesis Structure

Chapter 2: Background presents the history and current research strands of value related design. The concepts of societal values and societal value stakeholders are conceptualized in relation to HCI and design. The chapter also examines how cultural interventions have been, and continue to be, utilized as a means of pursuing societal values.

Chapter 3: Theoretical Framework presents the theoretical framework through which insights from the individual studies are synthesized and analyzed. It expands on the view of design as the semantic mediation of values, which forms the basis of the framework comprising the dimensions of value elicitation, value translation, and value manifestation.

Chapter 4: Method and Methodology presents the Research through Design approach and explains how it is applied in this thesis. The chapter also outlines the designerly worldview that underpins the research, describes the methods used for analysis, and discusses ethical considerations and researcher positionality.

Chapter 5: The Studies presents the individual studies and their findings, and recounts the design processes of each project. The chapter also provides an overview of the articles included in the thesis.

Chapter 6: Design for Societal Values brings together and synthesizes insights from the design projects based on the theoretical framework. The chapter builds on a retrospective analysis of the projects and also draws on follow-up interviews with the SVS collaborators.

Chapter 7: Summary and Conclusion sums up insights and findings, presents the contribution, and proposes directions for future studies.

Chapter 8: Postamble provides reflections on the thesis and my PhD journey.

Chapter 2. Background

The thesis builds on the assumption that societal values can be manifested through design. The challenge of incorporating human, moral, and societal values into design has long been a focus of design-oriented research, particularly within HCI (Kensing and Blomberg, 1998). This chapter provides an overview of how these challenges have been described and addressed by scholars within HCI and related fields through research traditions such as Participatory Design (Bannon and Ehn, 2012), Value Sensitive Design (Friedman and Hendry, 2019), Design for Values (van den Hoven et al., 2015), and Design for Social Innovation (Manzini, 2015).

The chapter begins with a historical overview of the relationship between design and values, followed by a review of HCI strands that engage with values and design.

The chapter then introduces and conceptualizes societal values as explicit values that benefit society and communities in a way that can transcend individual values. It also conceptualizes societal value stakeholders (SVS) – organizations such as municipalities and NGOs whose primary purpose is to pursue societal values – in relation to HCI and design.

Finally, the chapter presents the design domain addressed in this thesis: social innovation projects that involve the design of cultural interventions, such as interactive museum exhibitions, public playgrounds, and public art installations. It also examines how such projects engage with policymaking and the pursuit of societal values.

The chapter provides a foundation for the theoretical framework presented in the subsequent chapter.

Design and Values

The idea that human and societal values can be embedded in artifacts and innovations has been around since the earliest civilizations. The Mesopotamian *Epic of Gilgamesh* reflects on how city walls provide values such as safety while offering a form of immortality through a shared cultural heritage (Sandars, 1977). Japanese Zen aesthetics considers how everyday artifacts,

such as teapots and rakes, can cultivate values such as humility and contemplation (Suzuki, 2019). In the Western tradition, Aristotle (350 BCE/2004) argued that future technologies, such as automated looms, could create more opportunities for civic engagement, while the medieval scholar Roger Bacon reflected on ethical technologies and envisioned how flying machines could improve human well-being (Ingham, 2015). In the 19th century Carlyle (1829) argued that mechanical inventions transformed human morals and social relations.

When design emerged as both a profession and a research field during the 20th century, reflections on values became an integral part of the discipline. During the early decades of the century, the Bauhaus movement stressed the democratization of design by combining aesthetics, functionality, and human needs (Dutch Art Institute, 2019; Hameed, 2024). The movement became highly influential and remains an inspiration for designers and design scholars today (Hameed, 2024).

The relationship between human needs and design was further developed by Dreyfuss (1955) in *Designing for the People*. Out of this work grew Human-Centered Design, a research tradition that places users' needs and values at the center of the design process, creating a foundation for what later became the research field of Human-Computer Interaction (Norman and Draper, 1986).

While Human-Centered Design focused mostly on the usability of individual artifacts, other scholars put greater emphasis on the relationship between society, values, and design. In *Do Artifacts Have Politics?*, political scientist Langdon Winner (1980) argued that all artifacts possess political qualities and that institutional decision-makers and designers shape society by embedding values in the artifacts they create. Design scholar Victor Papanek (1985) emphasized the ethical responsibility of designers to contribute to positive societal change and to co-design with stakeholders beyond direct users. He criticized an excessive focus on consumer-driven user needs, proposing that designers should engage more actively with broader societal values.

In Scandinavia, design was also closely connected to human needs through an explicitly political framing. An important early actor was Hemmens Forskningsinstitut (the Home Research Institute), founded in 1944 by women's organizations. Rather than developing commercial products, the institute focused on improving women's living standards through design (Svensk Filmindustri, 1951; Redström, 2017). During the 1970s, this early focus on human needs evolved into collaborative design approaches sometimes referred to as the Scandinavian School of Design (Göransdotter, 2022), which later developed into Participatory Design, discussed later in this chapter.

These historical examples demonstrate that the relationship between design and values has been a central concern across multiple fields of practice and research. This broader historical context predates the growth of HCI and has significantly influenced how values have since been conceptualized within the field.

HCI Strands of Values in Relation to Design

With the emergence of Human-Computer Interaction as a research field, values in relation to design became a research focus. This section expands on the development of value-focused strands of design-related research within HCI and neighboring fields, that have all influenced this thesis work. The strands are thereafter brought together and related to how values are approached in this thesis.

Design Thinking and Values

Design thinking, the formalization of procedures used in the design process, was first described by Arnold (1959). In the 1980s, design thinking spread quickly and was further developed by authors such as Lawson (1980), Cross (1982), and Rowe (1987). It became, and has remained, highly influential across sectors (Dam, 2023). In parallel with the development of design thinking, scholars increasingly argued that its methods could be applied to value-oriented purposes beyond commercial considerations (Papanek, 1985; Buchanan, 1992). Today, design thinking is increasingly applied in value related domains (Starostka et al., 2022), for instance through approaches such as *Design Thinking for Greater Good* (Liedtka et al. 2017) and *Design Thinking for Social Innovation* (Brown and Wyatt, 2010).

While design thinking has considerable potential, applying it to value-driven projects without modification has been criticized for transferring a model created for commercial purposes to projects that would benefit from approaches more aligned with their value goals (Hernandez and Goñi, 2020). Design thinking has also been criticized for insufficient attention to how moral values become embedded in design and for leaving too little room for reflection (Sengers et al., 2005). Scholars have therefore called for a rethinking of design thinking (Kimbell, 2011) to move beyond its de-politicizing and commercially oriented assumptions. While acknowledging the benefits of design thinking, this thesis also adopts a critical perspective. Following other value-related traditions within HCI, such as Reflective Design and Value Sensitive Design, it seeks alternative approaches to integrating design

with moral and societal value concerns. Several of these research traditions are expanded on below.

Participatory Design

During the 1980s, two major approaches to involving stakeholders and their values in design emerged within Human-Computer Interaction: user-centered design and Participatory Design (Bannon and Ehn, 2012). User-centered design, as formulated by Norman and Draper (1986), focuses mainly on ergonomics and usability from a design engineering perspective, influenced by Dreyfuss. While much of user-centered design and user experience research has focused on marketable products, scholars have argued for a broader focus on user values that includes, for example, social values, rather than treating user values simply as a list of design requirements (Kujala and Väänänen-Vainio-Mattila, 2009).

Participatory Design (PD) emerged during the 1980s from the Scandinavian tradition of collaborative design. PD strives to move beyond usability, advocating the democratization of design processes through the active involvement of stakeholders (Kensing and Blomberg, 1998; Lundin, 2005).

Early PD research involved collaboration with labor union organizations (Ehn et al., 1987), but the field has since concentrated primarily on collaboration between designers and end users (Simonsen and Robertson, 2012). In their book on participatory design, Bødker et al. (2022) state that "Participatory design is concerned with the needs, wants and desires of users" (p.5), and define PD as "activities where users, designers and researchers collaborate toward shared goals" (p.3). More recently, however, researchers have argued for broadening the scope of stakeholders involved in Participatory Design (Vines et al., 2013).

One challenge identified early within PD, and one that this thesis engages with, concerns how to create shared understanding between participants and designers and overcome design language barriers (Ehn, 1992). According to a recent review of PD, there remains a need to develop additional tools and methods that help establish common ground between designers and other stakeholders (Elsayed-Ali et al., 2023).

Participatory Design has also been criticized for reinforcing unequal power relationships between designers and other stakeholders. As a result, terms such as *co-operative design* and *co-design* have emerged to emphasize more balanced and reciprocal forms of collaboration (Bødker et al., 2000; Sanders and Stappers, 2008).

Value Sensitive Design

Value Sensitive Design (VSD), formulated in the mid-1990s, is an important source of inspiration for this thesis. VSD is closely related to, and partly emerged from, Participatory Design (Cenci and Cawthorne, 2020). However, in line with the approach taken in this thesis, VSD places greater emphasis on the values pursued through stakeholder engagement than on participation itself.

Over the past several decades, VSD has generated a substantial body of publications and research projects, becoming an established approach within HCI (Winkler and Spiekermann, 2021). The principal scholar associated with VSD is Batya Friedman (1996), who co-developed the approach to foreground moral and ethical values in design, criticizing an overly narrow focus on economic and efficiency values (Friedman, 1996). VSD is a technology-agnostic approach in which values, rather than specific technologies or domains, are central (Friedman and Hendry, 2019), a perspective that aligns closely with the focus of this thesis.

The values addressed within VSD can vary widely and include welfare, trust, privacy, autonomy, and environmental sustainability (Friedman and Hendry, 2019). VSD defines values primarily in terms of human values, described as "what is important to people in their lives, with a focus on ethics and morality" (Friedman and Hendry, 2019, 4). While VSD intentionally adopts a broad definition of values, the types of societal values that are the focus of this thesis largely fall outside its scope. Even though some VSD research has addressed issues such as sustainability scholars have acknowledged that design projects intended to promote environmental and social good rather than self-interest might require different design approaches. (Sze-man Mok et al., 2016)

Within VSD three main categories of values are identified: project values, designer values, and stakeholder values (Friedman and Hendry, 2019). Of these, project values receive comparatively little attention. However, Friedman and Hendry acknowledge that "when the divergence between designer and project values is significant, then additional methods to manage those differences may be warranted" (Friedman and Hendry, 2019, 67–68), without elaborating on what those methods might entail. This observation is pertinent to the present thesis, which examines how such divergences can be addressed and argues that explicitly articulated project values can still be insufficiently developed within design processes.

VSD adopts a broad understanding of stakeholders, encompassing both direct and indirect stakeholders and, at least tentatively, non-human stakeholders. Furthermore, VSD recognizes that different actors within a design project hold different sets of values that can be relevant to study (Friedman and

Hendry, 2019). However, it does not clearly explain the nature of these differences, particularly with respect to commissioning stakeholders.

VSD has been applied across a wide variety of design contexts, including the cultural and creative sector. For example, Muntean et al. (2017) applied VSD to the interaction design of museum experiences, expanding on how cultural values can be translated into interaction design. Their study found that collaborative exploration of values was crucial and that tangible, embodied interactions supported a greater understanding of values than discussion or written descriptions alone.

Although VSD has been highly influential and widely applied within HCI and beyond (Winkler and Spiekermann, 2021; Cenci & Cawthorne, 2020), it has also been the subject of criticism. One major critique is that values are treated as too abstract and poorly defined, resulting in a lack of conceptualizations that support implementation beyond simplified instrumental approaches (Manders-Huits, 2011). Design researcher Taylor Stone (2021) argues that designers should create context-specific value interpretations for each design project, rather than relying upon broad, abstract definitions such as those employed within VSD (Stone, 2021). While this thesis builds on VSD, it places greater emphasis on the contextual values of individual design projects than on the more overarching values often emphasized in the VSD literature.

VSD was developed within a North American liberal tradition. Although it explicitly identifies values to promote, it largely neglects the fact that values such as autonomy and privacy might be understood differently in more collectivist societies (Borning and Muller, 2012).

Another criticism of VSD is that it fails to clearly identify and distinguish among the different types of stakeholders involved (Manders-Huits, 2011). It has also been pointed out that the relationship between designer agency and the nature of normative values that shape and inform design processes remains underexplored in VSD (Donia and Shaw, 2021).

VSD proposes a specific methodology for studying values in relation to design: the tripartite methodology. VSD as a methodological approach, and its application within this thesis, are discussed further in Chapter 4.

This thesis expands on VSD while addressing several of the critiques mentioned above. It provides a clearer value conceptualization of societal values, emphasizes context-specific values, looks beyond individually oriented human values, and contributes concrete methods for translating values into design practice. Furthermore, it extends VSD by foregrounding project values and applying the framework to the design of cultural interventions together with SVS.

Reflective Design

Reflective Design, formulated by Sengers et al. (2005), is inspired by VSD but places greater emphasis on critical reflection on values in relation to design. It proposes that designers should re-think their own roles by adopting a more reflective approach, and emphasizes that every stage of the design process involves value-based decisions. Reflective Design promotes a collaborative approach that brings stakeholders into value-oriented design processes as more equal participants. Reflective Design also draws on ludic design, a specific form of Research through Design focused on playful exploration developed by Gaver. Sengers et al. argue that this approach is especially meaningful for value-oriented design because playfulness can help non-designers feel more included in the design process. Reflective Design also emphasizes engagement with values through the act of designing itself.

The principles of Reflective Design include encouraging designers to adopt a reflective stance in order to identify unconscious values and assumptions and to reconsider their own values. Furthermore, designers should support stakeholder reflection, dialogic engagement, and critical perspectives. Reflective Design argues that reflection should be embedded throughout both the design process and the resulting designs, rather than being treated as standalone activities.

Reflective Design proposes design strategies that focus on the relationships among users, designers, and systems. These strategies allow users to shape the meaning-making process, providing rich feedback and employing probes and metaphors.

This thesis aligns with Reflective Design in its emphasis on collaboration, playful approaches, and shared reflection. However, it departs from Reflective Design by moving beyond a user-centered perspective and by placing greater emphasis on explicit values than Reflective Design currently does.

Design for Values

Expanding on Value Sensitive Design (VSD), Design for Values (DfV) focuses on the actual process of translating explicitly articulated values into design requirements (Van den Hoven et al., 2015), and draws inspiration from Schön's (1992) reflective practitioner approach. Like this thesis, DfV is practice-centric, examining how abstract values become manifested in design processes and results. It aligns with the approach taken in this thesis through its focus on explicit values and on the perspectives of commissioning stakeholders.

DfV was developed within a European context in which governments and the European Union pursue explicit value-oriented policies, resulting in a relatively top-down approach to value formulation. Within DfV, stakeholder inclusion is viewed as a means of identifying and articulating values through

shared reflection. DfV focuses mainly on values promoted by organizations or external policy frameworks rather than on users' perceptions of values (van den Hoven et al., 2015). Project goals are formulated and translated into design requirements to ensure that specific ethical values are reflected in the designs. In this way, abstract values such as sustainability or democracy are translated into concrete design requirements (Van den Hoven et al., 2015). Value articulation is a defining feature of the approach, intended to avoid the tendency for non-commercial values to remain implicit in design processes (Van de Poel, 2013).

DfV philosophy professor van de Poel (2013) argues that designers should be sensitive to the moral values within design endeavors and distinguishes between intrinsic values – values important in themselves – and instrumental values – values that are important because they serve other ends. This distinction is developed later in the thesis.

DfV has been criticized for reducing complex ethical and political questions to engineering optimization problems (Donia and Shaw, 2021). This thesis opposes the simplified approach to values in DfV, but draws inspiration from DfV's emphasis on organizational values, its conceptualization of different types of values, and its focus on how values are translated into concrete design features.

Responsible Innovation and Socially Responsible Design

Responsible Innovation emerged from VSD and DfV (Van den Hoven, 2013). It focuses on innovating responsibly in relation to moral values (Stilgoe et al., 2013), and gained significant momentum during the 2010s, particularly within the European Union (2022), where it is now integrated into official policy. A central intervention-oriented activity within Responsible Innovation is socio-technical integration, which situates design within broader societal contexts. From this perspective, innovation should be contextualized and implemented in ways that align with the needs and values of local communities (Owen et al., 2013).

A specific type of Responsible Innovation is Socially Responsible Design, formulated by design scholars Melles et al (2011). This approach focuses on the use of design to pursue social, humanitarian, cultural, environmental, and economic values beyond consumerism. Melles and de Vere argue that designers have a moral responsibility and point to a lack of design training that prepares them to negotiate diverse and conflicting societal interests or to engage in policymaking. This thesis aligns particularly with Socially Responsible Design in its emphasis on moving beyond user-centered co-design to address broader societal needs (Melles et al, 2011), and in its proposal to step away

from design methods such as the double diamond by extending design processes to allow greater space for moral reflection (Hernandez and Goñi, 2020).

Design for Social Innovation

Design for Social Innovation emerged from the broader concept of Responsible Innovation and focuses specifically on addressing social challenges, such as well-being, community building, cultural heritage, and learning (Cibin et al., 2020). All the projects included in this thesis can be understood as social innovation projects.

The EU defines social innovation as "the creation and implementation of new ideas, practices and policies designed to address social challenges and enhance the well-being" (European Commission, 2025). Social innovation is a substantial sector in its own right. For example, the EU's Social Innovation Fund alone has a budget of 197 million euros (European Commission, 2025). During the last decade, social innovation has received increasing attention within design-related research and has moved from the fringe to the center of political agendas and public debate (Manzini, 2015).

Design for Social Innovation was formulated by design scholar Manzini (2015), who argues that designers participating in social innovation projects need to rethink established design practices. Rather than focusing on problem-solving, designers should facilitate value conversations, embrace cultural diversity, and foster creativity. Studies on social innovation highlight the importance of intermediaries who can bridge different actors and domains within a project (Cibin et al., 2020), a role that is of particular interest for this thesis.

Manzini's approach to social innovation has been criticized, most notably by design scholar DiSalvo (2012), for paying insufficient attention to power structures within design projects.

While acknowledging this critique, which is expanded on in later chapters, this thesis aligns with Design for Social Innovation in several respects: through the type of projects it examines, the stakeholder groups it engages with, and its view of design as grounded in value conversations.

Bringing Together the Strands of Values and Design

The previous sections have reviewed value-oriented design traditions within HCI research and practice. This section synthesizes these perspectives and positions the thesis in relation to them. While acknowledging the potential of Design Thinking, the thesis takes the position that design methods need to be adapted to better support design projects aimed at pursuing societal values.

The thesis draws on PD through its emphasis on collaboration with non-designer stakeholders but aligns more closely with VSD in viewing collaboration not as an end in itself but as a means of engaging with values. The thesis extends VSD by addressing several of the critiques directed at the approach. Specifically, it emphasizes context-specific values, provides a clear conceptualization of societal values, and focuses on how values are translated into design solutions.

The focus on translating abstract values into design solutions is shared with DfV. However, the thesis does not adopt the engineering-oriented and instrumental treatment of values that characterizes DfV. Instead, its approach to values is more closely aligned with RD, which emphasizes that engaging with values calls for reflection, that values are inherently contextual, and that they should be explored through long-term co-reflective practice. The connection between RD and Ludic Design is also relevant to this thesis, particularly in its use of playful approaches to support value-oriented design work.

All the design projects included in this thesis can be understood as social innovation projects. Consequently, the thesis draws on Design for Social Innovation, particularly through its emphasis on value conversations and its focus on collaborating stakeholders, such as municipalities and NGOs.

While PD, VSD, and RD largely focus on users, this thesis aligns more closely with DfV and Design for Social Innovation in its emphasis on commissioning stakeholders as the primary focus of investigation.

The thesis thus builds on HCI's long-standing focus on the relationship between values and design processes while contributing several new perspectives. These include its focus on societal values, its emphasis on explicitly articulated values, its grounding in long-term peer-level collaboration, and its prioritization of SVS over a user-first perspective. Furthermore, the thesis applies these ideas within the domain of cultural interventions through engaging with real-world design projects. Having established how the thesis relates to existing value-oriented design traditions, the following sections present a more in-depth investigation of the value concepts in focus.

Societal Values

This section conceptualizes societal values in relation to design and HCI by drawing on established definitions from outside HCI alongside a collection of related concepts. This conceptualization forms the foundation for how the term is used throughout the theoretical framework.

Outside HCI, it is well established that values can be conceptualized into different typologies and taxonomies and that different categories of values can

possess distinct characteristics (Dancy, 2005). The distinction between human values and societal values is well established outside HCI research and is used by international organizations. The OECD defines societal values as values that "define the priorities of societal cultures, the shared principles and guidelines that frame social order and institutional life. These values endure when they are enshrined in social and institutional structures, documents and democratic practice and when they are endorsed through public opinion" (Organisation for Economic Co-operation and Development, 2021). From this perspective, societal values are closely tied to policymaking, as organizations strive to promote and realize certain societal values (Appelstrand, 2002).

Both VSD and DfV emphasize the importance of detangling the types of values in value-oriented design (Friedman, 1996; Van den Hoven et al., 2015). Still, the specific types of values under consideration are seldom clearly conceptualized within VSD or DfV (Manders-Huits, 2011; Stone, 2021).

HCI scholars Détienne, Baker, and Le Bail (2019) problematize the definition of values within VSD, instead describing values in design projects as existing between individuals, groups, and the societies from which those values emerge. They argue that many design projects engage with value-based communities that are held together by shared value systems and ideologies. Thus, values should not be understood solely as attributes of individuals and their needs.

Related Concepts to Societal Values

The conceptualization of societal values draws on a collection of related concepts, including collective values, public values, and the Swedish concept of värdegrund. This section examines these related concepts in relation to the conceptualization of societal values adopted in this thesis.

Collective Values

The first related concept is *collective values*. Psychologist Schwartz (1992) defines collective values as values pursued by a collective such as a nation, society, or community. He argues that collective values are intrinsically different from individual values: "Values that serve individual interests are postulated to be opposed to those that serve collective interests" (Schwartz, p. 13, 1992). Schwartz illustrates this distinction through the tension between pursuing personal security and national security, where the latter may be pursued at the expense of individuals who are perceived as threats. The distinction between individual and collective values is also widely acknowledged outside academia. For example, the United Nations distinguishes between the values in the Universal Declaration of Human Rights, which focuses on individual

rights (United Nations, 1948), and the Declaration on the Right to Development, which emphasizes collective values (United Nations General Assembly, 1986).

Social sustainability can also be understood as a form of collective value, reflected in the UN's 2030 Agenda for Sustainable Development and its 17 Sustainable Development Goals (United Nations, 2015). Social sustainability is gaining increasing attention in design research and is concerned with the well-being of societies (Corsini & Moultrie, 2021), including issues such as equality, social justice (Woodcraft, 2012), cultural heritage (Murzyn-Kupisz, 2012), education, and the development of sustainable communities (Rockström and Sukhdev, 2016).

Public Values

Another concept related to societal values is *public values*. Public values are well established within public administration and management, where they are foundational for public value theory. This perspective argues that the values pursued by the public sector cannot be reduced to cost-benefit analyses or customer orientation, but rather follow a distinct logic centered on the public good (Meynhardt, 2009). Public administration scholar Moore (1995) defines public values as values that create benefits for society, citing examples such as keeping streets safe and clean, providing library services, and ensuring access to education. According to Moore, public values are upheld through collective responsibility, even when doing so conflicts with individual preferences, such as requiring children to attend school or enforcing the obligation to pay taxes. Public value initiatives are usually funded through public resources and pursued by government organizations, but they can also be advanced by other organizations dedicated to community improvement, such as NGOs (Stewart, 2009; Moore, 1995). Public values are challenging to measure directly, yet they permeate policymaking institutions and their decision-making, functioning as implicit or explicit guiding principles for collective action (Stewart, 2009).

In recent years, public values have also attracted growing attention within design research (Junginger, 2017; Armstrong et al., 2014). Scholars have argued that such values call for specific design considerations that would benefit from more research. Design projects can become arenas for engaging with both the formulation and implementation of public values when these perspectives are integrated into the design process (Junginger, 2017).

Värdegrund - Societal Values in the Swedish Context

Since the thesis is set in a Swedish context, the concept of *värdegrund* is of particular importance. The thesis conceptualization of societal values draws

heavily on värdegrund and can, in part, be understood as a translation and adaptation of the concept.

Literally, värdegrund translates as "value fundamentals", but there is no direct equivalent in English. The term is widely used in Sweden to describe the foundational ethical and moral values explicitly articulated by institutions and organizations (Bringselius, 2019; Brytting, 2010).

According to a Swedish governmental report on värdegrund, the concept encompasses ethics, virtue, anti-discrimination, social responsibility, integrity, equality, sustainability, equity, loyalty, morale, compassion, dignity, norms, duty, and conscience (Brytting, 2010). Värdegrund further implies that such values are contextualized and translated into actionable organizational policies and practices (Brytting, 2010; Colla, 2018). Rather than focusing on individual values, värdegrund primarily concerns societal values (Brytting, 2010). It can therefore take precedence over individual preferences, requiring organizations to actively promote particular values. This can include fostering acceptance of diverse sexual orientations, advocating for gender equality among groups where such values are less established, encouraging anti-racist attitudes among students expressing racist views, and upholding democratic values among people with more authoritarian perspectives (Colla, 2018). Today, värdegrund is often used in both public debate and official documents (Bringselius, 2019; Brytting, 2010).

Sweden has a long tradition of public policymaking centered on moral values (Colla, 2018), and värdegrund emerges from this tradition. The concept became widely established in 1994, when Sweden introduced a new school curriculum built around an extensive värdegrund that included democracy, human dignity, environmental sustainability, freedom, equal rights, tolerance, justice, integrity, cultural heritage, compassion, knowledge, well-being, freedom of opinion, and solidarity with vulnerable groups (Englund & Englund, 2012; Skolverket, 1994). The current national curriculum still builds on a similar värdegrund (Skolverket, 2022), reflecting a high degree of consistency over the past three decades.

The concept soon expanded beyond the education setting to become a seminal component of Swedish public policymaking, Today, employees throughout the Swedish public sector are expected to adhere to their organization's värdegrund (Bringselius, 2019), which directly shapes the day-to-day work (Björkvall and Nyström Höög, 2019). In 2010 the Swedish Agency for Financial and Public Management (Statskontoret, 2010) published a report on värdegrund to ensure that it permeates the entire public administration, emphasizing the importance of *värdegrundsarbete* – ongoing reflection and work to uphold a shared organizational value foundation.

Related concepts have emerged in the other Nordic countries, with some variations. In Norway, the equivalent concept is *verdigrunnlag* (Plesner, 2005), literally meaning "value foundation." In Denmark, the term *værdigrundlag* is used by some organizations but is not embedded in the national curriculum (Børne- og Undervisningsministeriet, 2024). Swedish-speaking schools in Finland also use the term värdegrund (Vasa Övningsskola, n.d.). However, while these concepts exist in the other Nordic countries, none seem as integral to the public debate as värdegrund is in Sweden.

The concept of värdegrund operates on several levels. It appears in national policy documents, while individual organizations also formulate their own specific värdegrund. For instance, preschools throughout Sweden develop local värdegrund statements by interpreting and contextualizing the national curriculum, often together with the children (Heimer and Pihlgren, 2023) (figure 4).



Figure 4. A *värdegrundsvägg* ("value foundation wall") in the cloakroom of a Swedish preschool showcasing their own värdegrund in a way accessible to the children.

Although värdegrund is particularly central within the public sector, Swedish NGOs also typically formulate their own värdgegrund to capture the societal values they seek to promote (Kings, 2018; Lindberg, 2021). Furthermore, Sweden has established formal värdegrund agreements between the public sector and NGOs (Reuter, 2012).

Although värdegrund has not been explicitly integrated into discussions about design, some design institutions articulate their own design-oriented värdegrund. As an example, Umeå Institute of Design states that its

värdegrund promotes open, curious, and collaborative exploration of design, with the goal of helping people grow and creating sustainable solutions for global societal challenges (Umeå Universitet, 2024).

Conceptualization of Societal Values in the Thesis

As conceptualized in this thesis, societal values are values concerned with the needs of the collective, communities, and society at large. They are non-commercial in nature and are distinct from – and can take precedence over – individual human needs and values. Societal values are explicitly articulated in policy documents and are pursued by public stakeholders, NGOs, and other societal institutions. The core traits of societal values are outlined below.

Societal values may transcend individual human needs. This characteristic draws on insights from scholars such as Schwartz (1992) and Moore (1995). One example is how democracies, in order to protect democratic institutions and principles, may restrict the rights of those who seek to undermine them (Müller, 2016). In an increasingly polarized political climate, such restrictions become particularly relevant and can be addressed either through regulation or by encouraging individuals to adopt democratic approaches (Knobloch & Gastil, 2015).

Societal values are soft and primarily intrinsic values. They include moral values that are hard to measure, such as integrity and respect, which are particularly prominent within the public sector (Wæraas, 2018). Societal values are primarily intrinsic (valuable in themselves) but can also serve instrumental purposes by reinforcing other values. Intrinsic values can be inherently challenging to measure because of *incommensurability*: they can function as inseparable wholes that cannot easily be split up into measurable components (Zimmerman and Bradley, 2025).

Societal values are interconnected. This is related to the idea of incommensurability and draws inspiration from VSD's concept of *webs of values* and tensions in such webs, making it possible to create opportunities for design decisions that can simultaneously support multiple values (Friedman and Hendry, 2019).

Societal values can be made explicit, operationalizable, and actionable. Rather than remaining broad abstract values, such as autonomy or security, or implicit values such as the kind often discussed within VSD and RD, societal values can be pursued through top-down policymaking as well as through bottom-up initiatives led by NGOs and grassroots movements.

Societal values introduce a societal perspective into design processes, a perspective often overshadowed by an overtly narrow focus on users (Kensing & Blomberg, 1998; Norman, 2005). In design projects that address complex

societal problems, an exclusive focus on users can become limiting. Such contexts need design methods that are adapted to foreground the values pursued by, for instance, public organizations, an aspect that is not traditionally taken into account in human-centered design (Van der Bijl-Brouwer, 2016). Bringing this perspective to the forefront is a central contribution of this thesis.

These characteristics all have implications for designing with societal values. Although societal and individual values often overlap and are not necessarily in conflict, this thesis argues that they are not identical and might require different approaches within design processes.

Societal Value Stakeholders in Design Projects

The term *societal value stakeholders* (SVS) is used in this thesis to refer to stakeholders who, regardless of organizational form (e.g., public or civic), pursue societal values rather than consumerism as their primary aim. SVSs are thus conceptualized based on the purpose of an organization rather than its organizational structure. SVS, in a Swedish context, largely correspond to organizations driven by värdegrund (*värdegrundsdrivna organisationer*). This section expands on the concept of SVS in relation to design and HCI.

Within HCI, SVS have not previously been clearly distinguished as a specific category of stakeholders in relation to design projects. In organizational and business studies, however, there is a well-established distinction between profit-driven and mission-driven organizations (Komatsu Cipriani et al., 2020). According to public administration scholar Wæraas (2018), there is an inherent difference between the private sector, which focuses on its own interests, and the public sector, which are tasked with balancing complex political, economic, and social concerns in the pursuit of the public interest. The concept of SVS, as used in this thesis, builds on this distinction.

SVS representatives can also be understood as a form of *street-level bureaucrats*, following Lipsky's (1980) definition of frontline workers who provide services directly to citizens. These actors, according to Lipsky, not only implement public policy, but also shape it through the ways in which they interpret and realize policy in practice. This can also be related to how Durrer et al. (2023) describes that stakeholders engaging with cultural placemaking can also be seen as engaging with local scale policymaking. Building on Lipsky's definition of street-level bureaucrats and on Durrer et al,, SVS representatives are in this thesis seen as engaging with both policy implementation and policymaking on a local scale.

Public organization representatives are increasingly expected to adopt economic and customer-service perspectives, a problematic trend that can

undermine the societal values these organizations seek to pursue (Moore, 1995). Rather, actors collaborating with mission-driven organizations should adopt to the values and logics of these organizations (Komatsu Cipriani et al., 2020).

Design has increasingly expanded into domains concerned with societal challenges (Buchanan, 1992), and design thinking, originally developed primarily for commercial contexts, has started to gain popularity among nonprofits and other mission-driven organizations (Junginger, 2017; Starostka et al., 2022). While design methods can have transformational potential for social innovation, they must be applied in ways that align with the values and modes of operation of these organizations (Brown & Wyatt, 2010).

Schön (1992) was among the first to problematize the use of design to address public challenges. Using the example of developing an inclusive public playground, he cautions that "Professionally designed solutions to public problems have had unanticipated consequences, sometimes worse than the problems they were designed to solve" (Schön, 1992, 4). He therefore argues for greater reflection-in-action to develop a deeper understanding of the value implications of design decisions.

In design projects with SVS, these often have the authority to determine which design solutions are implemented. Based on their organizational values, they can act as gatekeepers who decide whether a proposed solution should be implemented (Krippendorff, 2014). Kavathatzopoulos (2008, 2009) describes how working within 'värdegrundsdrivna organisationer' involves professionally engaging with moral dilemmas, developing a high degree of ethical competence, and internalizing the organization-specific societal value framework of the organization one represents. In relation to design, the complex and often emotionally charged moral issues faced by SVS differ fundamentally from technical problems, which often have clearer solutions and can be addressed more pragmatically.

Soft values play a prominent role within the public sector, and public-sector employees often rely on such values when making decisions. However, a problematic trend has emerged in which these organizations are expected to pursue measurable, hard values instead (Wæraas, 2018). This shift towards efficiency metrics and quantifiable results creates tensions. Consequently, there is a need to strengthen the soft-value identities of these organizations when designing with and for them, rather than imposing performance metrics, measurements, and evaluations.

A study by Chen et al. (2022), based on a value-sensitive evaluation, demonstrated that users had vastly different value priorities compared to practitioners representing the design-commissioning organizations. This suggests

that co-designing with SVS calls for a different approach than co-designing with end users.

Most public-sector representatives involved in design projects have not been formally trained in design (Junginger, 2017; Starostka et al., 2022). They are what Manzini (2015) refers to as diffuse designers – individuals who engage in design-like activities without formal training in design methods. Unless design processes actively support their participation, their perspectives are easily overlooked. A lack of designerly training can make it difficult to envision things that do not yet exist or to generate innovative design ideas. (Sanders & Strappers, 2008)

At the same time, designers might struggle to understand the value context of a project. As described in the previous section, SVS are typically deeply immersed in the context-specific values of their organizations, whereas designers often move between projects. Design researchers Margolin and Margolin (2002) argue that designers need to immerse themselves more fully in the value context of design settings – for instance, by working as interns in hospitals before designing for and with healthcare professionals. They also argue that designers working on projects related to societal values should have a stronger background in public policy. Building on this argument, Melles and Misic (2011) argue that designers who remain detached from the value context are more likely to produce unsuccessful designs. Instead, they propose that designers should engage deeply and over time with stakeholder contexts in order to gain a nuanced understanding of the cultural, environmental, or societal issues that the designs aim to address. The combination of SVS' limited designerly expertise and the designer's distance from the value context suggests that the design process needs to be scaffolded in both directions.

Studies have also found that co-design with public-sector stakeholders requires collaboration around common goals while bridging public policy, user needs, and broader societal needs in ways that current design practices are not adapted to support (Poleac, 2023). Junginger (2017) describes how non-designers can easily be silenced by excessive design jargon and that this must be addressed to enable more equitable collaboration between designers and representatives of societal organizations.

In multisectoral design collaborations it is crucial to bridge differences in priorities, expertise, and modes of communication if projects are to succeed (Eckhardt & Rosenkranz, 2010; Jain et al., 2014). Otherwise, conflicting stakeholder values may undermine the design process. In her HCI dissertation *Articulating the User*, Rajkowska (2022) observes that designers, particularly external consultants, tend to take the lead in creative processes. Influenced by pragmatic approaches, they might overlook important values. She captures

value negotiations in an innovation project involving museum experiences, where museum stakeholders viewed interaction design as a means of promoting broader goals, such as cultural heritage and reflection, while the designers regarded playful and engaging interaction design as valuable in its own right (Rajkowska, 2022).

In the article *Design Research and Practice for the Public Good: A Reflection*, Junginger (2017) argues that digitalization has increased the need for collaboration between designers and public institutions and that representatives from public organizations must be actively involved to that design process remain focused on the public good and societal values. She stresses the importance of looking beyond traditional design methods by placing greater emphasis on human relations, using approaches such as playfulness and storytelling to create shared understanding and collective visions. Junginger also advocates for a new type of design leadership that respects the dedication and inventiveness of public-sector professionals. Finally, she problematizes how many contemporary Design Thinking methods are too embedded in a consumerist mindset and are therefore not suited to co-designing with and for public institutions. She concludes by calling for more research into the distinctive characteristics of design for the public good (Junginger, 2017), a challenge that this thesis directly addresses.

This section has argued that designing for and with SVS requires developing and implementing design processes, considerations, and strategies that differ from those commonly used in commercial design practice. It also highlights how not doing so can undermine both the design process and its results.

Cultural Interventions to Pursue Societal Values

This thesis focuses on pursuing societal values within the context of designing cultural interventions. This section conceptualizes the term *cultural interventions* and relates it to the pursuit of societal values.

Societal institutions have created cultural interventions for policymaking in many forms throughout history, from medieval passion plays (Britannica Editors, 2024) to 19th-century world expos (Carter-Lome, 2017), and the emergence of public museums (Gray, 2015). Cultural interventions are an important policymaking tool used by governmental organizations and NGOs and are gaining increasing attention (Yúdice, 2003). The EU, for example, engages in cultural governance (Bailey et al., 2024), and EU funding for culture and creativity through programs pursuing explicit policy values has increased substantially. The current Creative Europe programme has a budget of 2.44 billion euros (European Commission, 2023).

Cultural policy research is a growing field that examines the use of arts, cultural experiences, and pop culture as policymaking tools (Yúdice, 2003). Cultural values are soft values, something that has implications for cultural policymaking, for instance because results can be challenging to measure (Kaszynska, 2025). Cultural policymaking can be viewed as a situated practice, shaped by local values, contexts, and conditions when creating cultural experiences (Durrer et al., 2023). The supporting article *Larping with Erasmus* (Johansson & Morganti, 2026) provides a background understanding of how the EU engages in cultural policymaking through funding live-action roleplaying (larp) projects that promote policy values such as tolerance, migration, and democracy.

Cultural policymaking is rapidly changing due to increasing digitalization (Durrer et al., 2023). One example is the possibility to redesign public spaces by utilizing digital technologies (Drucker and Gumpert, 2012), such as through interactive museum exhibitions (Chang, 2006), and interactive public artworks (Nobel Prize Museum, 2025). By designing for meaningful social interactions, cultural interventions can promote increased well-being, encourage social reflection, and spark meaningful value discussions (Sakel et al., 2025).

Cultural interventions can serve not only as design outcomes but also as approaches within the design process itself. Bringing cultural, artistic, and creative elements into design processes can open new design spaces (Gaver et al., 1999; Candy et al., 2018), and different creative, playful, and cultural methods have been utilized to engage with values in design within traditions such as Participatory Design (Ehn, 1992; Ehn & Sjögren, 1991; Ehn & Kyng, 1991) and Reflective Design (Sengers et al., 2005).

As discussed above, cultural interventions can serve as a tool for pursuing societal values, and they are receiving increasing attention. Therefore, it becomes important to study this as a value-oriented design domain. Cultural interventions can function both as design outcomes and as approaches within the design process, with creative and playful methods themselves becoming forms of internal cultural intervention.

Concluding Remarks on the Background Chapter

This background chapter provides a foundational understanding of HCI research on design that engages with values. Relevant traditions are presented and situated in their historical context. Further, the chapter expands on how these traditions relate to each other and to the thesis topic.

The chapter distinguishes between societal values and other forms of values, a distinction not clearly articulated in current HCI research on value-oriented design. This distinction, and its implications for design, will be developed further in the next chapter.

The chapter establishes that there is plenty of prior HCI research on incorporating stakeholders in design processes and on integrating stakeholder participation with value-oriented design approaches. This thesis complements these approaches through its focus on the collaboration between designers and societal value stakeholders (SVS), grounded in the distinctive characteristics of SVS and their role in value-oriented design projects. The thesis thus addresses an identified gap in HCI research on design collaborations with SVS in relation to societal values within social innovation projects.

Chapter 3. Theoretical Framework

This chapter introduces the concept of design as semantic value mediation. Three dimensions – value elicitation, value translation, and value manifestation – are conceptualized and expanded on. The theoretical framework presented in this chapter provides the foundation for the analysis and discussion chapters that follow.

Design as Semantic Mediation of Values

Based on ideas by scholars such as Latour (1994), Verbeek (2011), Manzini (2015), Krippendorff (2014), and Flanagan, Howen and Nissenbaum (2008), this framework proposes viewing design in social innovation projects as *semantic value mediation*.

A foundational assumption for this thesis is that designs can mediate societal values. Designed artifacts and activities have not only physical properties, but also built-in values shaped by designers, commissioners, policymakers and other stakeholders. This stance aligns with how Latour (1994), in *On Technical Mediation*, explains how humans mediate their relationship with the world through artifacts that are designed with moral intent. Latour exemplifies how traffic safety values are mediated through a speed bump, where the intentions of designers, commissioners and lawmakers are embedded in, and enacted through, its design. The stance that design is about value mediation also draws on design scholar Verbeek (2011) and his argument that designs mediate human experiences as well as moral and ethical values, as people engage with the world through the moral lens of designed artifacts. Verbeek argues that this has implications for design processes but does not expand on how, a question this thesis addresses. Value mediation through design becomes a fusion of technology and culture, where policies are pursued through both designs and designing. Whether intentional or not, all design is in this sense political, making designers policymakers in a broad sense through their design work. This reasoning aligns with Flanagan, Howe, and Nissenbaum's (2008) *Embodying Values in Technology: Theory and Practice*, which stresses that moral values in design deserve greater attention. They likewise describe

design as political, designers as having moral obligations in relation to their work, and design processes as the translation of abstract values into concrete design outcomes.

The focus on *semantic* mediation of values draws on Krippendorff's (2006) proposal to shift from viewing design as the creation of artifacts, to focus on the creation of meaning through a design's appearance, affordances, metaphors, and material properties. Sematic aspects of design, as described by Krippendorff, include the intentional shaping of meaning-making. To engage with semantic mediation of values, designers need to hone what Krippendorff refers to as second-order understanding – the ability to understand and respect stakeholders' understanding of what is meaningful. Viewing design as sematic mediation of values brings about a shift away from engineering-oriented concerns, such as usability and efficiency, towards semantic concerns, where designers could and should aim for forming dialogical relationships and shared understandings of meaning together with stakeholders, rather than treating them merely as sources of information. Design as semantic mediation of values ties into the idea that design projects consist of an expressive communicative chain between designers, artists, producers, and users, as described by Flangan et al. (2008). For this thesis, the role of the designer and the SVS in this chain are of particular interest.

Value mediation goes beyond a focus on users and customer needs. This aligns with Krippendorff's (2014) argument that meaning-making through design has implications not only for users but also for other stakeholders, communities, and society at large, and with his emphasis that designers should focus on a broad spectrum of stakeholders. Commissioning stakeholders such as SVS when engaging in design endeavours, can function as what Krippendorff describes as value gatekeepers, determining whether and how a design is implemented. How this value gatekeeping is upheld or breeched affects the design process.

Semantic value mediation is here approached as a profoundly complex and contextual process, calling for designers to avoid oversimplistic solutions and interpretations of design situations. Engaging with values becomes a core challenge in all design projects, but especially in collaborative ones, where different values and priorities need to be incorporated, and in projects with explicit value aims, such as designing cultural interventions with SVS. This standpoint aligns with previous work, including Flanagan et al. (2008), who emphasize the complex social, historical and political contextual aspects of values in design. It also aligns with Stolterman and Löwgren's (2007) *Thoughtful Interaction Design*, in which they argue that designers should draw on the psychological concept of authentic attention to understand the complex

particularities of each situation while balancing their own designerly agency with attentiveness to the needs and wishes of other stakeholders, such as commissioners. The importance of context also relates to the type of organization with which a designer collaborates. Khan & Matthews (2019), in *The Semantics of Design and Why They Matter*, argue that it is crucial for designers to adopt different roles depending on the design situation and the organizations involved. Their proposal that designers should develop context-sensitive ways to represent, communicate and articulate values in ways that resonate with collaborating organizations, closely aligns with the approach taken in this thesis.

As the term is used in this framework, mediation does not occur only through the artifact, which departs from how it is described by Verbeek (2011) and Latour (1994) due to their focus on factual designs rather than the process of designing. Mediation is also understood as a process in which designers can function as mediators. This aligns with the use of the terms *mediation* and *mediators* in human relations research: individuals who help parties communicate, manage emotions, foster collaborative problem-solving, and work towards mutually acceptable solutions (Brummans et al., 2022). The framework also draws on Manzini's conceptualization of the designer's role in social innovation, where he states that "designers' role includes the role of mediator (between different interests) and facilitator (of other participants' ideas and initiatives)" (Manzini 2014, p. 65). Manzini argues that designers should prioritize, facilitate, and design for value conversations, and that designers in social innovation projects must rethink their professional practice by shifting away from a designerly culture rooted in industrial design towards a view of design as a collaborative sense-making process (Manzini, 2015). Alternative terms have been used to describe this role. For example, Kimbell (2011) characterizes designers as cultural interpreters. Dennington (2018) describes designers as translators, both advancing a similar line of reasoning to Mazini. The choice of the term *mediation* in the theoretical framework of this thesis emphasizes that values are not regarded as fixed entities that can be directly translated or interpreted. Translation is understood as just one aspect of how designers can and should engage with values. Rather, the concept of mediation highlights that values are continuously renegotiated and reshaped throughout the design process, within the expressive communicative chain of design.

Building on the works discussed above, this thesis adopts the position that design can be understood as semantic value mediation. This perspective acknowledges that moral values are embedded in designs and that designers play a key role in the process that shapes such values. Designers therefore need to engage with design as meaning-making, develop strategies for engaging with values, foster dialogic relationships, build second-order understanding

with different types of stakeholders, practice authentic attention, resist simplistic interpretations of values, explore alternative approaches to design, and make value engagement a top priority.

Design as semantic value mediation includes both the ways in which values are mediated through design and the ways in which designers can act as mediators. These two aspects interact as design projects progress towards increasingly refined design proposals. Together with stakeholders, designers engage in a practice of mediation that transforms abstract values into concrete designs, which in turn seek to mediate the intended societal values.

Having established the framing of design as semantic value mediation, the next section introduces the specific dimensions of the theoretical framework developed in this thesis.

Dimensions of Design as Semantic Mediation of Values

There are many different design-related frameworks, and among these Design Thinking (Dam, 2023) has been one of the most influential. Design Thinking typically describes four iterative phases: empathize, define, prototype, and test. These phases inform each other throughout the design process. As discussed in the background chapter, however, applying Design Thinking unmodified to value-oriented design endeavors can be problematic. While acknowledging the benefits of Design Thinking, this thesis also takes a critical stance towards it. In line with other value-related HCI traditions, such as Reflective Design and Value Sensitive Design (VSD), the thesis seeks alternative approaches to combining design with moral value endeavors – approaches that might draw inspiration from Design Thinking while introducing additional dimensions.

The dimensions of the theoretical framework developed in this thesis draw inspiration from the framework proposed by Flanagan, Howe, and Nissenbaum (2008) in *Embodying Values In Technology*, but with some important deviations. There are similarities between VSD and *Embodying Values in Technology*, and Nissenbaum, one of the latter's co-authors, also helped to formulate VSD together with Friedman. Flanagan et al. propose three activities in a design process aimed at embodying moral values in technology: Discovery, Translation, and Verification. These activities resemble the phases of Design Thinking but focus on the relation to values in each phase.

Discovery is about identifying values and is divided into four categories: project values, values in instrumental design features, designer values, and user values. Project values are regarded by Flanagan et al. as predefined and therefore of less interest. Values in instrumental design features refer to

functional design requirements that influence values. Designer values concern the values held by members of a design team, an area the authors argue has received insufficient attention. User values are about identifying which values are important to users.

The second activity, *Translation*, focuses on converting values into design specifications, similar to the concept of translation within DfV (Van de Hoven et al., 2015). Translation is divided into operationalization – articulating values in concrete, designable terms – and implementation, which specifies the design features.

Verification concerns determining how successfully values have been embodied in a design. The authors suggest that verification can employ the same methods used to evaluate other design qualities, such as user testing, prototyping, and interviews. However, they acknowledge that evaluating values introduces a new layer of complexity beyond assessing usability. This complexity stems from the nature of the values: interconnected, abstract, and often difficult to measure. However, the authors do not expand on how to address this complexity.

This thesis draws on Flanagan et al.'s (2008) framework but introduces several important modifications. Rather than describing a design process as consisting of phases or activities, the framework conceptualizes dimensions that are all continuously present throughout a design process. This distinction is important because it implies, for instance, that value translation is never fully complete. The dimensions are not intended to be prescriptive; instead, they function as sensitizing constructs (Blumer, 1954) for analyzing a design process. The role of sensitizing constructs is discussed further in the methods chapter.

From the lens of design as the sematic mediation of values, the limited attention given to project values and already-explicated values in both VSD and Flanagan's et al.'s approach becomes problematic, as such values can still require mediation. To emphasize this distinction, the term *value elicitation*, rather than *discovery* or *explication*, is used throughout this thesis. Additionally, the concept of translation is interpreted somewhat differently from both Flanagan et al. and DfV. Finally, *value manifestation* is introduced to describe the dimension through which values become manifested in a design. This dimension also encompasses evaluating how successfully values have been manifested in relation to the design aims. The three dimensions of the theoretical framework in this thesis are thus value elicitation, value translation, and value manifestation. These dimensions are conceptualized and discussed in the following sections.

Value Elicitation

Value elicitation, as it is used in this work, encompasses activities and approaches for understanding and discussing values in design endeavors. The use of elicitation is connected to an interpretative perspective on values, in which values are seen as fluid, their interpretation might change over time, and interpretations can differ not only between organizations but also among individuals and roles within the same organization.

Elicitation is a well-established approach in ethnography (Geertz, 1980; Kyololo et al., 2023). Elicitation in a design context is discussed in the article *Elicitation of Situated Values: Need for Tools to Help Stakeholders and Designers to Reflect and Communicate* by Pommeranz et al. (2012), who describe value elicitation in design as engaging with designers in interpreting, formulating, explicating, and clarifying situated values unique that are to each design context and stakeholder. Pommeranz et al. formulate value elicitation in design as an alternative to requirements engineering, which they criticize for lacking clear methods for addressing soft issues.

Pommeranz et al. (2012) partly build on VSD but argue that VSD methods lack strategies for handling misinterpretations of values between designers and stakeholders. They also argue that values develop throughout a design process, and that stakeholders' envisioning of new technologies can in turn influence their values. They call for more tools to support value elicitation in design processes and suggest giving greater room to expressing and sharing value deliberation. They build on how elicitation is used within ethnography, focusing on long-term engagement with stakeholders, the use of multiple modalities to illuminate insights, and the exploration of interpretations and feelings connected to values. This constant renegotiation and reinterpretation of values mean that elicitation is never a fixed process. Pommeranz et al. (2012) propose methods for value elicitation such as value scenarios, which focus on emotions, and prototyping aimed at sparking reflections on values. Value elicitation can also be advanced through speculative design and design fiction, as proposed by Wong et al. (2017). Based on these perspectives, value elicitation is understood in the framework as a creative practice, with values, in line with Dewey's arguments (1934), regarded not as static but as taking shape through expressive practice. From this lens, value elicitation is not only about articulating values but also about reshaping and forming them as a design project progresses.

Further value elicitation within the framework is also conducted in relation to intrinsic values to create a solid ethical foundation for technology design, aligning with proposals by Bednar and Spiekermann (2024).

As discussed earlier in the thesis, SVS possess a high degree of value competence, which has implications for value elicitation. While users might struggle to articulate their feelings and priorities and relate them to underlying values (Kujala & Väänänen-Vainio-Mattila, 2009), SVS discuss complex values in their professional roles (Kavathatzopoulos, 2008). Designers can build on this when approaching value elicitation in co-design projects.

Value Translation

The value translation dimension, in which values are turned into design requirements, is well-formulated within Design for Values (DfV) (van de Poel 2013). This thesis draws on DfV's approach to value translation while also problematizing it. DfV focuses almost exclusively on value translation, but the translation dimension in this thesis should be understood, following Flanagan et al., as just one aspect of incorporating values into designs.

According to van de Poel et al., in the 2013 book chapter *Translating Values into Design Requirements*, translation processes happen in all design projects, but they often remain implicit, particularly with respect to non-commercial values. The process is made more explicit in VSD, but van de Poel et al. argue that VSD lacks clear methods for how to engage in value translation. Such methods are instead the focus of DfV, where they argue for the need for more conceptual tools to support value translation. One such tool is the value hierarchy, in which values are translated into design specifications and then followed through to ensure that those specifications are meaningfully incorporated into the finished designs (van de Poel, 2013).

The design translation process can also be studied using an RtD approach, as design researcher Claire Dennington (2018) has done, describing that designers should focus on understanding the sought-after values in design processes and then translate them into functioning design choices. Flangan et al. (2018) describe design processes as the translation of abstract values into concrete designs.

Communication designer and professor Baule (2016), in his article *Towards Translation Design: A New Paradigm for Design Research*, discusses a translational paradigm emerging within design that can generate new design sensitivities. He proposes that designers utilize strategies such as visualizations and tangible, multimodal practices during the translation process. Furthermore, he argues that designers should engage with semiotic transformation and specific cultural and social connotations to aid the translation process. Design translation becomes a form of value storytelling. Baule states that the translator is required to be an expert in understanding the thoughts of others, interpreting nuances of meaning, adapting to context, and expressing

values in understandable ways (Baule, 2016). Baule's approach aligns with Krippendorff's proposition that second-order understanding can be achieved through design processes that employ alternative strategies, including the use of narratives, metaphors, and scenarios, to bridge abstract values and concrete design outcomes.

Cibin et al. (2020) expand on the role of translators in social innovation projects. They argue that, to translate values into designs, values must also be translated between local stakeholders and design researchers in the projects. Being a translator is a distinct role, but one that designers can fulfill if they have the right skills and competencies.

In summary, the translation dimension concerns turning abstract values into something designable. This process is understood as interpretative, with the designer acting as a cultural intermediary and translator who employs strategies for mediating values into concrete designs.

Value Manifestation

As design projects progress, values become manifested and embodied in prototypes and designs (Flanagan et al., 2008). The concept of value manifestation draws on how Latour (1994) and Verbeek (2011) consider designed artifacts as mediating values.

Flanagan et al. (2008) acknowledge that artifacts can be proactively designed to manifest values. Within HCI, DiSalvo et al. (2014) engage with how designers can pursue societal and political values through their designs. Drawing on Latour, they expand on how designers enact social conditions and imagined futures in relation to democratic values. They point towards a traceable process through which design choices make objects political, and how designs can change people's subjective experiences in relation to political issues. Designed objects thus become part of policymaking by manifesting values. DiSalvo et al. argue that acknowledging the politics of design changes the role of the designer, from one who provides solutions to one who engages in articulating political issues through design. This requires the designer to scaffold between values and design throughout the design process (DiSalvo et al., 2014). This stance influences the way value manifestation is approached in the theoretical framework.

In strands such as DfV, no clear distinction is made between design requirements and the values that are actually manifested in the design. However, this thesis takes the stance that there is a difference between design requirements and actual design outcomes, building on Flangan et al. (2008), who describe the potential distortions between requirements and realized designs.

The thesis also acknowledges that unintended societal values may be manifested in designs – values that were never addressed during translation.

Analyzing whether values have been manifested in designs can be tied to artifact analysis, which is further developed in the method chapter. There is a need to verify whether the values elicited and translated are actually manifested in the designs. Examples of value verification methods, according to Flanagan et al. (2008), include user testing, interviews, and surveys. Value verification introduces additional layers of complexity compared to other forms of evaluation, due to the complex webs of moral values, which makes them harder to measure. Flanagan et al. (2008) do not expand further on the consequences of this difference between traditional design evaluation and value verification, and their proposed methods are closer to conventional design evaluation techniques such as usability testing.

Krippendorff (2006) argues that user testing is only one form of validation and can be insufficient for assessing whether values have been incorporated into a design. Rather, the meanings of artifacts can also be validated through how stakeholders communicate and through their discourses around the artifacts. Further validation can be done by tracing how stakeholders respond to design proposals and analyzing which ideas persist across iterations. Krippendorff argues that the meaningfulness of an artifact might not be observable directly. Focusing on semantics and meaning in design, according to Krippendorff, means that such properties cannot be measured in the same way as engineering-oriented design qualities.

As values are manifested in prototypes and designs, they can in turn spark new value elicitation. Each translation also leads to new manifestations. Thus, as already noted, value manifestation is not a final stage; rather, even the first sketches already participate in the manifestation of values. Value manifestation therefore also concerns how abstract values become concrete design features within the complexity of design processes.

The dimensions presented here will be used in the next chapter, where insights from the design projects and studies are analyzed in relation to the framework and existing literature.

Chapter 4. Method and Methodology

This chapter presents the overarching research method and methodology of the thesis. The thesis is methodologically grounded in Research through Design (RtD), and this chapter expands on the specific RtD approach used in this thesis, relates it to other methodologies, describes the sources and analysis used in the thesis, and presents and discusses ethical considerations as well as my positionality in relation to the thesis research.

Research through Design

RtD is a common methodological approach within Human-Computer Interaction (HCI) that combines research with active engagement in the design of artifacts and services while reflecting on the design process (Isley & Rider, 2018). At the core of RtD is the assumption that knowledge is created through the act of designing (Godin & Zahedi, 2014). Typically, RtD projects explore different concepts through the design endeavor and generate design exemplars, design implications and other insights that go beyond the specific project (Höök and Löwgren, 2012). The RtD projects in this thesis engage with different types of societal values in relation to designing playful cultural interventions.

RtD was first formulated by Frayling (1993) at the Royal College of Art and Design, proposing design as a way to explore concepts and generate knowledge. In his formulation of RtD, Frayling draws on Action Research, a methodology first developed in the 1940s in which researchers intervene in real-life situations while simultaneously studying them (Adelman, 1993).

The idea of Research through Design was introduced into HCI by Zimmerman, Evertsson and Forlizzi (2007), who argued that design practice should receive more attention within HCI, particularly in interaction design. RtD was formulated to be distanced from design science, an approach drawing on Simon (1969) and further developed by Hevner et al. (2004), which focuses primarily on improving engineering designs through systematic study. RtD instead approaches design as a creative practice and draws on Schön's (1983) concept of reflection in action – combining practitioner expertise with

reflection as a mode of inquiry. Schön in turn built on Dewey's (1889) concept of learning by doing. This foundation ties RtD to pragmatism (Dixon, 2019; Servillo and Schreurs, 2013).

Since 2007, RtD has grown and branched out as a methodology within HCI and can even be viewed as an emerging research paradigm (Isley & Rider, 2018). The following paragraphs outline different strands of RtD and relate them to the approach taken in this thesis.

An ongoing discussion within RtD concerns the type of knowledge it produces and how that knowledge should be conveyed. Stolterman argues that each design is an ultimate particular, possessing unique properties (Stolterman, 2008). This raises a central challenge for RtD: if each design situation is unique, how can it produce insights that are relevant beyond the specific case? Engaging with this issue, Höök and Löwgren (2012) describe the concept of intermediate-level knowledge – knowledge that is generative beyond a specific case without aspiring to become overarching theory. Intermediate-level knowledge can be in the form of design implications or strong concepts that can inform future design or highlight conceptually significant aspects of a design.

Intermediate-level knowledge can be understood as produced through the interplay between design practice and theory. RtD scholars argue that theories and concepts either emerge from the design practice (Gaver et al., 2022), intermediate with it (Redström, 2017), or precede conceptually driven design work (Stolterman and Wiberg, 2010). The RtD projects in this thesis include emergent, intermediary, and conceptually driven design investigations, while the overarching thesis contribution has developed alongside the design work.

One important distinction within RtD concerns whether the focus of study is the design process, the design outcomes, how those outcomes are received – for instance, by users – or a combination of these. RtD typically emphasizes the designed artifact, accompanied by a concise description of the process that led to it (Oogjes & Desjardins, 2024). Gaver (2012) established that RtD can convey knowledge through the development of conceptually rich artifacts, and Pierce (2014) further showed how RtD can focus on knowledge embodied in designs rather than derived from user studies. According to Godin and Zahedi (2014), the generation of such artifacts, combined with the use of complementary methodologies and validation by the research community (Volonté & Colombo, 2015), can contribute to validating RtD research. In this thesis, the individual designs are presented in the articles as conceptually rich artifacts, while the thesis as a whole is process oriented. This aligns the thesis with what Dalsgaard calls *research in and through design*, an approach that "aims at

exploring what happens in design processes through the researcher's engagement in experimental design projects." (Dalsgaard, 2010, p. 1).

Regarding what constitutes a design output, artifacts are a common focus within RtD (Oogjes & Desjardin, 2024). This thesis instead draws on Waern and Back (2017), who argue that designed activities, rather than artifacts, can be understood as ultimate particulars of interaction design.

An important discussion within HCI design research concerns the distinction between critical or pragmatic approaches to design (Bardzell, 2019; Forlizzi et al., 2018). RtD often adopts critical perspectives (Pierce et al., 2015), but it can also be closely connected to practice by proposing alternative design solutions and implications (Gaver, 2012). This thesis leans towards a pragmatic approach, aiming to develop implementable design solutions together with SVS.

Facing challenges and failures in design processes can also provide valuable knowledge contributions within RtD (Gaver et al., 2009; Howell et al., 2021). Design projects are inherently messy, and it is important to identify which insights generated through that messiness can become useful for others (Gaver et al., 2022). This thesis therefore seeks not only to describe successful design outcomes, but also to examine the breakdowns, challenges, and failures in the projects.

RtD can be understood as an abductive methodology that engages with futuring through design (Redström, 2017). While some RtD projects investigate speculative or distant futures (Bertran et al, 2021), this thesis instead focuses on near-futuring by exploring implementable design solutions. Previous research has shown that this emphasis on the near future can generate knowledge that bridges current and future practices (Charisi et al., 2018; Laouris, 2015; Lupetti et al., 2018).

RtD is inherently interdisciplinary (Isley & Rider, 2018) and has even been described as post-disciplinary (Wilde, 2020). This is especially true of this thesis, as the projects are themselves both multidisciplinary and multisectoral. They engage with diverse issues, such as community building, sustainable development, and cultural heritage, while drawing on theories and perspective on playfulness, education, psychology, landscape architecture, HCI and interaction design, and value-centered design.

Ethnographically Inspired RtD

An important methodological stance in this thesis is that design processes are complex, relational, contextual, and dynamic. This aligns with similar positions formulated by Wakkary (2005) and Stolterman (2008), who argue that it is problematic for interaction design researchers to borrow heavily from

scientific and systematic methods. Instead, they advocate developing a deep understanding of the human nature of design practices. To support this approach, the thesis draws inspiration from ethnomethodology, design anthropology, and autoethnography. The focus is on interpretation, thick descriptions of design situations, and the exploration of meaning-making rather than systematic or objective truths. This approach is expanded on below.

Ethnomethodology has been an important inspiration for interaction design studies in HCI (Dourish, 2006), stemming largely from the design lab at Xerox Park (Crabtree et al., 2012), with HCI pioneers such as Suchman (1987), who established that human interaction with technology is a situated practice. Design ethnography has since become an important foundation of interaction design research and practice (Crabtree et al., 2012). This thesis utilizes many ethnographic methods, such as observations and interviews, while also drawing on ethnographic perspectives through its emphasis on subjectivity and the social and cultural context of design. Traditionally, design ethnography focuses on observing and interpreting existing practices. By combining design ethnography with RtD, this thesis instead places greater emphasis on creating new design situations. This brings the methodology closer to Action Research (Avison et al., 1999).

Because the thesis draws on first-hand lived experience, it also leans towards autoethnography. During recent years, scholars have acknowledged a substantial overlap between RtD and autoethnography (Triantafylli & Bofylatos, 2019), and although still not common, the two approaches have been explicitly combined (Feng et al, 2026; Gamboa, 2022, Claisse et al., 2025). Autoethnography is often evocative and heavily focused on the researcher's personal experience. This thesis instead leans on analytic autoethnography, as formulated by Anderson (2006), which emphasizes theoretical analysis of the situations the researcher is immersed in, rather than focusing on personal emotions.

The thesis also builds on long-term engagement in design processes, making it compatible with an anthropological approach to RtD. Design anthropology (Gunn et al., 2013) is based on researchers becoming immersed over an extended period in a social setting, in a sense becoming an integral part of that setting. Design anthropology, as described by Otto and Smith, draws on Friedman's work and is compatible with VSD.

Taken together, this thesis positions RtD in a way that emphasizes subjectivity, first-hand lived experience, interpretation, and long-term active engagement in design processes. It draws inspiration from ethnomethodology, analytic autoethnography, and design anthropology, but treats these as

complementary perspectives integrated into the RtD approach rather than as separate methodologies.

RtD and Its Relationship to the VSD Tripartite Methodology

This thesis draws on RtD as a methodology, but since it also builds on Value Sensitive Design, it is important to clarify how the thesis relates to VSD as a methodology. While VSD is primarily a design approach and theoretical lens, it also offers a methodological framework (Friedman and Hendry, 2019).

Previous research has shown that VSD and RtD can be effectively combined. For example, Conversano, del Conte and Mulder (2019) drew on VSD to explore value design activities across seven RtD projects, demonstrating how joint reflection and value consolidation helped bring values to the forefront of RtD projects.

VSD proposes a tripartite methodology in which empirical, conceptual and technical investigations of values are carried out iteratively (Friedman and Hendry, 2019). Technical investigations may involve either retrospective analyses of existing designs or the proactive creation of design solutions that embody desired values. Technical investigation remains underrepresented in current VSD studies (Gerdes & Faber Frandsen, 2023), particularly the proactive design approach, which aligns closely with RtD and therefore forms a central part of this thesis. Conceptual investigations have been carried out in the thesis through the conceptualization of societal values and through literature analyses of project values such as community, belonging, and loneliness. Empirical investigations have been conducted through observations, interviews with SVS, and other ethnographically inspired methods. While the thesis engages with all three components of the VSD tripartite methodology, it develops its own framework for analyzing the design projects.

Long-Term Real-World Design Implementation Projects

The design studies in this thesis span several years and can be understood as longitudinal in situ studies. Such studies are underrepresented in HCI, according to a literature review by Kjærup et al. (2021). For RtD, participating in a design project over an extended period of time and allowing it to unfold naturally can be crucial for generating meaningful insights (Godin & Zahedi, 2014).

RtD commonly involves the development of prototypes (Godin & Zahedi, 2014; Oogjes & Desjardins, 2024), but prototypes risk carrying simulated properties that are not aligned with how designs function in real-world implementation (Salovaara et al., 2017). According to Wiberg and Stolterman (2014), isolated prototypes seldom capture real-world design practices, which

creates a gap between research and design practices. This thesis addresses this gap by engaging in real-world design projects aimed at implementation. Focusing on the full design process through to implementation and beyond is increasingly relevant, as digitalization has radically altered when a design process can be considered concluded (Scherling, 2024). Design processes now extend into implementation and beyond inauguration, through reiterations and reprogramming in line with design-after-design (Fry, 2017). The blurring of when a design process ends has implications for RtD, calling for a move beyond prototype-focused RtD (Giaccardi, 2019). One response to this has been increased interest in RtD for deployment studies (Odom et al., 2020), but these studies typically engage with artifacts developed within and for a research context, rather than with authentic design implementations.

This thesis bridges RtD-based research and authentic interaction design practices by engaging in real-world design projects and by following design processes beyond the prototype phase, towards design solutions that are implemented long-term.

Research through Co-Design

All the design projects in this thesis are carried out as co-design projects, where societal value stakeholders have been full members of the project teams. In recent years, there has been a call for RtD to be more closely integrated with co-design, and various forms of participatory RtD have been proposed (Wilde, 2020). According to Sanders and Strappers (2008), collaborative design approaches change both how design research is conducted and what knowledge is produced. Busciantella-Ricci and Scataglini (2024) argue that Research through Co-Design provides unique insights by engaging with what they call the disagreement space – where effort is made to produce shared outcomes despite differing perspectives and priorities. A current challenge for co-design research is evaluating the real-world impacts of co-design processes (Slattery et al., 2020). Understanding the nature of design practice is crucial in interaction design research (Stolterman, 2008). Outside academia interaction designers typically collaborate with clients, and a core aspect of their practice is balancing between their own needs, others' needs, wishes, and design priorities – something that, according to Stolterman and Löwgren (2007), should receive more attention in interaction design research.

RtD Based in Multiple Design Projects

The thesis builds on several different design projects, forming a collection of empirical insights from work on different design topics, in different venues,

and with different project partners. This approach aligns with Gaver's argument that a single design solution represents only an isolated point in design space, and that collections of designs and projects can be brought together to generate RtD knowledge and identify concepts that extend beyond specific contexts (Gaver, 2002).

The wide array of disciplinary and sectorial perspectives in this thesis has been synthesized by applying Redström's (2017) approach to making design theory – where an eclectic collection of theories, design inspirations, stakeholder input, and other inputs are brought together into a designerly worldview as a foundation for formulating design programs. The approach involves identifying core concepts in the RtD process, and investigating and probing them through the act of designing. RtD knowledge is thus generated in an intermediary way through the design process.

As the design projects progressed, new conceptual perspectives emerged, allowing for a reframing and re-examination of the RtD studies. Rather than resisting the reframing or ignoring insights in order to remain aligned with the initial research plans, this thesis draws on Zimmerman et al. (2022), whoemphasize that reframing in RtD can generate important retrospective insights. Similarly, Gaver et al. (2022) highlight emergence as a key aspect of RtD, where new insights arise during the design process and subsequently inform further design activities.

There is an important chronology in how insights and design approaches have developed within and between the design projects in this thesis. As new projects were initiated, we carried forward insights from previous projects. Some projects were also conducted in parallel, informing one another by providing perspectives from different stakeholders and design contexts. This mirrors Redström's (2021) idea of making design theory through probing and the development of a designerly worldview across multiple design projects.

To provide an overview of this chronology and how it informed the thesis: PlayIT was the first initiated project. Towards the end of the project, we wrote the supporting article on value-driven playground design (Back, Johansson & Wireband, 2021), where an early foundation for the thesis topic emerged, though without a clear theoretical framing. In parallel with PlayIT, I was also involved in GIFT, and the initial problem formulation of the thesis largely stems from insights related to a breakdown that occurred in one of the GIFT sub-projects, and how playful design activities in that situation helped to bring value tensions to the surface. When the ABC project began, one year after PlayIT ended, ideas on exploring methods and aspects of designing for values were refined. The Castle Museum was initiated midway through the PhD and could from the outset build on insights from the other projects as well as

findings from the PhD work itself. These projects can be seen as iterative, where theoretical insights and design practices evolved both within and across the projects.

Furthermore, in the beginning of the PhD, I placed extensive focus on design outcomes and how they could embody values. This was later reframed towards a stronger focus on values in design processes. This shift was informed both by insights from the design projects and by theoretical engagement with DfV, VSD, PD, value mediation, and other value-related approaches. This led to a more nuanced understanding of values in relation to design compared to how values were approached in the early PlayIT publication.

By combining insights from the different projects and the literature, the framework for the thesis emerged. Gaps were also identified in existing approaches – for example, how explicitly stated project values in VSD are typically treated superficially, which did not align with experiences from the real-world design projects, where project values were continuously reinterpreted and negotiated. As the theoretical framework became more established, this led to further iterations and refinements of the design approaches.

Designerly Worldview of the Thesis

In adherence with Redström's approach, a designerly worldview has been formulated throughout the thesis work. While the designerly worldview is, in a sense, highly personal and case-specific, as an articulated design methodology it can also be generative for others. The designerly worldview should be considered both a type of design methodology and a result. As it has evolved, the designerly worldview has informed design activities and methods in the design projects as well as the analysis. In Chapter 7, aspects of how the designerly worldview has been enacted in, and informed, design practice is expanded upon. In this methods section, the methodological aspects of the designerly worldview are addressed and related to literature, grounded in core assumptions about how to successfully engage with societal values together with SVS in design processes.

Deliberative Design Practice

Co-design in this work is approached as a deliberative practice. Several influences in the literature support this orientation. Laurel (1991), in *Computers as Theatre*, argues that designing interactive experiences should involve transferring power in the design process from those with technical expertise to those who understand human interaction and values. Détienne, Baker and Le Bail (2019) describe the importance of acknowledging different cultures when

engaging with values in design, arguing that sustained collaboration fosters a shared culture around values through negotiation and argumentation between designers and stakeholders. Manzini (2015) emphasizes that social innovation design projects are highly dynamic processes involving complex consensus-building through social interaction. Tjahja and Yee (2021), in *Being a Sociable Designer: Reimagining the Role of Designers in Social Innovation*, propose that design is a socially embedded practice, and that sociality and facilitation of value conversations are crucial to engaging with values in design processes. They further suggest that designers should prioritize building trust and personal relationships.

Busciantella-Ricci and Scataglini (2024) describe co-design as a "disagreement space," while this thesis instead proposes engaging with design as a deliberative space. Differences between stakeholders can be considered as an asset – a space for generating new ideas and inspiration, and for merging perspectives. To support this, design processes should foster curiosity, deliberation, joint exploration and perspective-taking, and aim not necessarily for consensus, but at least for shared understanding. This standpoint draws directly on deliberative theories (Jennstål and Öberg, 2019; Jennstål et al., 2021) and is also inspired by my own work on deliberation and role-play design (Johansson et al., 2026; Bowman et al., 2026).

Acknowledging and Bridging Competencies

Manzini distinguishes between diffuse and professional designers. However, this thesis problematizes the associated power dynamic, which positions designers as more "professional" in a one-directional hierarchy. If SVS are understood as diffuse designers, they are not inferior to designers; rather, they possess different competencies in their professional capacity, such as professional moral value competence (Kavathatzopoulos, 2009). Conversely, designers may be understood as having diffuse societal value competence. This balance must be addressed in social innovation projects and is as important – if not more important – than ensuring that SVS understand design methods. This stance aligns with Tjajha and Yee (2021), who describe how the position of stakeholders has shifted towards greater equality with designers, rejecting the view of clients and users as passive. They propose that designers should be willing to engage with ways of knowing and doing beyond the designerly, something I aim to embrace in my design endeavors.

Balancing Adherence with Agency

A challenge for designers is balancing adherence to stakeholder priorities with pursuing one's own designerly perspectives. According to Manzini (2015), social innovation can include elements of both radicalism and activism. Still,

he positions social innovation design as part of human-centered design, even though activism can involve challenging public opinion. In contrast, Tjaha and Yee argue that it is problematic in social innovation to lean into human-centric design, instead proposing a situational and social-centric focus (Tjahja & Yee, 2022).

Manzini criticizes how designers can get stuck in either of two extremes in social innovation projects. One extreme involves becoming process facilitators who mainly gather input and synthesize it into design suggestions, in what he calls "Post-it design." The other extreme, which he calls "Big Ego Design," involves designers pursuing their own ideas without adequately engaging with input from the broader design network. He argues that designers must use creativity to transform their personal design culture into visions and proposals, while remaining responsive to the broader design network in the social innovation process, thereby maintaining dialogic collaboration. The designer has to balance critical, creative, and dialogic approaches, gradually introducing more mature proposals into the design conversation while mixing action and reflection (Manzini, 2015). This aligns with Thoughtful Interaction Design, as described by Löwgren and Stolterman (2007), in which designers balance integrity in their own designerly views with responsiveness to stakeholders and a reflective mindset.

In my approach to Design for Societal Values, I seek to balance adherence and agency – being sensitive to societal values and SVS perspectives while also exercising designerly agency to advance design ideas that enable compelling interesting interaction design. This becomes a mediation between designerly perspectives and societal value perspectives.

Furthermore, I seek to tap into the creative, playful, and social aspects of design while also acknowledging the specific properties of societal values. This calls for developing novel design approaches that differ from design thinking and are also partly distinct from current value-oriented design approaches and evaluation frameworks.

Creative Design Approach

Design in the thesis is approached as a creative practice, drawing on scholars such as Löwgren (1995) and Laurel (1991). The approach to creative design in this methodology particularly aligns with Buchanan's (1992) argument for viewing design in relation to societal challenges as a liberal art, where adaptability and empathic sensibility – rather than purely technological solution proposals – are central skills. When engaging with design in relation to complex societal challenges, it is as important to unpack the problem space and underlying assumptions as it is to develop solutions. Furthermore, this should

be done through empathic sensibility towards the stakeholders a designer collaborates with, in order to find creative ways to engage with values and to sensitize them to the situational values of each design endeavor.

Playfulness

Playfulness is central to all design endeavors in this thesis. In the design work, I have explored how playfulness can support the pursuit of societal values, both in the design process and through the resulting designs. This builds on insights from previous literature on how playfulness can be utilized in interaction design (Gaver et al., 2004; Tieben et al., 2014), and specifically how it can support value conversations in design contexts. There is a bridge between play and semantics in how Gadamer (1960/2013) conceptualizes play as a collective expression of meaning and as an exploration of worldviews. Within participatory design, Ehn and Sjögren (1991) advocated for design-by-playing and demonstrated how playful approaches – such as role-playing exercises, playful workshops, and games – can make design ideas more accessible to stakeholders, support the creation of shared visions of the future, bridge language barriers, and spark design discussions. The playful design approach draws on ludic design (Gaver et al., 2004), emphasizing playful exploration, open-endedness, and reflection in both design outcomes and processes. Ludic design is, as described in the background chapter, integrated into Reflective Design (Sengers et al., 2005). Playfulness does not equate to fun but is understood in the thesis as a mindset that can encompass a wide range of emotions. Of particular interest in the design work of this thesis is how playfulness can be used when engaging with SVS in their professional capacity.

Tangibility and Embodiment

An important aspect of designing with and for values concerns how to translate abstract values into concrete design outputs. This is relevant both in the resulting designs and as an approach in the process itself. Tangible and embodied design exercises, such as mock-ups and cardboard prototypes, have long been recognized as a way to make futuring and value exploration in design processes more hands-on (Ehn & Kyng, 1991). Within VSD, tangible exercises such as using small wooden figures, game-like boards, and building small models are commonly used (Friedman and Hendry 2019). Physical artifacts can function as boundary objects for discussions on values (McLeod et al., 2012; Schoormann and Kutzner, 2020). In the design projects, physical objects, crafting simple prototypes, and testing design solutions are explored as a way to enhance societal value conversations. The framework places particular emphasis on how values are manifested in designs throughout the design processes.

Storytelling and Metaphors

Values can also be approached through storytelling, metaphors, and the formulation of visions of the future. Several scholars support this approach. Junginger (2017) proposes utilizing creative storytelling methods when engaging public stakeholders to explore values. As described in the theory chapter, Krippendorff (2006) advocates the use of metaphors and scenarios to make abstract values more concrete in design processes. Ehn and Sjögren (1991) also promote storytelling metaphors as tools for engaging stakeholders in value discussions. Reflective Design (Sengers et al., 2007) similarly highlights how designers can elicit value-rich feedback by using metaphors. In the design work of this thesis, storytelling and metaphors have been used in several ways, both in the design process and the design outputs.

Defying Measurability

While the thesis does include some measurable aspects, such as the number of participants and the duration of their engagement with a design, the designerly worldview builds on an overarching stance of defying measurability in relation to societal values. This position draws on several value-related theories. The soft and intrinsic properties of societal values, as well as the problems of imposing measurability onto SVS, are discussed in the background chapter. Dewey (1934) was among the first to argue that the meaningfulness of an experience cannot be broken into measurable units without destroying the character of that experience, and that economic or scientific approaches to experiential values should therefore be avoided. Buchanan (1992) similarly emphasizes the inherent indeterminacy to design problems that engage with societal challenges. This stands in contrast to, for instance, how DfV focuses on translating values into measurable design requirements (van de Poel, 2015). Rather than setting design requirements and measuring against them, the designerly worldview is based on a designer's engagement with the creative task of interpreting societal values. This involves engaging with all the soft and intrinsic properties of values and manifesting them in design.

Active Value Mediation

The thesis focuses on social innovation projects that pursue societal values in policymaking contexts. This stance can be understood as a form of active value mediation, as formulated by Verbeek (2006). He makes a distinction between a minimal morality approach – where design avoids undesirable value mediation – and an approach in which design actively embeds specific

forms of value mediation (Verbeek, 2006). Building on Verbeek's distinction, this thesis engages with active value mediation.

A striking insight into the difference between active and passive value mediation emerged when engaging with different value-oriented design traditions. When first encountering DfV, it initially appeared to align in several ways with the perspectives of this thesis. However, this alignment shifted when I more closely examined how "values" are operationalized. In a seminal DfV example, van de Poel (2013) described a case in which animal welfare is translated into design specifications for chicken coops in industrial farming. A hierarchical structure of values is constructed, translating overarching values into specified requirements, such as nest size (at least 1 m2 per 120 hens), automatic egg collection, and low-cost systems. From an animal ethics perspective, this raises concerns about how such values are interpreted in practice. The value translation is performed from an engineering perspective and does not include any deeper reflections on whether 120 hens sharing 1 m2 of nest space can be meaningfully considered animal welfare beyond compliance with regulations. Nor does it address whether caged systems can be considered compatible with animal welfare at all. Furthermore, no clear distinction is made between economic and non-economic values. In this framing, DfV does not engage with the nature of the values at the top of the hierarchy, but mainly focuses on minor improvements within existing constraints rather than pushing boundaries in animal welfare.

Drawing on Verbeek (2011), increasing cage sizes slightly to comply with welfare regulations represents a minimal morality option. Rather than merely following policies and regulations, design can and should also be used as a form of policymaking, as proposed by Kimbell and Bailey (2017). This thesis rejects purely instrumental and minimal morality approaches and advocates for active societal value mediation in projects involving SVS, where design becomes a form of policymaking. These are the types of design endeavors that are in focus in this thesis.

Sources and Analysis

This section outlines the sources this thesis draws upon and how analysis has been conducted. Details on sources, documentation, and analysis methods for individual studies are provided in the respective articles. This section focuses on the overarching sources and analytical approach of the thesis.

The thesis builds on a mixed-methods approach (Shorten and Smith, 2017). The foundation consists of first-person engagement in design projects, including the planning and participation in design activities such as meetings,

workshops, prototype building, prototype testing, and design evaluations. In addition, the thesis draws on insights from the designed artifacts themselves and their evaluation. This has been complemented by interviews with SVS, conducted both during and after the projects.

In-the-wild studies (Rogers & Marshall, 2017) have been conducted in relation to the implemented designs. However, as this thesis engages with real-world design projects, the design processes themselves can also be understood as taking place in-the-wild.

Building on established best practices in RtD (Prochner & Godin, 2022) key procedures in the design projects included detailed documentation of design processes, activities, and outcomes. Documentation was collected through memory notes, photos, audio recordings, sketches, design documents, and prototypes. These sources served as an aid for reconstructing what occurred in the projects. Observations were conducted qualitatively, using documentation methods such as thick descriptions (Freeman, 2014).

The thesis combines theoretical insights emerging from the design projects with theoretical insights informing them. In later project activities, such as in the Castle Museum project and later stages of the ABC project, both design activities and interview questions were specifically oriented towards the research topic. In earlier design projects and activities, such as PlayIT, there was instead a need to retrospectively analyze design documentation and conduct follow-up interviews in relation to the thesis topic.

RtD researchers have long struggled with finding suitable formats for illustrating how theories and concepts are woven into designs, as the written format of academic publication is inherently different from the embodied nature of design practice (Gaver, 2012). A widely used format for presenting RtD knowledge is the annotated portfolio, where collections of designs are annotated to highlight concepts and theories (Gaver & Bowers, 2012; Gaver, 2012). This method has been criticized for being vague regarding what constitutes "theory," but it can provide a useful bridge between design practice and academic research, as well as a way to present intermediate-level knowledge (Löwgren, 2013). In this thesis, annotated portfolios are used in the individual articles, while the thesis also provides a higher level of abstraction by synthesizing the design projects based on the theoretical framework.

The analytical approach to understanding how values are manifested in designs takes inspiration from artifact analysis. Artifact analysis is a well-established qualitative method, for instance in archaeology, used to analyze an artifact not through user testing but through systematic examination of its properties (Fleming, 1974; Norum, 2008). Artifact analysis can also be applied to design (Lee & Stolterman, 2014; Trausan-Matu & Slotta, 2021; Hanington &

Martin, 2012), and acknowledges the knowledge embedded in designed artifacts themselves. It involves a systematic examination of objects to understand the cultural, societal, and technological contexts of both the objects and their creators and users. The thesis takes inspiration from artifact analysis to examine how societal values are manifested in designed artifacts.

The synthesis of insights from the design projects is further inspired by narrative inquiry (Bell, 2002; Clandinin & Huber, 2010), which emphasizes rich recollections of events rather than attempts at objective description. Writing about design processes as narratives has been recognized as a way of conveying comprehensive accounts of often messy design processes (Gaver, 2012).

The analysis in this thesis builds on narrative descriptions of insights from design projects related to the conceptualization of societal value and how to design for it, grounded in the theoretical framework and its dimensions; value elicitation, translation, and manifestation. These dimensions should not be understood as themes in a rigid thematic analysis sense, but rather as sensitizing concepts. Sensitizing concepts were introduced in sociology by Blumer (1954), distinguishing between definitive concepts and open-ended, interpretive concepts that guide analysis. Sensitizing concepts have been used in a variety of studies (Bowen, 2019). Their use in this thesis is similar to bridging concepts – forms of intermediate-level knowledge used in design research to bridge theory and practice through generative, though not fixed, conceptual structures (Dalsgaard & Dindler, 2014). The sensitizing concepts of the theoretical framework provide a lens for analyzing insights from the design projects, but should not be viewed as fixed categories.

Vignettes are used in this thesis to introduce each dimension of the theoretical framework, capturing the lived experience of participating in the design projects. As described by Goodman, Stolterman and Wakkary (2011), vignettes are a way of conveying insights from a designer's perspective and can be understood as a form of micro-narratives (Skeggs et al., 2025).

Ethical Considerations

Ethical considerations can address both how ethics have played out in the actual research and the formal ethical requirements. Regarding formal ethics, two ethical approvals from the Swedish National Ethical Review Authority have been obtained. One covered the ABC project studies and was approved, while a more overarching application for studying playful interventions in public places and their design processes was deemed not to require ethical approval.

In some ways, the thesis has been relatively unproblematic from a formal ethical perspective. The SVS have acted in their professional capacities, and the designs have consisted of playful installations that are not heavily provocative. While the focus on societal values involves semi-political content, these values have been grounded in publicly formulated policy documents. However, there is a distinction between formal ethical approval and the enactment of research ethics in practice (Haggerty, 2004). When engaging with humans and real-world situations, ethical challenges appear in various and sometimes unexpected ways, calling for on-the-spot considerations (Murphy & Dingwall, 2001). This definitely applies to the co-design projects in this thesis. Some core ethical considerations are expanded upon in this section.

The Ethical Relation Between Researchers and Practitioners

Engaging in practitioner-close research raises ethical questions regarding the relationship between the researcher and the practitioners. One such question concerns authorship in academic publications. Authorship guidelines, such as the Vancouver recommendations (International Committee of Medical Journal Editors, 2026), are not fully tailored to RtD co-design projects. As practitioners engage in design work and reflection, their contributions may extend beyond being research subjects and instead constitute meaningful RtD knowledge contributions.

It is important to acknowledge the academic power dynamics involved in research and co-design with non-academics. Based on experience, non-academic practitioners might be unaware of the significance of authorship and author order. They might not even be aware that publications are being produced.

When I, as a practitioner, was invited to co-author, I did not initially understand what that entailed, but later came to recognize the importance of being included. In our projects, we have attempted to invite practitioners as co-authors. But this introduces another dilemma; without dedicated project funding, practitioners and their organizations are often not compensated for co-authoring, while researchers benefit financially and in terms of academic career progression.

Another tension concerns anonymity versus attribution. In creative practice, it is deemed ethical and standard practice to always credit the creators of designs, artworks, or games. In many research traditions, however, anonymization is considered the norm. In our projects, practitioners have sometimes preferred, or even required, to be named, either individually or through their organization. We have opted for offering pseudonymity in publications (Zerdick, 2021), while being open about organizational names in other communication channels.

Economic Interests in RtD Projects

Practitioner-close RtD projects have a short path to potential productification, which comes with ethical implications. The blending of commercial and research interests is common (Caulfield and Ogbogu, 2015) and can be ethically challenging, such as through conflicts of interest (Capps, 2016). At the same time, it can also contribute to positive societal impact (Slotfeldt-Ellingsen, 2023). In RtD, productification can provide opportunities for long-term in-the-wild studies (Rogers, 2007), provide showroom design cases (Koskinen et al, 2011), and open additional research opportunities. From a systemic perspective, current funding structures and university policies often support productification, but this can lead to conflicts of interest in academia-industry collaborations (Hillerbrant and Werker, 2019).

These ethical considerations have not been merely hypothetical in the projects. For instance, in the Castle Museum project, the museum wanted to implement the design proposals and was prepared to set aside a budget for this purpose. After discussions with the university's innovation support, we stepped in as consultants for the implementation. In the ABC project, a similar situation arose when a public real estate company sought to implement an interactive art installation. In that case, we chose a different path and commissioned an external art group. Both approaches came with opportunities and challenges, and we addressed them through reflexive ethical discussions and by drawing on relevant ethics literature. Our collaborators were mainly SVS, who do not pursue commercial interests, but this does not completely eliminate the ethical complexity. Such issues are more extensively explored in medicine and engineering research (Hillerbrant and Werker, 2019), but appear less developed in RtD literature, potentially because many RtD projects remain within academic prototype contexts (Godin and Zahedi, 2018).

Another ethical angle relates to the use of substantial public funding aimed not only at research, but also at improving citizens' lives. The RtD projects have largely been guided by community needs, although at times we have had to step back from aspects that were mainly of research interest. For example, in the ABC project, we initially proposed features for the artwork that were of research interest but costly, and not strongly supported by the art group. This became a discussion in the research team about whether it was ethically appropriate to pursue their inclusion.

There are economic incentives involved in the very structure of social innovation projects that have ethical implications. While paying large sums to research subjects is ethically controversial (Largent and Lynch, 2018), giving non-academic project partners salaries for years for their participation in RtD projects is standard practice. Drawing on my own experience, being given the

opportunity for our small social enterprise to be part of an innovation project, that meant secure income for years, adhering to being a research subject under those conditions was an easy choice. With such powerful economic incentives, this could be considered undue influence (Largent and Lynch, 2018). On the other hand, it is built into the funding structure as a systematic problem, rather than a reason to avoid doing a specific RtD collaboration.

Another ethical dimension of social innovation projects is that they typically require strong consortiums to secure funding. In an RtD innovation project I was part of as a research assistant, discussions in the research group included concerns that it might be unwise to describe the many collaboration challenges too openly in research articles, as this could affect future collaboration opportunities. This form of self-censorship felt ethically problematic to me and points to the need for greater discussion and awareness of these issues.

Consent in Long-Term RtD Projects

The third major ethical consideration concerns consent in long-term real-world RtD projects. While participants provide informed consent, there are many ethical gray zones regarding what constitutes data collection in long-term, ethnographically inspired studies (Rysst, 2025). In a traditional interview setting, participants are typically well aware of being recorded. In contrast, in our projects, collaboration with SVS has extended over years, with hundreds of hours of interaction. In routine meetings years into a project, participants may no longer have the same top-of-mind awareness of being part of a study. In informal situations – when wine is shared during project dinners and silly design ideas are discussed – I have to assess in the moment whether such interactions constitute research data. When meetings end up in harsh words or even tears, the research must navigate whether to act as an observer documenting tensions or as a fellow human being, colleague, or friend prioritizing care and support. As an active project member, the researcher may at times experience emotional involvement, making it difficult to remain unbiased. To mitigate these challenges, decisions must be made as to what is necessary to record and include in the research materials. To ease the pressure of being long-term research subjects, only selected sessions were documented, and participants were explicitly informed when documentation occurred. This relates to the ongoing balance between respecting research subjects and contributing to research insights (Vetenskapsrådet, 2024).

The ethical stance has been informed by insights from agile ethnography ethics (Mara et al., 2017), which emphasizes avoiding harm, maintaining ongoing consent, and keeping a dialogue with project members as research focus

evolves (Atkinson, 2009). These approaches align well with the abductive and emergent nature of RtD projects.

SVS representatives were informed that we only pseudonymity (Zerdick, 2021) could be guaranteed and that they might still be indirectly identifiable. It should be acknowledged that it takes courage to participate on these terms, as not all research insights might be favorable to participants. This also places an ethical responsibility on researchers to provide an honest account of project developments, while being aware of potential professional and personal consequences for the participants.

Implementing designs has enabled in-the-wild studies but has also introduced new ethical dilemmas regarding consent in open settings such as parks, museums, or public streets. Informed consent has been obtained for most user studies and tests, but not for all non-intrusive in-the-wild observations. In such cases, we have relied on best practices for ethical covert research (Spicker, 2011).

Real-World Consequences of RtD Implementations

Engaging in real-world implementation projects comes with consequences for thousands of citizens. The stakes are therefore high, so we have not been able to be as experimental as would be possible in prototyping contexts. As discussed earlier in the methods section, failures can generate valuable RtD insights, but in implementation context such failures have real-world consequences. Many concepts that would have been interesting to pursue from a research perspective had to be deprioritized in favor of safety and feasibility. While one might argue that responsibility lies with the SVS as implementers rather than researchers, this underestimates the agency of the RtD researcher. In the PlayIT project, we became aware of serious potential risks associated with our design suggestions. For example, we proposed technology-enhanced treasure chests that children could open. Municipal representatives assessed that such a design would violate EU safety regulations for playground equipment and described risks such as children losing fingers if one child tried to open the chest while another jumped on it. Another playground feature, a birdhouse opening in a story hut, required specific dimensions because there was a risk of children getting stuck and potentially strangled. Such discussions were very sobering, as they highlighted just how high the ethical stakes were in real-world design. At the same time, engaging with these constraints prevented us from pursuing certain designs that could have been safely trusted as prototypes.

As always with ethical considerations, it is important to look at both sides, including the ethics of not conducting this type of research. Beyond producing

designs that directly benefit local communities, the work also contributes to broader research insights. Despite the ethical challenges, we have aimed to follow sound research ethics not only formally, but also informally in how we have approached projects, topics, and people.

Positionality as a Designer, SVS, and Researcher

In line with the ethnographic approach to this research, this section outlines my positionality in relation to my roles as a designer, stakeholder, and researcher.

In all three main design projects, I have been actively involved throughout the entire design processes. The projects were all conducted in small teams, which means that my influence on the processes and their outcomes has been substantial.

I have been involved in writing grant applications, participating in the first tentative project kick-offs, and planning timelines and budgets. I have made crucial design decisions, attended crisis meetings, engaged in email conversations, facilitated brainstorming sessions, prototyping, completed final fixes before inaugurations, and conducted follow-up studies. I have collaborated with project team members in weekly online meetings, site visits, informal dinners, and academic conferences. I have planned and hosted design workshops and most of the design activities within the projects. I have walked through muddy construction sites trying to visualize a playground among the bulldozers. I have taped out the floor of a castle to create a Dogma-film-like mock exhibition. I have walked with my eyes closed between office buildings to listen to the city, in search of sounds for an interactive art installation. I have participated in abstract value discussions, but also in nitty-gritty detailed conversations about the color of plastic eggs or which fabric should be used for a canopy (red looks royal, but what colors did Queen Kristina actually use? And doesn't blue make people look weird when taking selfies on the throne?). I have taken on tasks I never anticipated, such as directing voice actors in sound studios and painting last-minute dioramas at the castle. Together, these experiences reflect the messy reality of carrying out design projects.

The studies extend beyond the core project teams. I have collaborated and had discussions with scenographers, technicians, neighbors, municipality representatives, graphic designers, copywriters, authors, musicians, UX-designers, programmers, builders, landscape architects, miniature makers, politicians, city gardeners, curators, and many others. I have had try-out sessions with stakeholders ranging from children to CEOs.

I have stood alongside the rest of the teams during emotional inaugurations. Together, we have experienced crises, tears, laughter, a pandemic, after-work

gatherings, project breakdowns, and epiphanies. The members involved in these projects are not just participants in something I study; they have become close collaborators and, in many cases, friends. This has been both a strength and a methodological challenge. During the final stages of the PhD, an important task has been creating analytical distance from the projects in order to better understand what actually happened. This has required shifting perspective from designer to researcher. My extensive involvement has inevitably made me biased, but it has also provided deep knowledge of the design endeavors, internalized through a first-person lived experience developed across several projects spanning multiple years.

I have brought my own design philosophies, societal and political engagements, demographics, and personal background into these projects. I have had a strong voice in shaping both the design processes and their outcomes. As years have passed, I have also been shaped by the people I have met, the contexts in which I have worked, and the experiences we have shared.

Being part of those design projects has also been a journey through different professional roles. In the PlayIT project, I was not a researcher but represented an SVS in the expert role of a playful sustainability educator. At the beginning of the project, I was what Manzini would call a diffuse designer, with limited familiarity with design thinking and similar methods. During the project, I became acquainted with these approaches, recognizing both their strengths and limitations. I was introduced to RtD and co-authored my first academic publication. I was then asked to be a research assistant in the GIFT sub-project and was tasked with co-designing sensitizing scenarios and facilitating workshops. In parallel, I founded my own design company, designing educational games and interactive experiences, primarily for SVS clients. An interest grew in understanding the mechanisms involved in designing for societal values. After PlayIT and GIFT, I participated in writing the application for the ABC project. When the project began, I represented the NGO partner and served as project lead. I later applied for and was offered the PhD position in the project, transitioning into the role of an RtD researcher. My background and my changing roles throughout these projects have shaped both the design work and the research presented in this thesis.

Chapter 5. The Studies

This chapter presents a summary of the findings from the individual studies. More details can be found in the corresponding publications. The first three studies are tied to the three main design projects, and for each of them the design process is described. The fourth study focuses on concrete workshop methods for engaging with values in design processes. The fifth study is a meta-analysis of the design projects and is presented in Chapter 7, Design for Societal Values, where insights from the projects are synthesized and analyzed in relation to the overarching thesis topic and research question, based on the theoretical framework.

Study 1: Design Study, PlayIT – Playful Biodiversity Awareness

This study centers on the design process and outcomes of the PlayIT project, with a focus on co-design with a municipality and a sustainability organization. The values in focus include empowering children's play through technology and increasing biodiversity awareness. The findings include how the playground was designed to promote biodiversity awareness through playful interactions. The study also provides insights into the design process and how values were handled throughout it. The study is presented in Articles I and II, together with supporting article *Value Driven Design for Playful Technology Enhanced Installations in Public Settings* (Back, Johansson & Wireband, 2021).

Design Process Description

In PlayIT, the core project team consisted of around 5-10 people, with some variation over the course of the project. Weekly online check-in meetings were held throughout the project, complemented by joint workshop days and other design activities. Midway into the project, Covid-19 hit, moving all project meetings online. Since the playground was outdoors, some studies could still continue as planned.

Very early on, a possible synergy emerged: the municipality was planning to build a new playground, and its timeline aligned well with that of PlayIT. It was decided that the project should try to develop a playful solution for this playground. This also meant that the municipality could add extra resources to the project, making it possible to aim for a fully implemented design.

During the first months, however, the project ran into problems. It quickly became clear that the tech company and municipality did not share the same vision for the project. The tech company wanted to use its existing solution – a gamified fitness watch for children – while the municipality did not want to promote a specific commercial product. In addition, the tech company relied on external programmers in Romania rather than having in-house developers. The municipality became concerned about issues relating to employment conditions and intellectual property. This friction ultimately led to a project breakdown and a subsequent restructuring of the consortium, which proved to be a lengthy and complicated process.

Meanwhile, the RtD researchers, sustainability organization, and municipality progressed with ideation and prototyping. The group came up with several design concepts that the municipality wished to implement. Eventually, another local tech company was brought in, although by then about half of the project period had already elapsed. The new company immersed themselves in the project, working to understand the underlying values before refining the proposed design concepts.

Two main design concepts, with the same underlying technical platform, were selected: the Magical Eggs and the Story Nest. The Story Nest later evolved into the Story Hut. The project entered an intensive period, with almost daily meetings and lots of decisions to make. Over the course of one year, the concepts were transformed into blueprints, built, and integrated into the playground environment. The designs had to accommodate numerous external requirements, including construction schedules, cable infrastructure, EU playground safety regulations, municipal data security requirements, and much more. In October 2020, the playground was inaugurated. Follow-up studies and iterative tweaks to the digital layers continued throughout the rest of the project, which concluded in the summer of 2021. As of 2026, the playground remains open and fully operational.

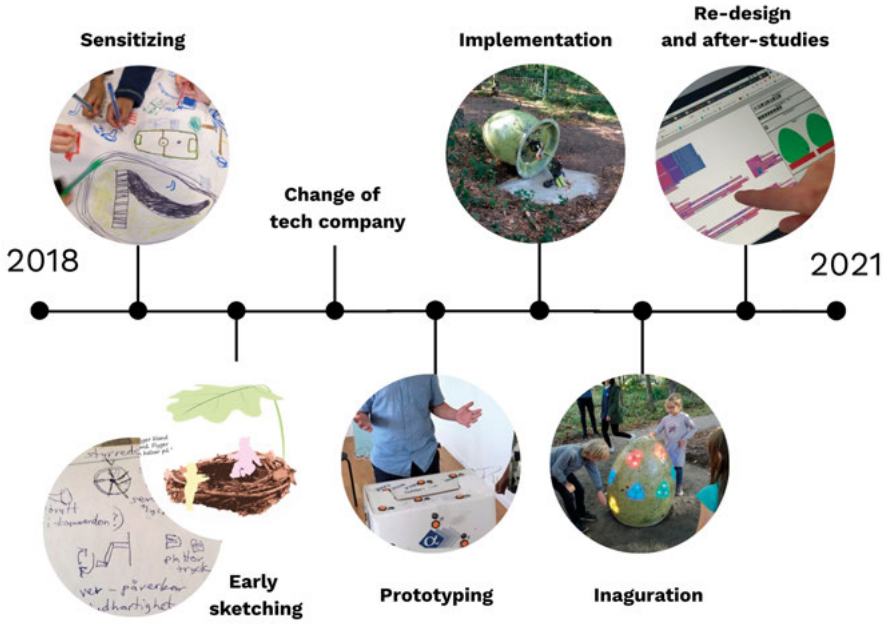


Figure 5. Timeline for the PlayIT Project.

Article I: *Daddy, You Can Be the Fox: Designing Hybrid Playgrounds for Nature Integrated Storytelling*

Johansson and Back, NordiCHI 2024. Published 2024.

The main article from Study 1 focuses on the IoT playground installation, the Story Hut. It examines a specific value –biodiversity awareness – and how this value was incorporated into the design. The article presents an annotated portfolio analysis of value-oriented design features and includes in-the-wild observations spanning six years. The results show that the installation encouraged extended and recurring use, and that the biodiversity-themed story content engaged children, who incorporated these themes into their play and reflected on biodiversity in their conversations. The article concludes with design implications for encouraging biodiversity awareness through nature-integrated storytelling in playground design.

Article II: *Let's Play Something New! – Designing for Digital Malleability in Outdoor Playgrounds*

Back, J., Johansson, K., and Hangvar, J. Mindtrek 2023.

This article examines how digital layers were utilized to support malleability and ongoing redesign of the playground. It relates digital malleability to increased play value, democratic play, and the municipality's ability to renew the playground for new target groups, educational purposes, co-creation activities, and new forms of meaningful play for children and families. The article adopts an RtD approach.

Study 2: Design Study, the ABC Project – Designing Interventions for Community Feeling

Study 2 followed the ABC project. In the project design, probes and prototypes were developed to promote meaningful encounters in public spaces and strengthen the sense of community. The project included several design strands, such as the development of an interactive co-creative art installation, a recurring sustainability festival, and both digital and physical platforms for citizen dialogue. A new evaluation method was also developed and applied in the project in the form of a psychological observation protocol for studying engagement at social gatherings. The project's insights were brought together in a *Handbook for Meaningful Meetings* (Sankt Kors, 2025), aimed at professionals working with place activation. One of the design strands – the interactive art installation – has been particularly important to this thesis. The concept progressed beyond the prototype stage into implementation, and is scheduled for installation in 2026. The social innovation project officially ended in October 2025, followed by post-project studies. Key findings include how societal value considerations shaped the entire design process, from planning to implementation. The study also demonstrates how process activities can be designed and structured to promote value conversations. Finally, it highlights how societal values become embedded in both prototypes and the final design of the interactive art installation.

Design Process Description

The core project team consisted of seven members. The university was represented by two HCI researchers (myself and my principal supervisor) and one researcher in psychology. The sustainability NGO was represented by its

director, who also served as project leader. The public real estate company was represented by a community manager, a sustainability manager, and a communicator. The NGO experienced some staff turnover during the project, so several different individuals participated as its second representative at different stages. Most of the core group had helped to write the project application. Other than the core team, around ten additional people contributed to varying degrees.

The project included weekly online Monday meetings. Different working groups also met around specific design strands and activities, while each organization had internal meetings related to the project and reported back during the weekly meetings. Approximately two or three on-site project days were organized each year, combining workshops with joint lunches, dinners, and other social activities. The project maintained a shared online workspace for documents, meeting notes, and photographs.

The project kicked off with environmental scanning and initial workshops to better understand the project topic. A reference group was put together, consisting of representatives from different stakeholder organizations within the city district. Early in the project, the team decided to explore several parallel design strands rather than concentrating on a single comprehensive solution.

Some recurring friction emerged between the NGO and the public real estate company. They described differing views on how social change should be created – whether through bottom-up or top-down approaches. The NGO wanted to work directly with residents and local communities. The real estate company shared this ambition but was more concerned with internal branding and anchoring activities, and was cautious about making promises to the community that it might later be unable to fulfill.

One design strand focused on arranging co-creative sustainability festivals. This idea was brought up early in the project as one of the initial design concepts. About six months into the project, a local climate NGO (not the project partner NGO) approached the real estate company with a proposal to organize a sustainability festival. The idea was brought up at a project meeting and incorporated into the project. This resulted in a recurring co-created sustainability festival involving around 500 visitors and 30 collaborating organizations.

Another design strand, driven by the project NGO, focused on implementing "offers and needs" markets that enabled local actors to share circular resources. These activities combined physical meetings with an app for collecting ideas, which in turn resulted in initiatives such as local art walks.

During the second year of the project, a master's thesis was conducted under my supervision, focusing on creating concepts for an interactive installation. These concepts were developed into prototypes, and the student was then

employed as a research assistant for six months in the project. The real estate company decided to implement the concept. An art collective was brought in to further develop the design proposals into a co-creative public artwork. The implementation was jointly funded by the project and the real estate company. The initial plan was to inaugurate the installation in early fall 2025. However, delays related to groundwork and underground infrastructure postponed the installation until later in 2026.

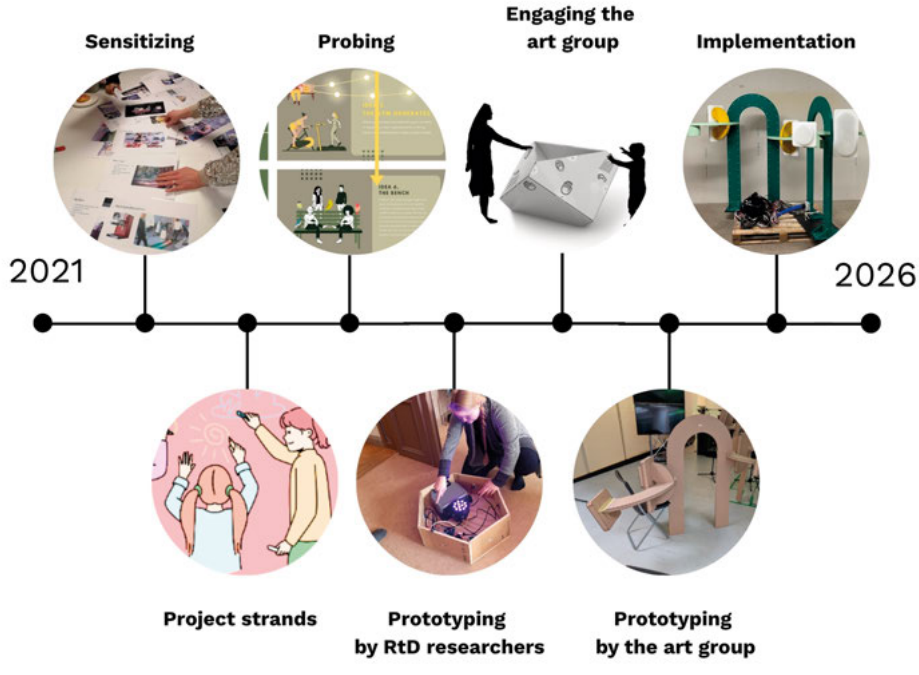


Figure 6. Timeline for the ABC Project.

Article III. *Valuable Design Conversations – Value Perspectives on Co-Designing a Public Interactive Artwork with Mission-Driven Stakeholders*

Johansson, Lindström. Manuscript.

This article examines value conversations with the societal value stakeholders throughout the ABC project, from the application process through to the design process. Particular attention is given to the development of the interactive art installation. The article traces how these value conversations influenced the final design. The findings show that creative and playful methods supported rich discussions about values, and that the SVS demonstrated

sophisticated reasoning about societal values when designers created opportunities for such discussions.

Study 3: Design Study, Castle Museum – Co-Designing Interactive Museum Installations

This study focuses on the co-design of interactive museum installations together with museum professionals. It builds on two design projects at the Swedish Museum of Technology, as well as the design process of creating a new museum exhibition at a castle. The study focuses on values surrounding cultural heritage awareness.

Design Process Description

In 2023 we successfully applied for funding for a one-year collaboration project with a municipal museum development team. The project centered on developing a new local history museum at a castle, where several rooms were empty. The collaboration explored how museum development could be combined with design research. Since few decisions about the new museum had been made when the collaboration began, we got involved throughout the entire design process.

The collaboration included two HCI researchers (myself and my principal supervisor) together with the museum development team, consisting of a digital producer, a project lead, and a museum educator. We had the most contact with the digital producer. The team met in biweekly check-in meetings, and we jointly planned and participated in design workshops and explorations. We also conducted a study visit to the Munch Museum in Norway, with which we had previously collaborated during the GIFT sub-project. In addition, we visited Tartu both for inspiration and to present the project.

After the official collaboration project ended, we continued contributing to the development of the new museum. The museum development team decided to fund the implementation of two design concepts that had been co-designed during the project.

One of the design concepts was a Quest Room, inspired by educational escape rooms but adapted to the specific conditions of museum environments. This concept built upon a previous project our research group had carried out at another museum. The second concept was an augmented throne that allowed visitors, when seated, to hear quotations attributed to the 17th-century ruler Queen Kristina. Beyond these two designs, we also came with input for additional technology-enhanced exhibition installations and broader design

perspectives on incorporating interactivity and digital technology into museum exhibitions.

During this period, I reached out to other researchers working at the intersection of HCI and museum exhibition design. For NordiCHI 2024, I co-authored a workshop paper and hosted the workshop *Interactive Storytelling for Communities, Heritage and Public Engagement*, which is included as a supporting article in this thesis. The workshop explored how interactive storytelling can be used to foster engagement with public values. Among other activities, participants experienced and discussed the Castle Museum installations, including the Quest Room and augmented throne.

The museum was inaugurated in May 2025, and both installations have been open to the public since then. Follow-up studies of the installation were conducted throughout 2025 and during spring 2026.

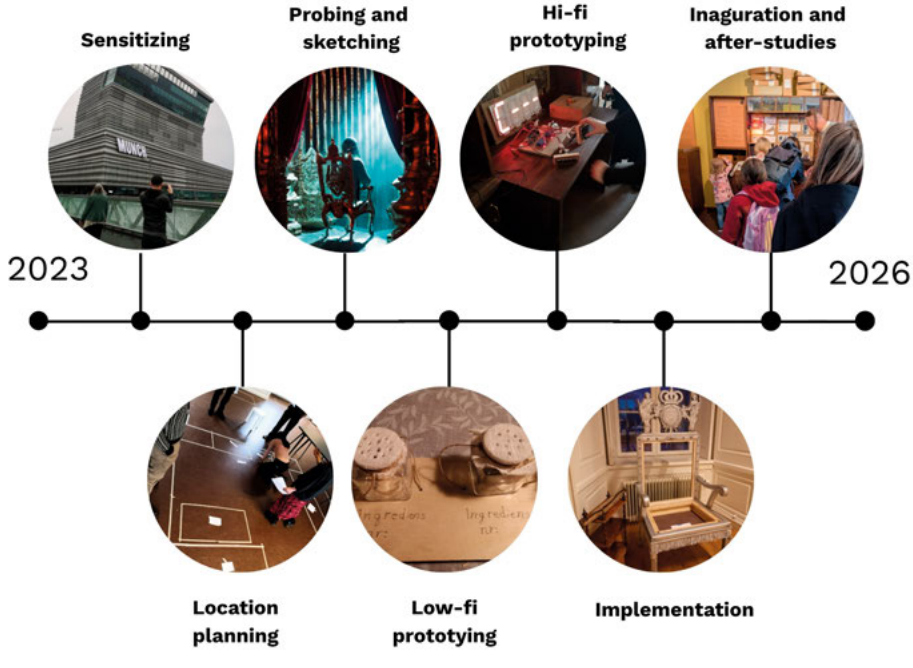


Figure 7. Timeline for the Castle Museum Project.

Article V: *The Quest for Learning – Pedagogical and Interaction Design Perspectives on Quest Rooms*

Johansson, Bandelin, and Back. ECGBL 2025.

This article builds on the Castle Museum project as well as a similar design project at the Swedish Museum of Technology. It presents the quest room concept and its development across design iterations. The perspectives of museum professionals are combined with user studies to examine the design. The article highlights, through annotated portfolios, how pedagogical and interaction design concepts were incorporated into the installation to promote cultural heritage learning. The study proposes Quest Rooms as a format for playful interactive installations, inspired by educational escape rooms but adapted to museum contexts in order to encourage engagement with cultural heritage. It also presents an edu-interaction design framework that combines five aspects: playification, embodiment, sociality, game mechanics, and interactive storytelling. The framework can be used both for analyzing and designing cultural heritage experiences in museums.

Study 4: Design Process Methods for Societal Value Elicitation

This study focuses on concrete design process tools and methods for value elicitation, emphasizing perspective-taking and reflection in relation to complex societal values.

The study explores approaches that extend beyond design thinking and traditional design process activities. It consists of two articles. The first examines how live-action roleplay (larp) can be used in design processes and is also tied to the supporting GIFT project. The second article investigates how design thinking, futures thinking, and systems thinking can be combined to foreground sustainability in design processes. Unlike the first article, the second is not tied to any of the specific design projects, but was developed together with collaborators from outside HCI to explore creative approaches to facilitating design workshops.

The first article examines larp in a broader sense as a design method and explores how it can promote perspective-taking and reflection in a variety of settings. It discusses the mechanisms that make larps effective for exploring values and design solutions that extend beyond technical considerations and individual users. One of the design exemplars is the Sensitizing Scenarios developed as one of the design strands within the GIFT sub-project. The supporting publication (Waern et al., 2020) describes these scenarios in more

detail. The scenarios were run in two museum innovation projects – one in Serbia and one in Jönköping, Sweden – and in both cases they were found to stimulate value conversations and enable SVS to voice concerns and perspectives. Larp as a design method is further discussed in the supporting publication *Larping (Live Action Role Playing) as an Embodied Design Research Method* (Márquez Segura et al., 2019).

Article V: *Why Larp? A Synthesis Article on Live Action Roleplay in Relation to HCI Research and Practice*

Johansson, Robinson, Back, Lynne Bowman, Fey, Marquez Segura, Waern, Isbister. TOCHI.

This article provides an overview of live-action roleplay (larp) and its application within HCI research and practice. It is a synthesis combining a practitioner perspective, a scoping literature review, and design exemplars from RtD projects. One of the design exemplars presented is The Object, one of the sensitizing scenarios developed within the GIFT sub-project. The scenario was run as a design activity for museum innovation teams to sensitize participants to relevant theories, stakeholder perspectives, and societal values. Topics explored included cultural heritage storytelling, cultural appropriation, provenance, critical perspectives, and new museology. The scenario also encouraged meta-reflections on museum innovation teams, creating an alibi for discussing attitudes, roles, and collaboration within project teams. Techniques such as role switching and time mumps were incorporated to encourage perspective-taking.

The article shows that the larp helped museum professionals articulate challenges and values while lowering social barriers within the design process. Participants described the scenarios and roles as realistic and reported that the larp helped them understand the perspectives and arguments of others. One participant even described the experience as therapeutic. The study suggests that larp can be an effective design method for professionals with no prior experience of roleplay and can support value reflection in museum design teams.

The findings demonstrate how larps function as a playful means of enabling value conversations. The method was found to lower social barriers, create an atmosphere conducive to value discussions, support participants in voicing their value-related concerns, and foster perspective-taking. It also strengthened group cohesion and functioned as a stepping stone for subsequent design activities. The article also identified several drawbacks: larps require skilled facilitation, are time-consuming, and sessions can be difficult to document

systematically. Overall, the study suggests that larps combine emotional engagement, playful imagination, perspective-taking, embodiment, and opportunities for futuring.

Article VI: *Integrating Systems, Design, and Futuristic Thinking for Sustainable Complex Systems Design*

Rouchitsas, Johansson, and Gürdür Broo. ICIET 2025.

The article presents a workshop concept for promoting value reflexivity in design processes by combining design thinking, futures thinking, and systems thinking. The approach aims to foster innovative, sustainable, and socially just design solutions by addressing complex systems in which societal, economic, environmental, and technological systems interact. The workshop combined several complementary methods. The workshop was run with participants, PhD students, who engaged in hands-on mapping using structured worksheets, AI-supported image generation, and in-depth discussions. The workshop was designed to deepen participants' understanding of complex systems, value interdependencies, reinforcement loops, unintended consequences, and the effects of design decisions on marginalized communities.

The findings indicate that the workshop promoted empathy and insights, while helping participants to understand how a co-creative approach can support the design of complex systems that benefit society broadly, beyond merely satisfying traditional design requirements. The workshop also encouraged participants to question dominant paradigms of economic growth and techno-centric approaches to sustainability.

The article proposes that value-oriented design workshops should promote anticipatory design – considering long-term, complex consequences and discussing them during the design process. It further advocates stakeholder ecosystem mapping to move beyond traditional stakeholder analyses, and co-practicing futures literacy to collectively explore multiple scenarios and envision transformative design outcomes. Finally, the article argues that interaction design processes should extend beyond immediate user interaction, usability, and user satisfaction to also encompass interactions with communities, ecosystems, and society. Sustainability is therefore positions as a central design objective, not an add-on consideration, and design proposals should be evaluated not only in terms of technical feasibility but also according to how well they fulfill broader societal values, such as the UN Sustainable Development Goals.

Study 5: Retrospective Cross-Project Insights on Designing for Societal Values

This study is a retrospective analysis of the design collaborations with societal value stakeholders (SVS). It draws on insights across the design projects and reframes the project documentation from the perspective of how societal values were approached in relation to the SVS perspective. Two retrospective project team interviews were held as part of the study, one in the ABC project and one in the Castle Museum project, reflecting on the design processes and outcomes with a particular focus on societal values. Study 5 is presented in the Analysis and Discussion chapter.

The study synthesizes insights from the previous studies. Design activities across the projects were narratively mapped to the dimensions of the theoretical framework. The study included a retrospective revisiting of design process documentation, as well as follow-up interviews with SVS representatives from the three design projects. The interviews focused on retrospective value reflections on societal values in relation to the design processes and outcomes. The Analysis chapter includes quotations from these follow-up interviews.

The follow-up interviews were semi-structured. Questions addressed participants' reflections on the projects in hindsight, which values they believed that the projects had or had not captured as intended, their most important takeaways, their thoughts on the design process – including what worked well and what was challenging – and their reflections on the designs that were developed. For the ABC project, the interview was conducted as an on-site group interview. For the Castle Museum project, it was conducted as an online group interview. For the PlayIT project, two individual online interviews were conducted: one with the municipal city gardener and one with the social enterprise CEO. The interviews were conducted to complement the extensive RtD sources and to provide retrospective reflections from the SVS representatives in each project. They were recorded, transcribed, and translated. The interview material was then analyzed narratively through the lens of the theoretical framework, and quotations were selected to illustrate the SVS' reflections on the design activities and value dimensions across the projects.

Chapter 6. Analysis and Discussion – Design for Societal Values

This thesis builds on several years of active participation in different co-design projects together with societal value stakeholders (SVS). This chapter presents a retrospective, reflective analysis of the projects, their processes, outcomes, failures, and successes, in relation to how design can support societal value mediation when co-designing cultural interventions with SVS.

The analysis and discussion sections are structured according to the three dimensions presented in the theoretical framework: Value Elicitation, Value Translation, and Value Manifestation. Each dimension is introduced with a vignette that describes a situation from one of the projects. The dimensions are then discussed, drawing upon examples and insights from across the projects. Findings are exemplified through quotations, descriptions of design activities and situations, and are related to literature. Before discussing each dimension in detail, some overarching insights from the project processes are presented.

The project insights draw on both successes and failures, and there have een plenty of both. Together, all have contributed important research insights. There have been small-scale magical moments and disastrous sessions, as well as project-wide breakdowns and breakthroughs. Over time, however, project approaches and methods emerged that proved generative for creating design collaborations and design solutions that adhered to the societal value aims of the SVS.

Societal Value Elicitation

In November 2021, we were thrilled to find out that we had received funding for the ABC project. In the application, written under considerable time pressure, we had stated that the project should aim to mitigate loneliness. During the very first project meetings, we began discussing what we really meant by loneliness. Was loneliness a feeling, or simply the absence of social contacts? Could you be lonely among friends? Could admitting loneliness make you feel more lonely? Would a quick smile from a stranger mitigate loneliness? This

marked the start of a value conversation that lasted throughout the project. As the project proceeded, we encountered new perspectives from one another: the community manager, the interaction designers, the NGO representatives, the psychology researcher. We explored the values via Post-it sessions, workshops, presentations, and weekly meetings. Every time a new person joined the project, this process partly began again. As concrete design ideas developed – a sustainability festival, a platform for community sharing services, and a co-creative interactive artwork – each design strand brought up new perspectives on loneliness. Four years later, at the project closing meeting, the discussion of loneliness was still ongoing. We had reached neither consensus nor conclusion, but a richer and more nuanced understanding of the concept.

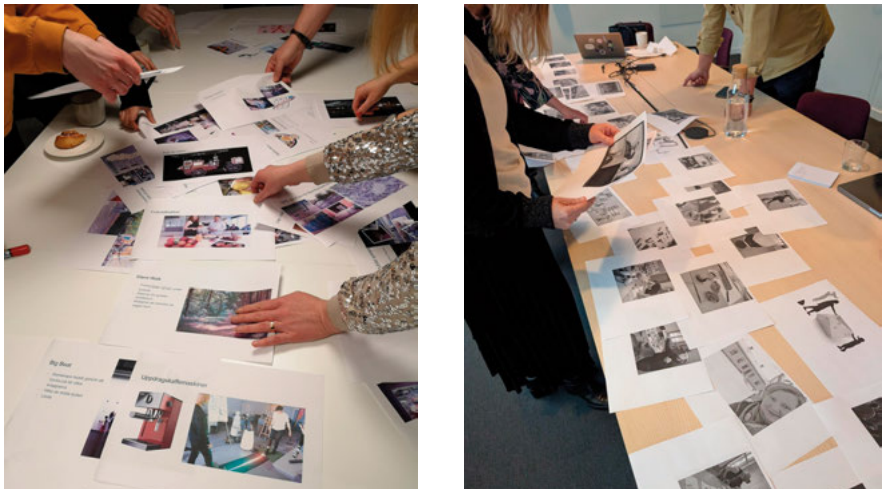


Figure 8. Left: From the first workshop in the ABC project, where we are using cards to find inspiration for what to do in the project. Right: From the last workshop, where we engage with retrospective reflection by looking back at photographs from the four-year project. At both sessions, and between the two occasions, we discussed loneliness in many different ways.

The discussions around loneliness described in the vignette provide a glimpse of how trying to understand just a single societal value – mitigating loneliness – can span several years. Despite this effort, it still felt like we had only scratched the surface of the value's complexity. Multiply this by all the other societal values that require similar consideration, and it becomes apparent how complex and challenging societal value elicitation can be. These observations align with previous literature describing how values are interconnected (Friedman and Hendry, 2019), how societal values often possess soft and intrinsic

qualities (Waeraas, 2018), and how values can be misinterpreted in design processes (Pommeranz et al., 2012). The way discussions of mitigating loneliness continued throughout the entire of the ABC project suggests that societal value elicitation is never truly complete but an ongoing and iterative process.

Many different societal values were discussed throughout the projects. For example, in PlayIT recurring conversations centered around democratic play and biodiversity awareness, as described in Articles I and II. In the Castle Museum Project, we engaged in conversations on cultural heritage experiences and authenticity, as described in Article V. What is apparent from the studies is that new angles and relationships between values continued to emerge throughout the projects.

The studies indicate that value elicitation is crucial when co-designing with SVS. Conversely, neglecting this aspect can prove detrimental. One example comes from PlayIT, which experienced struggles early in the process. The original consortium included a tech partner whose understanding of the project values differed substantially from those of the municipality and the other SVS. These differences ultimately led the municipality to discontinue the collaboration. One source of disagreement was that the tech company wanted to base the playground on a proprietary wearable device for children. The municipality rejected this proposal because requiring families to buy a specific product conflicted with the value of democratic play. As the municipal city gardener reflected in the follow-up interview: "We cannot promote someone's [product] for them to sell stuff. It is partly those things that are sensitive for a municipality – that simply does not work." These disagreements led to a complex process of replacing the technology partner, delaying the project by almost a year. While problematic, uncovering the value differences early made it possible to recruit a new tech partner, sensitize them to the project's values, and ultimately develop successful and functional design solutions. The rarity of finding a tech and design partner who genuinely understood those values was emphasized by the social enterprise CEO: "To find a tech partner who said they wanted to work with sustainability [...] that was a bit like finding a gold nugget."

The project breakdown of the interactive museum game developed as part of GIFT, illustrates another consequence of insufficient value elicitation. It was not until the project team got to engage in a value elicitation activity – in the form of a sensitizing scenario using live-action roleplay – that the museum professionals openly expressed their concerns about the proposed design. Had greater emphasis been placed on value elicitation in earlier stages, the breakdown might have been avoided, or at least recognized sooner.

Sometimes promising design ideas need to be abandoned because they conflict with societal value considerations. Extensive value elicitation during the early phases of a project makes it possible to identify which ideas should be pursued and which should be discarded. One example comes from the Castle project. An early concept centered on an intriguing cultural heritage object: a dictionary of a Romani dialect written by the 19th century prisoner Dal-Pelle while awaiting execution. His life story provided rich opportunities for interactive storytelling and interesting puzzle-based experiences, and the concept initially appeared highly promising. But rather than immediately pursuing the idea, the project team continued the elicitation. The first positive response was then followed by concerns. The Romani connection called for cultural sensitivity, while the risk of encouraging dark tourism by turning the lives of marginalized people into engaging visitor experiences was found to be problematic. As these perspectives accumulated, the project team decided to abandon the design concept.

Eliciting Explicit Values

An important insight from the studies in this thesis is that value elicitation is required even for explicit values, such as project values and pre-formulated organizational values. Such values are, as described previously in the thesis, often skimmed over in current approaches to incorporating values into design (Friedman and Hendry, 2019; Flanagan et al., 2008). The value described in the vignette – *mitigating loneliness* – was clearly stated in the project application and could be classified as what Friedman and Hendry (2019) describe as a project value. This value did not need to be discovered, as proposed by Flanagan et al.; it had already been established in the project application. Nor did it need to be explicated in the way VSD seeks to explicate values; it had already been documented and even agreed upon by the consortium. However, the value did indeed need to be discussed, as evidenced by the extensive discussions on loneliness that spanned several years. The example of loneliness highlights the complexity of interpreting values in collaborative projects, a challenge well documented in previous literature, such as by Khan and Matthews (2019), and mirrors how societal value elicitation unfolded in the other design projects as well.

Value explication, as it is approached in VSD (Friedman, 1996), and value discovery, as proposed by Flanagan et al. (2008), are typically concerned with uncovering values that have not yet been articulated – that is, implicit values. These approaches place less emphasis on values that stakeholders have already agreed upon as important. As described in the background chapter, Friedman and Hendry (2019) explicitly state that engaging with project values

– values specified in project applications and predefined documents – is outside the primary scope of VSD. A core insight from my research is that explicit values, such as project values, should not be skimmed over. Rather, the studies show that developing a nuanced understanding of agreed-upon values requires substantial effort, collaboration, and resources in a social innovation project, and that ongoing value elicitation throughout design projects focused on societal values is essential to their success.

Specific Properties of Value Elicitation with SVS

The importance of engaging stakeholders in value-oriented design endeavors is well established in the literature (Friedman and Hendry, 2019; Bødker et al., 2022; Ehn, 1992). However, as described in the background chapter, relatively little attention has been paid to how different types of stakeholders should be engaged. This thesis demonstrates the need to approach value-related sessions with SVS differently than sessions with other stakeholders, such as users. The distinctive characteristics of SVS are described in the background chapter, including their professional orientation towards values and their situated expertise in societal values. Throughout the design projects, it became clear that this called for a different approach to value elicitation than when engaging other stakeholder groups, such as users, in value-related conversations and activities.

Across the projects, value elicitation activities were carried out with SVS as well as with a variety of other stakeholders, including end users, and local representatives. The studies found that different stakeholder groups required different approaches to value elicitation. For example, in PlayIT we held workshops in which children painted their dream playgrounds, participated in try-and-talk sessions, and tested prototypes. Similarly, in the ABC project we had a reference group consisting of people who lived and worked in the city district, engaging them through workshops, group interviews, and feedback sessions on design proposals. In these activities, much of the effort focused on sparking value conversations and interpreting everyday reflections into broader understandings of values. This aligns with how Friedman and Hendry (2019) exemplify VSD methods, for example through work with homeless youth, where reflections on their lives became a basis for discovering values. For these stakeholder groups, expertise was grounded in lived experience rather than in the professional articulation of values.





Figure 9. The projects involved a wide variety of stakeholders. Left: Children drawing their dream playgrounds. Middle: From a reference group meeting. Right: A project session with the SVS.

While the value expertise of users and focus groups is grounded in lived experience, SVS approaches values differently. They engage in value reflection as part of their professional capacity and are, as discussed in the background chapter, experts in pursuing the situated societal values of their organizations. This was consistently confirmed by the studies in this thesis. The professional orientation of SVS towards values can be exemplified by this survey response from the community manager in the ABC project*: "For me, value words (värdeord) are something that is at the core of all my work."* This required a different role for us as designers. Rather than scaffolding end users to reflect on values, we instead sought to keep pace with the SVS' sophisticated moral reasoning and nuanced value discussions.

Across the studies, SVS repeatedly stressed the need to unpack value concepts in greater depth and resist shallow interpretations. Article III shows that, throughout the entire project, the SVS never once complained that value discussions got too long, too complex, or unnecessary. On the contrary, they wanted to deep dive even more into value conversations. The same pattern emerged across all of the other projects. One conclusion is therefore that SVS want to engage in sustained discussions of societal values during design projects. Furthermore, they possess sophisticated understanding of societal values that require considerable time and resources for designers to interpret and understand through value elicitation.

Creativity and Playfulness when Eliciting Values with SVS

An important finding regarding value elicitation activities concerns the importance of respecting the professionalism of SVS and providing clear framing and motivation for why creative exercises are conducted. An example comes from the GIFT sub-project and the development of the sensitizing scenarios, described in Article V and in the supporting article about the

sensitizing scenarios (Waern et al., 2020). The first iterations of the scenarios met some pushback from museum professionals, who found it difficult to relate the roleplay to their practice and therefore felt that the scenarios wasted too much work time. The scenarios were then tweaked to provide clearer framing and to align more closely with the project's themes. As a result, the next iteration was much better received by the SVS.

The studies also showed that even though project partners agreed on the importance of certain societal values at an overarching level, they had very different perspectives on what those values meant and how they should be pursued. As an example, in the ABC project, the psychologist, community manager, and NGO each brought their own professional background, terminology, and understanding of loneliness. The NGO lead explained how the value elicitation activities, and particularly the playfulness exercises during one of the seminal workshops, helped the group better understand the different value perspectives represented within the project:

> Making our different roles and functions and our different worlds visible I think has been good, especially when I got to be part of [the workshop days]. Because then, for the first time, I got into the project and kind of understood our differences and similarities in the project. That it became so clear like this; Aha! Do you think like that? And so, it became like that the playfulness became somehow cathartic in that.

The sustainability manager from the public real estate company agreed to this, adding:

> Yes! And it has been a huge deal! Because you think you have the same goal. […] it becomes this number 6 and number 9, depending on which way you look at it!



Figure 10. A metaphor presented by a societal value stakeholder representative explaining conversations about different roles, goals, and values in the project.

The quotes highlight that the SVS recognized the diversity of value perspectives within the project group and experienced the playful value elicitation activities as helping them better understand one another's perspectives, even to the point of describing the experience as cathartic. The value conversation activities conducted in the ABC project are further discussed in Article III.

One approach to value elicitation and perspective-taking explored in the GIFT project was live-action role-playing. The sensitizing scenarios developed in the sub-project provided participants with an "alibi" for expressing viewpoints and fostered an open atmosphere where differing opinions could be voiced (Waern et al., 2020). One museum representative even described the scenario session as therapeutic, as it helped surface underlying value tensions within one of the design strands of GIFT (Rajkowska, 2022). While a promising approach, designing and facilitating roleplays required substantial preparation and specialized skills from the facilitator. Consequently, these longer roleplay scenarios were not used in the three subsequent projects. Still, their potential makes it worth experimenting more with formats that can draw on live-action roleplay for value elicitation purposes in social innovation projects.

Lowering prestige also emerged as an effective way to encourage more open and honest value conversations. This could be achieved by removing the pressure to provide "good" design ideas. For example, as described in Article III, participants in the ABC project were asked participants to come up with not only feasible ideas, but also deliberately catastrophic and utopian ones. We were surprised by both the creativity this sparked among the participants and the rich value discussions that arose, particularly around the catastrophic ideas. The sustainability manager reflected that it was "brilliant! Like just, let's take away all the prestige. This doesn't matter; saving face or anything. Or showing off or walking the tightrope. But now you're just going to try to think a little outside your box. Which we all probably need to practice quite a bit." This reflection highlights several aspects of the exercise. It was appreciated, it demonstrated the importance of lowering prestige, it introduced an approach that felt novel to them, and it helped them consider perspectives they might not have otherwise explored. The situation is analyzed further in Article III. Similar observations were made across the other design projects, and over time, lowering prestige became an active strategy for encouraging meaningful value conversations. This finding also aligns with previous literature, where Manzini (2015) identifies lowering prestige and fostering an open discussion climate as important to social innovation.

The use of playful approaches to facilitate value conversations is well established in previous literature (Gaver et al., 1999; Candy et al., 2018; Ehn & Sjögren, 1991). Playful approaches are common in co-design with end users (Junginger, 2017), but introducing them into the highly professional setting of SVS presents different challenges. Across the projects, we found that with careful framing, playfulness effectively supports value elicitation with SVS.

Value Elicitation Pace

Throughout the projects, it became apparent that designers and SVS often worked at different paces of reflection. Design activities tend to be fast-paced, and it became clear that the SVS at times struggled to keep up and make their voices heard within that tempo. In the ABC project, we used multiple strategies to mitigate this, including designing preparatory exercises that allowed participants to reflect before workshops. One SVS commented on this exercise: "The structure with a preparation assignment and then brainstorming together and discussing freely - top grades!" Another participant emphasized the value of the preparatory activity, explaining "Sometimes it's hard to make yourself heard because I'm slow before I know what I think." These comments suggest that creating additional space for reflection was both needed and appreciated. The benefits of slowing down design processes and allowing more room for reflection align with previous literature (Stolterman & Löwgren, 2007; Sengers et al., 2005). Creating this extra space for reflection becomes especially relevant when working with SVS, who are typically used to a more slow-paced, careful value reflection (Wæraas, 2018; Kavathatzopoulos, 2008).

The individual articles provide additional concrete examples of design activities and exercises that can help promote value elicitation and conversations around values. Article III presents several examples of how value conversations were carried out in the ABC project, along with insights into the approaches and methods that supported them. Article V explains the use of roleplay as a method for value elicitation, while Article VI provides a set of workshop exercises for engaging stakeholders with values in design processes.

Value Elicitation as a Goal and a Means

In the context of social innovation, value elicitation can be understood as more than a means to achieving successful design solutions; it can also be considered a goal in itself. This notion is expanded on in Article III. It is well established that, in social innovation, the value conversations themselves might be just as important as the design output (Manzini, 2015). As the community manager in the ABC project said in the final group reflection, "I would have been happy and content even if we didn't get a sculpture, like with this project. And I feel like that now." As value-mediating designers, we can design activities for value elicitation in order to achieve design solutions, while also recognizing value elicitation as an important outcome in its own right.

Allowing space for value elicitation throughout the projects yielded valuable design input, brought tensions to the surface, and contributed to the

development of design solutions that resonated with both the SVS and the societal values they sought to pursue. But if we want the conversations to be turned into design output, value elicitation must be paired with value translation.

Societal Value Translation

It is spring 2019. After two years of design work in one of the design strands of the GIFT project, the local design company has developed and tested a hi-fi prototype for a mobile game designed for a museum exhibition surrounding the grave chamber of former dictator Tito. Unarticulated tensions have been building within the project. A colleague and I are on site in Serbia hosting a design workshop. The day begins politely, but with an undercurrent of apprehension. A timeline exercise reveals that the museum representatives and designers have barely spoken to each other for months. We then run one of our sensitizing scenarios. The participants become immersed in the creativity, and the room fills with laughter, energy, and vivid role-playing. During the discussions afterward, however, the museum representatives suddenly state that they do not like the proposed design at all. They argue forcefully that their museum cannot be turned into a banal, meaningless, gamified playground. The critique caught the local design company representative by surprise. He could not understand how the museum could dislike a design idea that users apparently liked, as demonstrated by the user tests. The museum representatives, in turn, could not understand how he could fail to recognize that cultural heritage values mattered more than user preferences. After the workshop, my colleague remarked that this was the first time the museum had felt comfortable expressing what they really thought about the design. The intended values, from the museum's perspective, had not been translated into the proposed design. Eventually, the museum discontinued the design endeavor.

The workshop described in the vignette was, in many ways, a starting point for the focus of this thesis. Something was clearly not working in the design process. This echoed a recurring problem I had encountered in other design projects. The proposed gamified solution was deemed banal, despite user tests showing that it increased user engagement (Waern et al., 2020; Rajkowska, 2022). It became evident that the SVS and the designers held fundamentally different views regarding how to translate societal values into design solutions. The designers emphasized designerly qualities such as interaction, usability, and functionality as indicators of meaningful design. The SVS, on the other hand, while also wanting to create engaging experiences, emphasized

meaningfulness in terms of values such as authenticity, narrative, and cultural heritage complexity.

The museum staff in the GIFT sub-project challenged the prioritization of user experience over other values:

> "You have to do really thorough research, and when you do that, then you can come to this part of how you are going to engage the visitors. [...] what's the goal, what's the purpose, where are you going to end up with visitors? What is the point for the museum to do that?"

This kind of pushback seemed confusing for the designers, particularly because the user tests were positive. It also illustrates that museums approach exhibition development very differently from interaction designers. Museum professionals can spend years developing narratives, looking into historical sources, and exploring alternative interpretations before considering how these should be conveyed to visitors. On a deeper level, this episode highlights that museums prioritize their cultural heritage mission above all else. The museum staff also expressed concerns about the design process itself:

> "If the museum is to implement something on a regular basis, they need to be much more involved in the development and curation of the content than they've been now. For the next prototype – somebody from the museum needs to be directly involved in the next creative process, attend the meetings when the ideas are developed, because this form of collaboration we had till now hasn't resulted in completely meaningful product."

Failing to develop strategies for translating the SVS' values into design proposals can, as the example shows, jeopardize an entire design endeavor, particularly because SVS often function as value gatekeepers (Krippendorff, 2014), as described in the background chapter. For designers, this requires developing one's so-called second-order understanding of the SVS' own perception of what is meaningful in relation to the values they seek to pursue.

Finding ways to gather feedback on design suggestions in relation to societal values is not an easy task. Previous research has shown that non-designers may hesitate to challenge design proposals (Manzini, 2015) and can become increasingly alienated as design processes become more technical (Sengers et al., 2005). In the example from the GIFT sub-project, the museum representatives were clearly uneasy with the suggested designs but, for many months, did not openly voice their concerns. The playful "alibi" provided by the sensitizing scenarios created space for critique and seemed to give the SVS confidence to challenge the design proposals (Waern et al., 2020), thus becoming more active participants in the value translation process. The sensitizing scenario method is described in Article V. The scenarios can be understood as a

tool for supporting both value elicitation and value translation. Our findings suggest that enabling SVS to voice concerns – and being prepared to adapt designs in response – should be a top priority when translating societal values into design.

The findings of this thesis also challenge the oversimplified way value translation is conceptualized in Design for Values (DfV), where the emphasis is on operationalizing design requirements (van den Hoven et al., 2015). DfV does not take into consideration the ongoing reinterpretation of values throughout a project or the relational and communicative processes through which value concerns are expressed. Across the projects, there was never a single point at which the interpretation of the societal values became static and ready to be translated into stable design requirements. Instead, interpretations continued to evolve throughout the process.

The project examples in this thesis therefore point towards complementary approaches to value translation. In the PlayIT project, we used strategies aligning with the DfV approach by developing value hierarchies and translating them into effect goals and design requirements. However, this alone proved insufficient. As described in the supporting article *Value Driven Design for Playful Technology Enhanced Installations in Public Settings* (Back, Johansson & Wireband, 2021), "The tool alone is not enough to ensure that the values are integrated in the design. The tool was complemented with a board of experts partaking in all major design choices, meeting at least once per week. A representative from the municipality, a play researcher, a sustainability educator and tech experts all gave their input on choices, to ensure that values were sustained throughout the design process" (p. 3). This approach to societal value translation aligns with Manzini's (2015) argument that social innovation projects require close cooperation among stakeholders. In practice, this meant that stakeholders revisited design requirements and discussed them in relation to the societal values on a weekly basis. This strategy proved fruitful and was incorporated into both the ABC project and the Castle Museum project.

Misunderstandings and communication breakdowns repeatedly emerged as obstacles to translating values into designs. In the ABC project, when tensions arose between the NGO and representatives of the public real estate company, we sought to avoid communication failures we had experienced in the GIFT sub-project. We therefore promoted sustained communication through informal conversations and relationship building, even during times of tension. As discussed in Article III, the SVS themselves highlighted relationship building as a key success factor, emphasizing that it fostered trust and strengthened collaboration. This finding aligns with previous research showing that group cohesion and a positive collaboration climate support social innovation aligns

(Howalt et al., 2019; Tjaha & Yee, 2024). These close relationships and informal communication channels enabled the SVS to influence, and have insight into, the value translation process.

A recurring insight from the design projects was that societal value translation occurs through all types of design decisions, including those that designers might not perceive as value-laden. This aligns with Stolterman and Löwgren's (2007) observation that even small-scale design decisions can embody values. For designers, it can be tricky to know which decisions are connected to societal values and crucial for the SVS, and developing this sensitivity requires both time and resources. One example comes from the PlayIT project, where the team decided that one of the installations would consist of interactive eggs. While everyone initially agreed on this concept, the tech developers based their blueprints on measurements from commercially purchased eggs. When they shared photographs of this work in the project's team channel, the SVS objected, arguing that basing the installation on store-bought eggs conflicted with the project's goal of promoting awareness of local biodiversity. Rather than dismissing this concern, the tech developers embraced it. This initiated a domino effect of tasks, such as identifying appropriate local bird species, measuring real eggs, developing techniques to mold plastic as to reproduce the correct dotted patterns and colors, and so on. This, in turn, led to complications, as the LED lights did not display properly through the colored egg. It took much effort on the part of the project team to prioritize biodiversity awareness over simply making a generic egg. This dedication to detail was considered just as important by the SVS, as noted by the social enterprise CEO: "[the tech representative] could like spend half a year thinking about the material to the eggs, this attention to details [...] asking around 1000 different places, I feel like there is some sort of loving care in going in with like, wanting it to become really good, instead of letting it break down in the next phase."

This seemingly small example – one of many encountered throughout the projects – illustrates how greater attention to societal value translation also increases project complexity and requires additional resources. A designer needs to constantly balance competing priorities (Stolterman and Löwgren 2007), and doing so requires sensitivity to which decision decisions matter most to the SVS. In practice, societal value translation and elicitation become deeply intertwined.

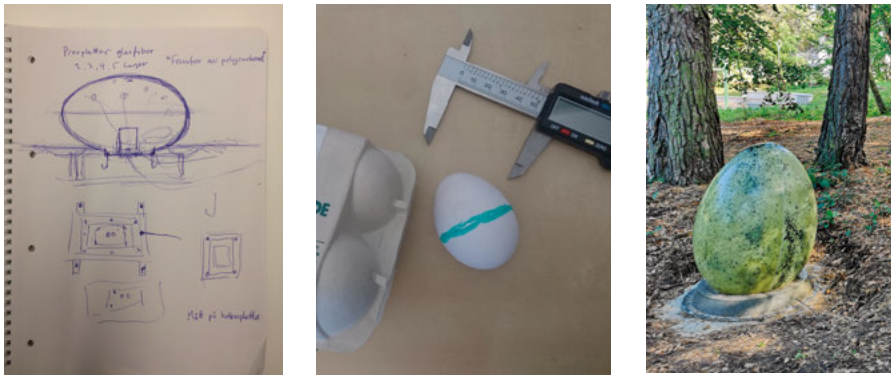


Figure 11. The shape, color and pattern of the egg installation in PlayIT sparked discussions about biodiversity, specifically as related to poultry and wild birds. Left: An initial sketch of the installation. Middle: A model egg, which met with pushback from the SVS. Right: The completed installation, speckled and colored to mimic a wild bird's egg.

Creating a design that truly resonates with situated societal values might require designers to engage deeply with those values themselves and, at times, deviate from standard design procedures, as was seen in the Castle Museum project and the development of the throne installation. Early on there were discussions about creating some kind of interactive chair, partly inspired by the seats in the story hut developed in PlayIT. A separate discussion focused on highlighting female historical figures, particularly 17th-century ruler Queen Kristina. Doing our research, we found that her throne had been preserved, along with her *maxims* – words of wisdom she had written down throughout her life. This gave us the idea of making a playful interactive throne where visitors could receive quotes from the queen. The SVS liked the idea and decided to implement it. Even after the design concept was established, numerous small translation decisions affected how well the installation mediated the intended cultural heritage values. Close collaboration with the SVS continued throughout implementation to incorporate their perspectives into as many design decisions as possible. We also sought to establish a co-creative process that combined the museum's curation and development processes with design methods.

Value translation activities included choosing which of Kristina's 1,583 maxims to incorporate into the throne. We could have selected a handful at random, but that would not have resonated with how the museum team approaches curation, narratives, and historical sources. The museum's digital producer and I each bought a maxim book (figure 12) and bounced quotes back and forth through private messages. We set up a joint Excel spreadsheet in which we evaluated and categorized the quotes. We formulated a set of

selection criteria based on both interaction design and heritage concerns. The quotes needed to be short and easy for visitors to interpret, capture the complexity of Queen Kristina's thinking, mix relatable and period-specific reflections, contain statements with which visitors might both agree and disagree, and balance serious with more humorous observations. We also categorized the quotes into themes, including ruling and politics, marriage, work, leisure, history, religion, gender, enemies and punishment, court life, money, health, knowledge, happiness, human nature, appearance, forgiveness, duty, hope, love, and friendship. Together, we created a shortlist of 78 quotes. We then conducted an online survey with the broader museum team, asking participants to rate the quotes on a scale of 1-5. After the voting, a more qualitative discussion was held. The museum staff primarily focused on the quote content from a cultural heritage perspective, while I sought to balance those concerns with quotes that were entertaining, unusual, and likely to engage and entertain visitors. Based on this, we selected 37 quotes. I recorded the selected quotes at home and built a portable prototype – a chest that played a quote whenever it was opened (figure 12). The prototype was tested over several months with the SVS, potential visitors, and audiences at events such as conventions. Based on insights from these evaluations, a final selection of quotes was decided upon. I also took on the literal role of translator, translating the quotes into English to make the installation bilingual.

The next step involved the voice recordings: what did Queen Kristina sound like? To explore this, we invited several voice actors to record sample quotes using different tonalities, from which the museum staff selected their preferred version. The museum team described wanting the voice that reflected Kristina's age and authority while also conveying the ambiguity of her writings, where some quotes expressed confidence and others sounded insecure or contemplative. During the actual recoding, I sat in the studio directing the voice actor on the intonation and emotional qualities that best suited each quote.

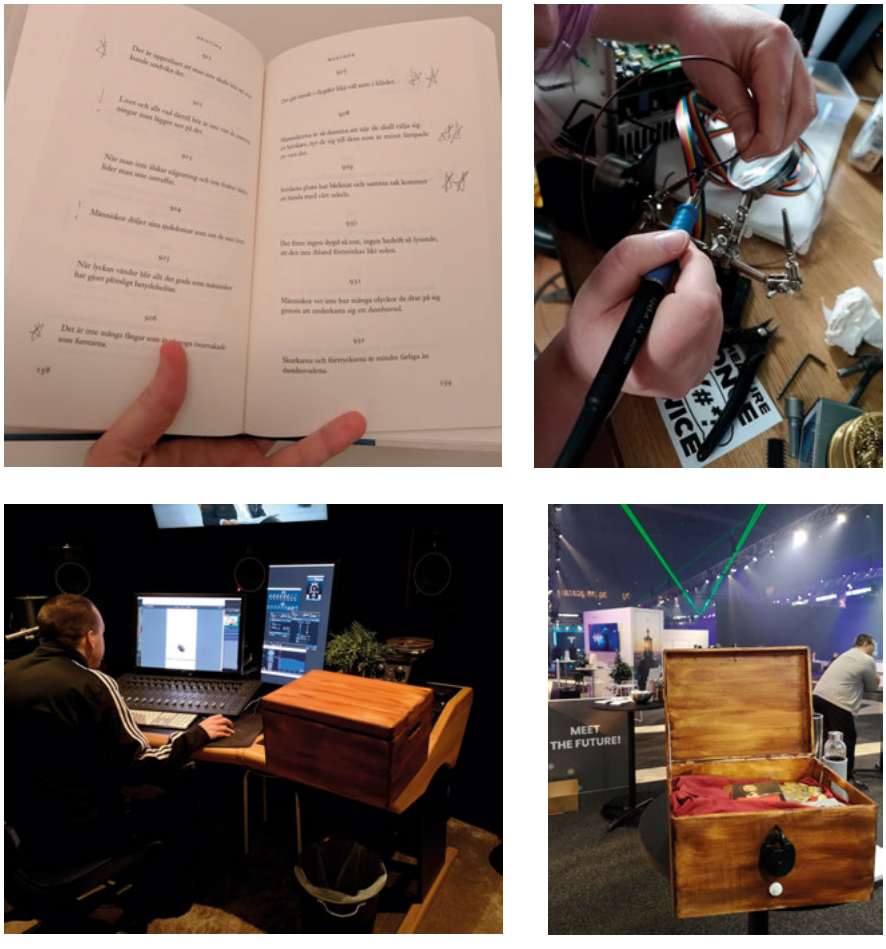


Figure 12. The Silver Throne design process, engaging with cultural heritage content and design intermediary. Top left: A page in my Maxime book, with my notes on quote 921-932. Top right: Building the prototype. Bottom left: At the sound studio recording Kristina's voice. Bottom right: One of the prototype tests at a convention.

When we installed the throne, I realized that I had become something of an expert on Queen Kristina's maxims – an expertise not typically associated with designers. This complex process ultimately boiled down to a 20-second interaction for each visitor, yet one containing deeply embedded cultural heritage content. Rather than bringing design methods to the museum in a one-directional way, we combined design methods with the museum's own curation and exhibition development process, including their emphasis on authenticity, rich engagement with sources, and careful construction of narratives.

The museum project manager reflected on the throne during the final project interview, conducted a year after the museum's inauguration. According

to her, it had become the most successful installation at the museum. She explains why by referring to both the interaction design values and its contribution to the museum's värdegrund and cultural heritage mission:

> I think that the most successful installation is definitely our throne, and it is just because the threshold for interaction is zero. You feel very invited to sit down—you sit down, and you receive wonderful maxims, or words of wisdom, from Queen Kristina. They're often very humorous, but there's also a seriousness to them, and they open up for a lot of discussion about ethics and värdegrund —from the 17th century to today. What has remained the same, and what has changed? That's really what you can explore around the throne: you can talk about the abdication that took place just a stone's throw from the museum, but you can also discuss broader, important questions that are more personal. I think it works extremely, extremely well for a very wide audience, including our primary target group. So I think it's a hugely successful installation!

As can be seen from the quote, the SVS considered the process successful in translating the sought-after societal values into the design. The quote also points to how the careful curation of quotes, in combination with interaction design, supported this value translation.

Article IV expands on the other design developed for the castle museum – the Archive Quest. The article proposes a framework for combining interaction design and learning theories to design for meaningful cultural heritage engagement at museums.

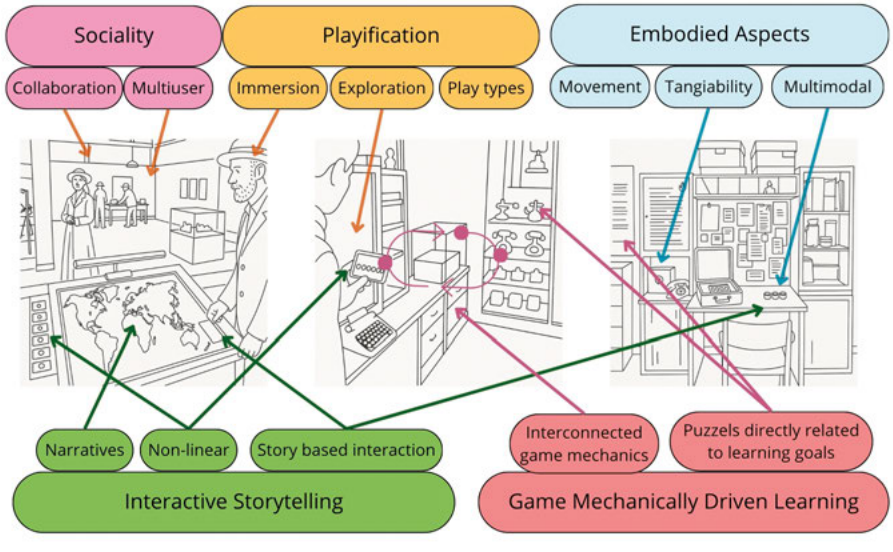


Figure 13. The Edu-interaction design framework from Article IV, based on insights from the Quest Room design.

A particularly interesting aspect of the framework, in relation to the thesis, is its emphasis on game mechanics as drivers of learning, where puzzles are directly connected to situated learning goals rather than being generic or superficial. This can be related back to how the museum staff in the GIFT subproject found the design solution banal, whereas the Archive Quest aligned more closely with the sought-after values. The museum staff consistently preferred quests that engaged directly with the museum content:

> When it comes to the archive quest, I think it turned out really nice. It was a process there and it's a bit like maybe... I might have wanted to do it even more so that we could have gone back to the exhibition so that it would have become like this is the essence of what we have, what you've experienced and learned. There I think the brush quest is a great example of that! [...] I might have wanted to see that we had worked even more with our, like, foundational narrative. But I still think that it works very, very well. Partly because the archives have a very important role for us. It becomes very important to me, it is, not a heart maybe, but it's a very, very important organ in our body, museum body and that's why I think it's important to put some effort into it. And I have to say - the quests feel very fun and very exciting, pleasurable, and it's a good final installation [in the exhibition] because it really sums up very much of the [museum] experience.

The brush quest mentioned in the quote centered on a particular artifact: a brush that had been manufactured by a prisoner at the castle and later used in the county governor's household on the other side of the castle. By engaging more broadly with the whole exhibition development process and immersing ourselves in the narrative and historical research preceding the exhibition, we discovered the brush and its year marking on the bristles. We recognized its potential as the basis for a quest that fulfilled the intended value goals for the quest room design, such as showcasing how preserved artifacts provide historical insight and highlighting lesser-known historical figures, such as the prisoner who made the brush.

The examples of value translation presented in this section point to the importance of listening to and understanding the priorities and ways of working of the SVS. As a designer, it may be necessary to immerse oneself not only in the technicalities of the design but also in the subject matter and situated values. It was found to be important to combine designerly competencies with value competencies during the value translation process, while remaining sensitive to the SVS' current practices when developing design solutions. Furthermore, value translation was found to be most effective when treated as something that permeates the entire process and informs all design decisions, rather than as a complementary set of requirements to satisfy.

The next section focuses on how those values are actually manifested in the designs.

Societal Value Manifestation

Spring 2026. I was at a dinner reception and was seated next to an older Brazilian couple. We tried to communicate through rather broken English. My friend had told me they mostly preferred shopping when visiting Sweden. They were on vacation, visiting relatives, so I asked what they had done in the city so far. They told me they had visited a museum at the castle. The man lit up, pulled out his phone, quickly scrolled through his photos and, with a big smile, showed me a picture of himself sitting on the silver throne, wearing a flamboyant cape and crown while striking a regal pose. They both laughed heartily, pointing at the picture and imitating the pose. They exclaimed that the queen had spoken to them. Then the woman looked at me intently and told me, in her heavy Portuguese accent: 'That queen, Kristina – she ruled, then left the throne. Very strong woman!' Then she nodded thoughtfully. When I told them that I had co-developed that throne, they were genuinely surprised.



Figure 14. Children trying on costumes before sitting on the throne.

The situation described in the vignette was particularly interesting because it captured a spontaneous, everyday interaction with one of the cultural interventions we had developed. It illustrates how the sought-after societal values we had tried to address through the design had been manifested in the artifact, and, through it, mediated to visitors in a way that remained with them after they had left the museum.

The vignette points to several intended societal values manifested in the design. The experience was designed to be accessible to non-Swedish visitors, to be social both on-site and through sharing photos on mobile phones, and engaging with history through playfulness. Beyond simply creating an enjoyable experience, the installation sought to encourage reflection and discussion about cultural heritage, something the woman in the vignette was apparently doing several days after her museum visit. We aimed to evoke curiosity about cultural heritage, and she had not only sat on the throne but had also read the exhibition signs about Kristina and her abdication.

The museum educator, who spends his days out in the exhibition, highlighted how the throne resonates with the museums' intended cultural heritage values, while remaining engaging for visitors. He described this vividly in the final project interview:

> I think that what is very good about these installations is that they are very—they are emotional. That is, they do something to us as human beings; they set our—well, they set our feelings in motion. You see the throne; you want to sit down. [...] Now, it is very anecdotal—when walking past. But it is, like, men and women. It is children, it is adults, it is groups of children and adults, it is individuals, [...] It is from all kinds of ethnicities, [...] the throne, there everyone participates. I think that's very… well, I think that's the key there: that it's very enjoyable, [...] I see it as a key to our relationship with history. Because—or the way I see it—is that when you've been to a museum, you should be transformed a bit as a person. You should feel that "yes, history concerns me, this is interesting, and it matters to me." And the very first step, to even reach that point where you believe it concerns you, has to be that you find it interesting, or enjoyable, or exciting. And then you can work with it being fun, or shocking—something brutal or uncomfortable, or grand and bombastic, or very beautiful or strange. This [the throne] becomes a way of lowering that threshold enormously, while still not compromising on the factual substance.

The quote highlights how the museum educator wanted the installation to mediate a specific kind of engagement – encouraging visitors not only to enjoy the experience but also to care about the cultural heritage it represents. It also suggests that museum professionals have a nuanced understanding of emotional responses that can promote cultural heritage awareness. Through the combination of their situated value competence and interaction design, societal values were manifested in the throne in a way that also engaged visitors.

The feedback we received on the throne installation stands in stark contrast to the museum staff's evaluation of the museum game in the GIFT sub-project. There, the sought-after values were not successfully manifested in the prototype. One of the SVS remarked, "This is fun but too shallow, it lacks the necessary complexity to be relevant from the perspective of the museum, silliness isn't a problem, it's the lack of meaningful follow up." Although the installation engaged visitors, it did not encourage meaningful engagement with the museum content and was therefore not perceived as successful by the SVS.

Determining whether societal values have been meaningfully manifested in a design is complex. As the example from the GIFT sub-project demonstrates, positive user studies alone are not enough to satisfy the SVS.

Our findings show that SVS also evaluate designs through their professional societal value competence, in the sense described by Kavathatzopoulos (2009). This evaluation might be expressed in terms of how an installation feels or whether it feels right. However, such judgments can be understood as expressions of a moral compass that has been developed through immersion in the situated values, missions, and organizational culture of their domain, in the sense described by Waeras (2018).

Furthermore, the SVS acknowledged that the societal values they seek to promote might not always align with the preferences of individual citizens but instead reflect a collective perspective. The municipal city gardener in PlayIT reflected on this in relation to the playground:

> It is always like this: An individual citizen, you have to understand that, an individual person looks after - me and my needs: outside my door. They don't think about that, but now we have quite a few doors and also environments outside people's doors that we have to think about, and then you have to think about a whole, and equity and gender equality and so on. [...] In the playground, you have to incorporate all these values, that it is valuable for several ages, that it is valuable for boys and girls [...] That's why I think a playground like this has quite a broad scope.

She further acknowledged that some decision decisions would inevitably disappoint certain groups while benefitting others:

> It is like this: you can never please all the citizens, so that everyone stands and applauds. There are always those who must say a lot of negative stuff, and usually it's like this: Well, what has it cost the taxpayers? It's a bit like that, but at the same time I think like this: no, but the children are really important. [...] that's the main thing, you know? I don't care what Nisse 86 thinks [...] He can actually diss it if he wants, but if little Lisa five years old thinks it's absolutely fantastic, then I think, then we've succeeded. [...] you're used to that, it's a bit like that with municipal citizens, it is not possible to get everyone aboard the train.

Finally, the municipality representative found innovation to also be a goal in itself, arguing that municipalities have a responsibility to pursue innovation solutions:

> It is like this, that the municipality wants to be at the forefront, right?! It's always like that with municipalities, we share best practices, there is no top secret, and we share good ideas, but then, as a municipality it is always fun to be first on the field, first on the ball.

We have found that implementing designs gives rise to new types of value conversations. In particular, the value gatekeeping role of the SVS becomes more pronounced as the stakes get higher. An example of this is brought up in Article IV, where in the design of the Quest Room, the SVS were not concerned about the inclusion of a fictional character until the design moved toward implementation. Likewise, in the GIFT sub-project, the SVS did not voice many concerns during prototyping but later stated they would never accept such a solution for an actual implementation. In both projects, unresolved value tensions thus became apparent as the design moved toward realization.

For PlayIT, the different SVS representatives emphasized different values that had been manifested in the design. The social enterprise representative highlighted the biodiversity theme: “To like, use the story world, to establish a relationship with nature both there [in the hut] and further out.”. The municipality’s city gardener instead emphasized the installation’s malleability:

> I think it was a really fun project and a good project because I think that this thing with playgrounds is so important to develop. It becomes even more important to involve children in the city and all this and with playgrounds; that we really have good playgrounds and this particular thing that they should be a bit changeable without having to keep changing equipment or adapting for age groups that this was so wonderful that you kind of included a broad spectrum, otherwise it would be very niche playgrounds and of course they have to be. But this was actually a very special playground.

These two value perspectives are explored further in Article I, which focuses on nature-integrated storytelling, and Article II, which focuses on malleability. While both representatives perceived important values as being manifested in the design, their reflections also illustrate that a single installation often needs to support multiple societal values. Different SVS bring different value priorities to a project, and each additional stakeholder contributes new value priorities and perspectives.

While the municipality’s city gardener was satisfied with the malleability functions, she repeatedly highlighted the need for continual renewal: “But what I also feel is the concern that you really have a plan so that they continue to develop and be fun, that you kind of don't stop there.” She explained that

municipalities often move slowly because they take a long-term perspective, focusing not only on implementation but also on the continued development and lifespan of public installations.

One design cannot do everything. Limited resources and other constraints mean that some ideas and values inevitably remain unfulfilled. One example comes from the PlayIT project, where there were proposals to integrate the experience with the nearby hospital, enabling hospitalized children to participate in the play remotely –for example, by interacting with an egg in the hospital that would affect an egg at the playground. The city gardener became emotional when describing how important she thought such a solution would have been, and expressed regret that this particular design idea had not been realized: "I thought it was so very beautiful, the connection to the hospital, to be able to join the play despite being indoors. [...] and I still think about that at times, could we not make that happen?"

Unintentional Value Manifestations

Not only intended values become manifested in designs. Because societal values are complex and interconnected, designs may also give rise to unforeseen value manifestations that in turn spark new value conversations. A striking example emerged during the final months of the PlayIT project. The playground installations had been inaugurated and well received, and project meetings had been reduced to biweekly check-ins. In one such meeting, the municipality representative brought up an issue that had come to their attention through complaints from citizens. The cozy story hut, with its enclosed walls, roof, and dim lighting, had become a place where homeless people slept during cold winter nights. This unexpected situation triggered a cascade of value reflections. We discussed hostile design, whether all citizens can be considered users, and how the situation should be addressed. While many of these issues fell outside the value scope of the project, they were very much part of the realities surrounding the implemented design.



Figure 15. Left: The Story Hut in the PlayIT project, which became a shelter for homeless citizens during dark and cold winter nights. Right: Some trash and a cigarette end found in the Story Hut.

Real-world implementation makes it impossible to isolate which societal values should matter. The unintended consequences of the story hut sparked a broader value conversation. Should homeless people be allowed to sleep there? Was there a risk of this negatively affecting children playing there? Should the municipality prevent people from sleeping in the hut? Or, conversely, should this be viewed as an unintended but valuable social function of the design, leading to adaptations that better supported its purpose as a temporary shelter? These discussions surfaced numerous competing values with immediate real-world consequences. In this way, the unintended value manifestation became the catalyst for new value elicitation.

Value Manifestations Throughout a Design Project

Values are manifested in final design outcomes, but they are also manifested in sketches, probes, and prototypes. Our findings highlight that progressing from design concepts to advanced prototypes means that the manifestation of societal values becomes increasingly granular and concrete. This gradual shift – from abstract project values to specific design features – can be used to iteratively promote value elicitation based on how values become manifested in the evolving designs. This aligns with Verbeek's (2006) view that technologies and designs can serve as tools for discussing moral questions.

In the ABC project, prototypes and probes were intentionally designed to function as springboards for societal value conversations. This process is described in more detail in Article III. The result was an ongoing interplay between manifestation and elicitation. The approach was appreciated by the project team, as exemplified through the following reflection from the NGO lead after discussing one of the probes during an interview: *"The interview she had with me. It felt like; now we're talking about what this is about for me, in my involvement. [...]. These different designs that she came up with, there were values in them that felt to me. [...] it feels like this is like: Yes! This is what you would want to integrate into the project."* The quote highlights how the combination of probes and interviews facilitated elicitation. The interviews also encouraged SVS to discuss very specific design features in relation to the intended values. One probe consisted of interactive ropes hanging from above (figure 16, right). The community manager objected to this proposal because of the *"risk connotating snares for hanging"* and argued that the design could therefore be counterproductive to the sense of community intended.



Figure 16. Left: A design probe that the SVS felt resonated with the sought-after values in the ABC project. Right: A probe with hanging ropes, whose connotations were not appreciated by the community manager.

Moving on in the ABC project towards prototyping, the sustainability manager reflected on a value-focused prototype session: "It feels pretty fun that we jump and dance around this paper-covered Pilates ball and it kind of ends up being, partly a work of art, but also like this, well, here we pull out like specific words and theories and here is a process and it fits into this part of our process in how we build a neighborhood community." The quote highlights how facilitated design activities helped the SVS to bridge the gap between abstract societal values and the values manifested in the design proposals.

Values are manifested in all designs. Therefore, it can be fruitful to evaluate how existing designs manifest values before developing new solutions. In the ABC project, this was done by creating a collection of inspiration cards, each

featuring an existing design related to a sense of community. The project team analyzed the cards, discussing how successfully each example manifested values that were important to the project.

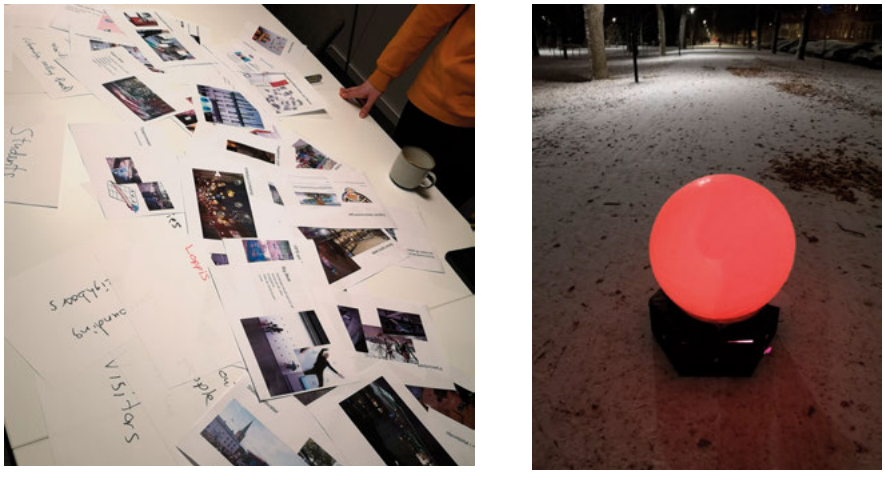


Figure 17. In the ABC project designs, manifesting values was used as a springboard for value conversations throughout the project. Left: Work on inspiration cards with existing design solutions, discussed in relation to societal values. Right: A high-fi prototype that SVS engaged with while reflecting on the values it manifested.

These examples showcase how value manifestation can be relevant from early phases and throughout the design process. Throughout the projects, we found that SVS sometimes struggled to come up with ideas for translating societal values into playful cultural experiences, particularly when technologies were involved. However, they were highly competent at evaluating proposed designs from a societal value perspective, provided they were given sufficient opportunities to do so.

Ultimately, it is important to think about both the intended and unintended values manifested in the design solutions developed in the projects, and whether the sought-after values are meaningfully realized through the resulting cultural interventions.

Value Manifestation and Evaluation

There is a need to identify appropriate ways of evaluating how societal values are actually manifested in designs. Determining whether a design promotes the sought-after societal values can be challenging, given the soft nature of such values (Wæraas, 2018). Throughout the projects, different methods were employed. In Article I, qualitative in-the-wild observations and thick

descriptions are used to reflect on how the Story Hut promotes biodiversity awareness among users. While many metrics and methods for measuring societal value impacts could be used, it is equally important to acknowledge the problematic side of over-measuring soft values (Wæraas, 2018; Flanagan et al., 2008).

When evaluating the values manifested in a cultural experience, designerly qualities such as engagement, usability, and user experience matter, but they must be balanced with societal value aims. As a designer, finding solutions that align both with societal value goals and interaction design priorities can create synergies. This is explored further in Article V, where the Quest Room design is analyzed through a combination of educational and interaction design theories to identify design solutions that create synergies between them.

One way to validate the manifestation of societal values in design is to rely on the value competence of the SVS (Kavathatzopoulos, 2009), who function as value gatekeepers by approving designs in relation to their values (Krippendorff, 2014). The findings suggest that obtaining honest feedback from SVS requires both relationship building and creating opportunities for them to voice concerns by reflecting on design proposals, prototypes, and implemented designs in relation to societal values. That the SVS recognize and approve the design solutions is important in itself for designers collaborating with such stakeholders. Moreover, their professional value competence can strengthen the incorporation of societal values into the final design solutions, complementing other evaluation methods.

Societal Value Stakeholders in Design Projects

An interesting observation from the studies is the extent to which the SVS representatives represent their organizations' values. This can be seen in how the municipality representative referred to "us municipalities" and how the museum project lead spoke about "our museum body." In this sense, the SVS can themselves be understood as manifesting and embodying their organizations' values.

An overarching observation across the design projects is that the SVS have a high level of value competence, but are generally not used to combining this competence with a designerly mindset. This is captured in how the NGO lead in the ABC project described the experience of working with design methods:

> It's been very stimulating [...] using design thinking in meetings, and in strategic planning of the project, I think it's really cool. But I would like to understand better what I'm doing, or what you're doing [pointing to designers in the room] too, how you're part of it. I let it happen; I expose myself to it and I think

> this is crazy fun, but I don't know the mechanism in the carousel, like, what makes it spin.

This reflects a recurring view expressed by SVS throughout the projects: while most found the design methods novel and appreciated using them, they found it a bit hard to understand what design entails and how design processes unfold.

The studies suggest that there is a need to familiarize the SVS with design methods and to be transparent about why and how the process is structured. This aligns with Manzini's (2015) observation that, especially when developing digital solutions, social innovation projects can easily develop power imbalances that favor professional designers. The challenge of achieving collaboration on equal terms, while utilizing design methods that fit the situated practices of the SVS, has been a central focus of the studies. The view that value competence and designerly competence should meet on a peer level was articulated by the social enterprise lead in PlayIT as a key success factor: "Maybe it was what made it feel so very good to work in this project – that we all were curious about each other's competences."

In the projects, several SVS representatives found the creative sessions and the design process both challenging but rewarding, as discussed further in Article III. The sustainability manager in the ABC project reflected:

> For me personally, it has been a challenge. It has been a lot of fuzziness for me, and I have been quite uncomfortable with it. And I have sort of learned to let go a little. I come from another world where meetings are strict and orderly. [...] it has been aaargh, hard at times but also incredibly educational about myself [...] And then I'm actually incredibly proud of what we have delivered [in the project] like oh shit, incredibly proud.

As the quote illustrates, the ambiguity of the process was experienced as both challenging and rewarding. Embracing uncertainty and staying with the trouble is something that most designers, at least within creative design practices, are relatively accustomed to. However, working in multisectoral design projects requires combining this mindset with other perspectives and ways of working, while remaining sensitive to how creative practices are implemented. The municipality's city gardener reflected on the process:

> It was a well-composed group too, because it's really fun that there are many who are creative and throw up as much as they want. But then there are these somewhat stable ones who can think about technology and technical feasibility, which may not be someone else's strong point. [...] It's also important in groups like this that there is this mix because you can't just sit and throw up ideas. Then something has to be boiled down and that was it. It was done too, and there were those who could, like yes: here we have something. This is technically solvable.

The quote suggests that the ability to synthesize ideas and contribute technical competence was sought after and appreciated. Interestingly, when asked what she contributed in the process, she replied that she did not quite know. This may indicate that SVS representatives do not always recognize the importance of their own value competence. The studies indicate that the SVS representatives possess a high level of value competence, and that societal values guide much of their work, informing everyday decisions and serving as a measuring stick through which they evaluate outcomes.

One project that was found to really hit the mark, both in terms of the process and the resulting design outcomes, was the Castle Museum Project. In the final interview, conducted one year after the museum was inaugurated, the museum project manager, drawing on many years of experience from the museum sector, reflected back on our collaboration and its outcomes:

> I think it's been an incredibly exciting and fun process, and I also think it is wonderful that it has been a kind of test-lab environment where things have been developed and evaluated [...] I think you can really see it in the results: these are very well thought-out, highly functional installations that we are very grateful to have in our museum. They feel very immediate and playful, but at the same time there is knowledge to be gained through them. You can use them in so many different ways, which I find really, really enjoyable. I also notice that when we present these installations, many people become very interested in the project and in how we developed them—what the processes looked like. So it is very, very nice, and I feel proud to talk about it. This collaboration feels really great.

There is no ambiguity in the quote – from the museum's perspective, both the project and its outcomes were a success. Something resonated with the SVS and with the societal values they sought to pursue.

The Castle Museum Project is the most recently initiated design project in this thesis. By the time this project began, insights from previous projects, the literature, and ongoing studies had already shaped the development of the designerly worldview. This meant that methods and approaches to designing for societal values could be implemented in the project from start to finish. While each design project is, in a sense, an ultimate particular (Stolterman, 2008), the insights presented in the thesis have proven generative across multiple design projects. This does not imply that every future project adopting a Design for Societal Values approach will succeed. But it does suggest that these approaches can, from an SVS perspective, contribute to successful collaborations and cultural interventions that meaningfully embody the sought-after situated societal values.

Seven years before the final interview in the Castle Museum Project, we participated as RtD researchers in the GIFT sub-project. Like the castle project, it focused on museums, cultural heritage, and technology-enhanced exhibitions. Although the collaboration began well, the museum representatives, at a late stage of hi-fi prototyping, concluded that they would never implement the proposed mobile museum game (Waern et al., 2020; Rajkowska, 2022). Their critique of the design solution was harsh, and the entire design endeavor was discontinued. Although the discontinuation meant that considerable resources had been invested in a design that was never implemented, the project generated many valuable insights for themuseum professionals, the designers, and the RtD researchers.

Between the breakdown of the GIFT sub-project and the encouraging final interview in the Castle Museum Project, much happened from an RtD perspective. The Play IT playground was developed and inaugurated, the ABC project was initiated and completed, with hundreds of meetings, sketches, workshops, user tests, interviews, prototype iterations, literature reviews, courses, conferences, and article writing sessions. Out of this inherent messiness of RtD (Gaver, 2022) and probing in different design projects, insights gradually emerged regarding the challenges of collaborating with SVS, pursuing societal values through cultural interventions, and addressing these challenges from a design process perspective.

Core Design Methods of Design for Societal Values

This section ties back to the designerly worldview presented in the method chapter. As previously described, the designerly worldview was developed iteratively throughout the design projects, and retrospectively, some core design methods have been identified as particularly generative for Design for Societal Values. Those core design methods, and examples of how they have been implemented, are presented here.

Deliberative Design Practices

The importance of a deliberative mindset among designers became evident throughout the design projects. This deliberation was ongoing throughout the projects as values and priorities were discussed and re-evaluated. While Buschiantella-Ricci and Scataglini (2024) describe a disagreement space, in these design projects this was instead approached as a deliberation space, where a shared goal of creating the design called for an understanding of different value priorities and the negotiation of joint decisions.

Balancing adherence with agency was found to be vital, and this balance was achievable in the design projects through sustained and transparent dialogue with the SVS, and by developing a deep understanding of each other's perspectives. In practice, this involved both specific design activities and an overarching collaborative approach to the SVS, in which the SVS were treated not as informants but as peer collaborators who contributed an equally valuable form of expertise.

Relationship building. Societal Value Mediation was not accomplished in the projects through a few stakeholder workshops followed by the production of a list of values. Instead, it relied on long-term relationship building, with the aim of genuinely understanding both the stakeholder organizations and the individuals involved through both formal and informal activities. Building trust and understanding over time proved fundamental to the process. Article III describes this relationship building in more detail.

Perspective-Taking. Understanding the complexity of societal values requires the ability to consider multiple – and at times contradictory – perspectives at the same time. Perspective-taking also fosters empathy among collaborators. Within a design team, it can support engagement with societal values through approaches such as roleplay, as explored in Article V. Article VI engages with a particular form of perspective-taking, and shows how combining the mindsets of design thinking, future thinking and systems thinking in design workshops can help address sustainability values. Perspective taking can also be embedded in the cultural interventions themselves. Examples include the throne installation, which invites visitors to adopt different historical perspectives, and the Story Hut, which encourages visitors to experience the world from the perspective of different animals.

Creative Approaches

Creative approaches were found to aid societal value mediation, if they took place in a way that respected the professionalism of the SVS.

Playfulness. Playful approaches, as suggested by previous research (Ehn and Sjögren, 1991; Senger et al., 2005), supported value conversations, provided they were framed in a way that felt meaningful for the SVS representatives. A recurring concept investigated in the projects is *playification* – the use of playful approaches to achieve purposes beyond play itself, in this case the pursuit of societal values. This concept is developed further in Articles I, II, IV and V. Playful methods were integrated fruitfully when approaching values together with the SVS. As the social enterprise lead in PlayIT reflected: "There was a big passion in the approach. We solved problems when we met them, without the passion becoming, what to say, unprofessional. It was a very

good feeling that everyone was open for playfulness and that there was a lot of upsides [to this].” This quote sums up how playfulness was appreciated when embedded within a professional approach.

Tangible and embodied approaches. These design activities helped transform abstract values into concrete experiences. Roleplay exercises, prototype building, and physical manifestations of values all facilitated societal value conversations. These findings align with previous research on boundary objects (Doolin and McLeod, 2012; Schoormann and Kutzner, 2020), suggesting that boundary objects can function as springboards for value conversations from early sketches through to implementation. As designs become more concrete, so too do the value discussions. However, realizing this potential requires creating space for and facilitating such conversations and remaining open to revising designs in response to the insights they generate.

Storytelling and metaphors. These were consistently used to facilitate discussions about societal values, aligning with previous research (Junginger, 2017; Ehn and Sjögren, 1991; Sengers et al., 2007; Krippendorff, 2006). Interactive storytelling, in particular, proved to be a powerful way to engage people with values, as discussed in Articles I and VI, and in the supporting publication *Interactive Storytelling for Community, Heritage and Citizen Engagement* (Paananen et al., 2024). Roleplaying and sensitizing scenarios likewise functioned as storytelling techniques that helped participants explore different value perspectives and build group coherence, as discussed earlier in this chapter and in Article V.

Chapter 7. Summary and Conclusion

The thesis is based on the research question, How can societal values be mediated in co-design projects with societal value stakeholders, particularly in the design of cultural interventions? This was studied through a collection of design projects that spanned over several years and involved a range of societal value stakeholders.

In order to approach different values in design, there is a need to differentiate between different types of values. This thesis shows how societal values can be understood as a distinct category of values pursued by societal value stakeholders (SVS). The thesis further demonstrates how the unique properties of societal values affect the design process. As shown throughout this work, placing too much emphasis on users alone may be detrimental to project outcomes, as illustrated by the GIFT sub-project, where the designer leaned on positive user evaluations, but the museum personnel rejected the design because it did not align with their societal value goals. This points to the need to complement user studies with other forms of validation. One such approach is to leverage the value competence (Kavathatzopoulos, 2009) of societal value stakeholders, provided that the design process enables them to voice their opinions and act as value gatekeepers (Krippendorff, 2014). While validation is important, this thesis also highlights the challenges of validating the inherently soft and intrinsic nature of societal values.

Current participatory approaches to value-oriented design typically engage people who are not used to reflecting explicitly on values (Friedman and Hendry, 2017; Bødker, 2017). When collaborating with SVS, the tables are turned: they possess high value competence and are professionally trained in navigating complex value landscapes, but they generally lack experience with designerly methods. As shown in this study, supporting their participation therefore becomes a matter of bridging their value competence with the design process. Approaches that proved valuable include deliberative design practices such as building personal relationships, maintaining sustained communication, helping stakeholders to understand how design processes unfold, and creating space for reflection throughout the project. Creative design approaches, including playfulness, roleplay scenarios, the use of metaphors and storytelling also aided value mediation. Further design activities such as probing and

prototyping could be leveraged to spark value conversations if the activities were adapted for value conversations.

Previous value-oriented design approaches, such as Design for Values, primarily focus on identifying values and translating them into design requirements. This thesis challenges that approach by showing that value mediation is an ongoing process in which values are continually deliberated – and thus cannot be fixed at a specific point in a design project. Instead, the model presented here advocates treating values as dynamic, evolving alongside the design process, its context, and its outcomes.

Social innovation projects are often carried out alongside, or as a complement to, the SVS' other responsibilities. Unless the SVS representatives find a project meaningful from a value perspective, they might disengage, as illustrated by the GIFT sub-project. This means that designers need to sustain the value relevance of the project throughout the process in order to maintain SVS engagement. This highlights the importance of relationship building and value conversations as core skills for designers. As demonstrated through the ABC project, value conversations in social innovation are not merely a means to an end – they are also valuable outcomes in themselves.

Contribution

This thesis contributes to HCI by providing insights into the perspectives of societal value stakeholders in creative design processes, based on real-world co-design projects involving the design and implementation of cultural interventions. The thesis conceptualizes societal values in relation to HCI and design, proposes a theoretical framework for analyzing Societal Value Mediation, and outlines the design approach of Design for Societal Values. The thesis further contributes insights into societal value stakeholders, showing that they typically possess high value competence while lacking experience in designerly methods. It proposes deliberative and creative approaches – including playfulness, tangibility, storytelling, perspective-taking and relationship building – that support societal value mediation in design processes. Furthermore, it demonstrates that value mediation is a fluid, ongoing interplay of elicitation, translation, and manifestation throughout a design process.

These insights can support the design of cultural interventions that better reflect societal values and help mitigate value-related project breakdowns in co-design projects involving SVS. The study also opens up avenues for future research on societal values and societal value stakeholders in design, while providing practical guidance for HCI researchers and practitioners seeking to strengthen design collaborations with these stakeholders.

Chapter 8. Postamble

Over the years, I have encountered so much frustration, communication breakdowns, and disappointment in design projects where designers have collaborated with SVS. These experiences have resulted in wasted resources, designs that failed to fulfil their intended purposes, and even the discontinuation of entire projects. But I have also seen the opposite: magical moments when collaborators found ways to convey societal values and produce design solutions that the SVS could not have imagined before the projects began. The IoT playground in Study 1, the outcomes of the ABC project in Study 2, and the Castle Museum installations in Study 3 all contain examples of how societal values were incorporated into design solutions in ways that resonated well with the SVS while also providing engaging experiences for visitors and other end users.

I have experienced both failures and successes while trying to convey societal values through design processes and design solutions. My eyes still sparkle when I get to witness the moment a group of people from different disciplines and backgrounds come together and begin to understand one another. Based on a shared purpose, a team transforms vague ideas, abstract values, and tentative sketches into manifested designs. I am still fascinated by how these processes change me, my collaborators, and our shared understanding. Seeing thousands of museum visitors, neighborhood residents, and children engaging meaningfully with the resulting designs makes the effort worthwhile.

Designing for Societal Values can be considered as the development of a designerly world view – one that is my own, but that can also be adopted and further developed by others in future research and in practice. During my PhD, I have been invited to serve as a speaker, workshop leader, and innovation consultant in several design projects beyond those included in this thesis. These collaborations have included organizations such as the Swedish National Heritage Board as well as museums, libraries, and municipalities. These invitations have come through recommendations and from practitioners inspired by the approach presented here. This suggests that, at least among SVS practitioners, the designerly worldview that has emerged resonates with them in ways that make them prepared to invest time and resources in implementing it within their own organizations. There are ripples in the water: others are

beginning to adopt and build on concepts and approaches developed through this work, both in research and among practitioners.

There are also so many small, personal moments that matter. An acquaintance is overjoyed to discover that I had a part in developing the Silver Throne, and she tells me she loves that installation so much that she has a printed photo of herself sitting on it, displayed in her office locker so she can look at it every day. An elderly visitor at the sustainability festival in the ABC project speaks with tears in her eyes about how this event gives her hope for her community. A young boy at the playground, together with his grandparents, finds comfort in the animal stories while waiting for his mom to receive treatment at the nearby hospital.

There is real-life magic in knowing that I helped make these moments possible through our design work and research. Designing for Societal Values, and finding ways to bridge designerly perspectives with those of SVS, is a challenging but worthwhile quest, and it remains a passion of mine.

After five years of PhD studies, I look forward to continuing to explore this topic. I hope this thesis marks not the end of that journey, but its beginning.

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